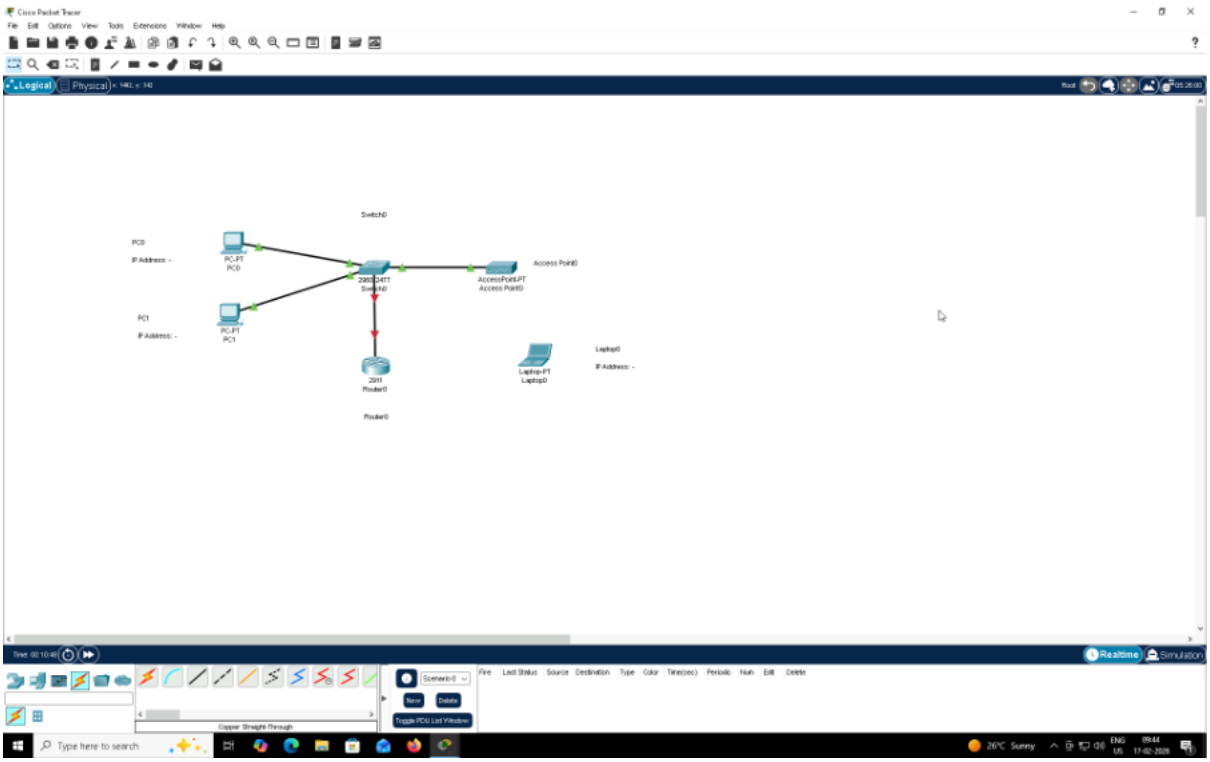
TOPIC – NETWORKING

Assignment 1: Network Device Configuration

As part of this assignment, you need to complete below tasks:

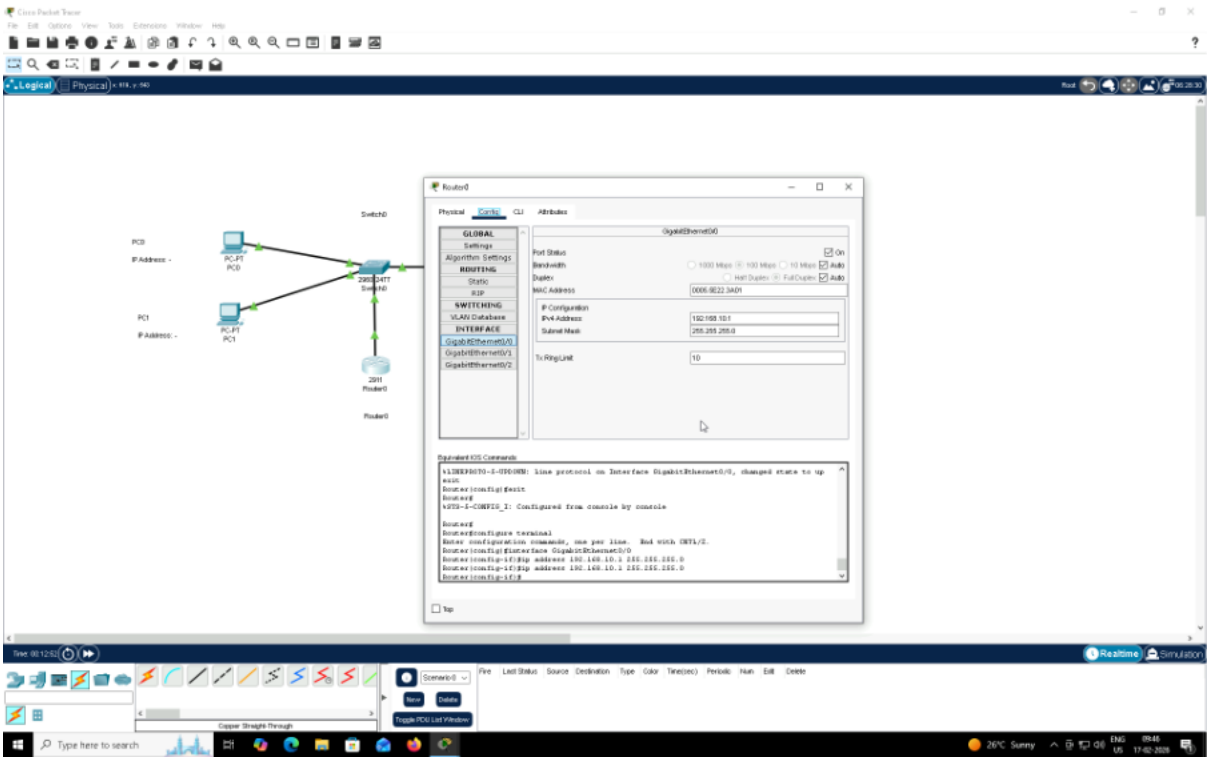
1. Set up a home/lab network with multiple devices (router, switch, wireless access point)
2. Configure IP addressing and basic network segmentation
3. Verify connectivity between devices using ping and traceroute commands



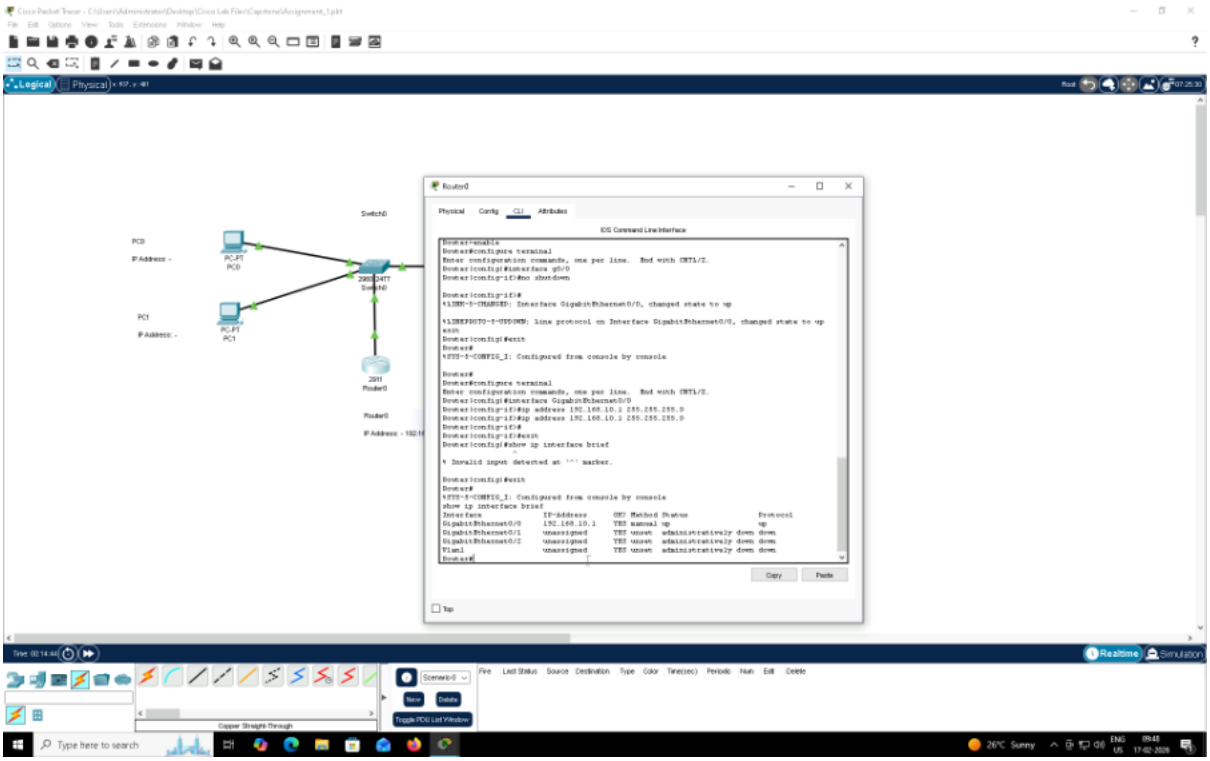
We will now assign IP addresses to devices and start the Router0 as well

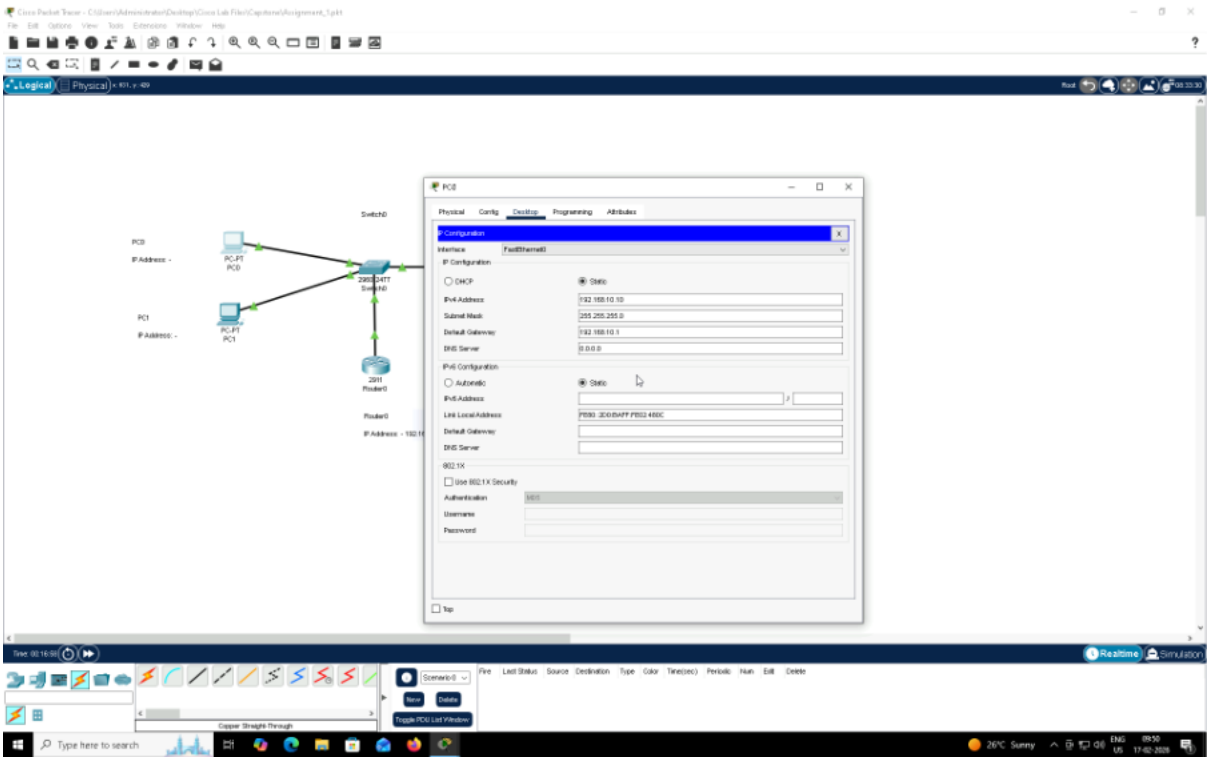
IP Addresses of: -

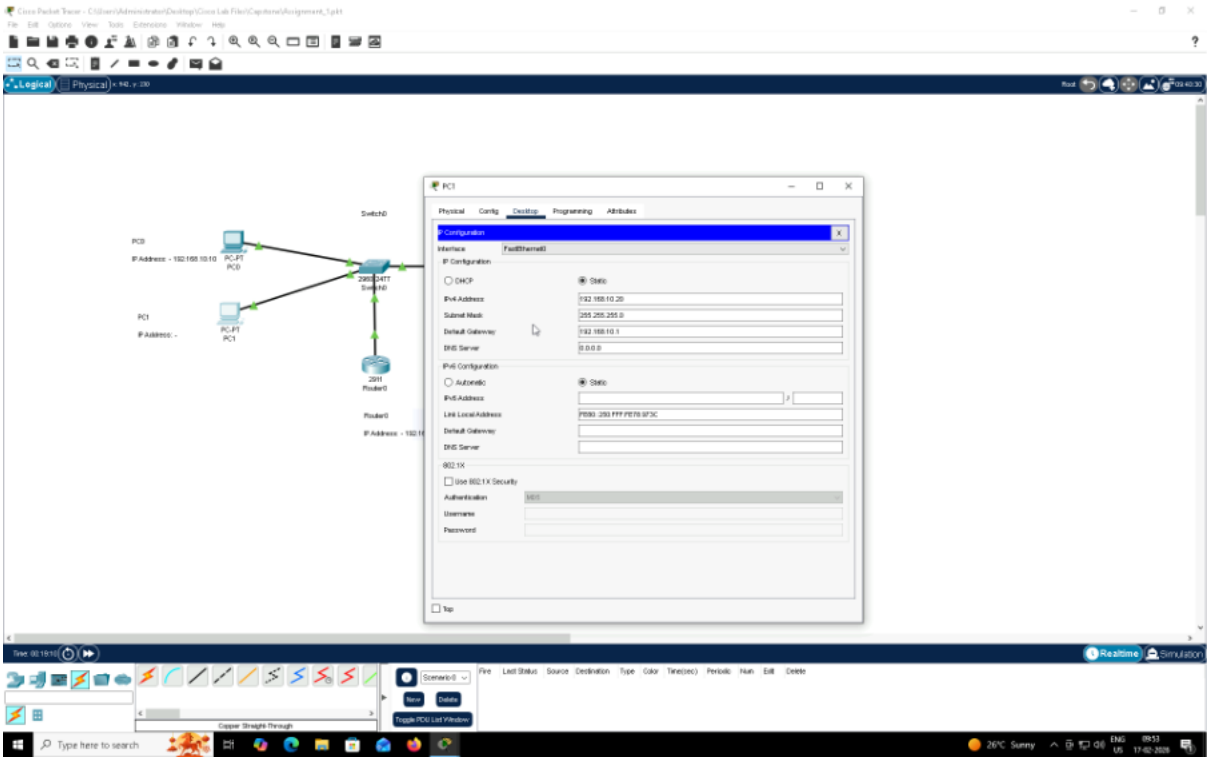
- Router0: - 192.168.10.1



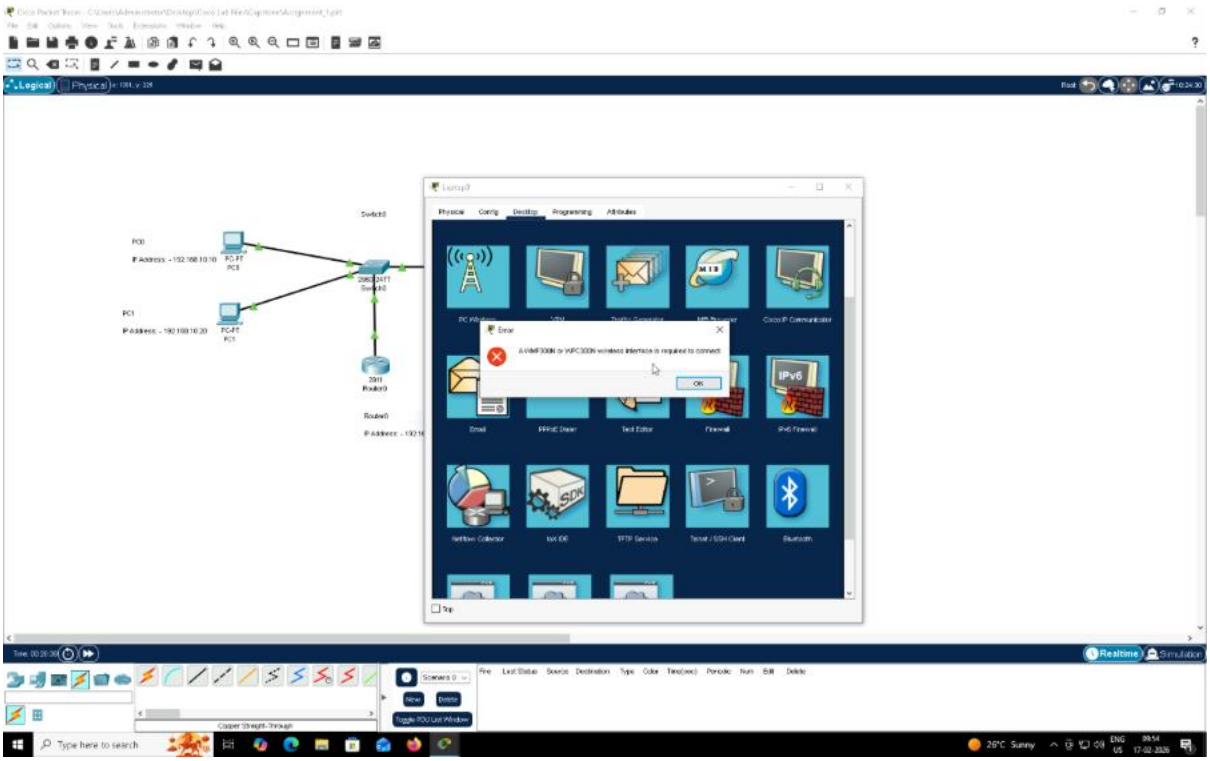
The Router Interface was configured with an IP Address to act as the default gateway for devices in the LAN Network.



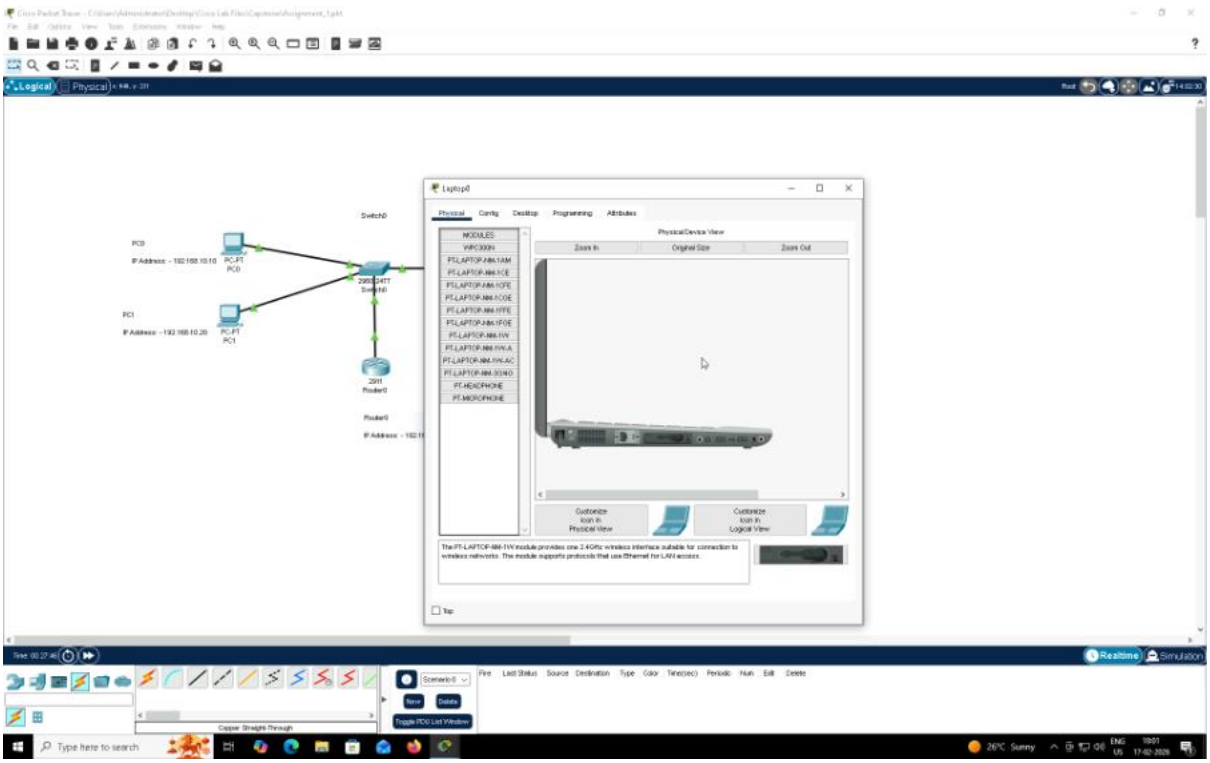


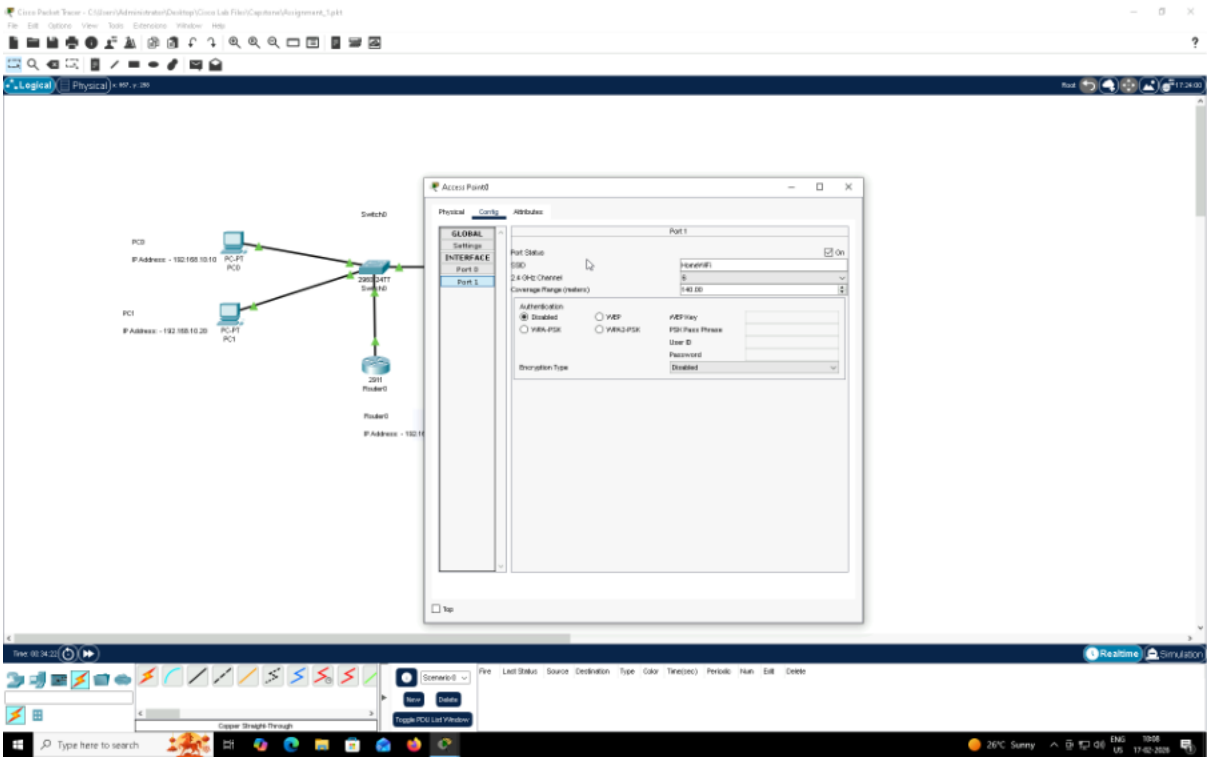


The IP addresses to both PCs have been assigned.

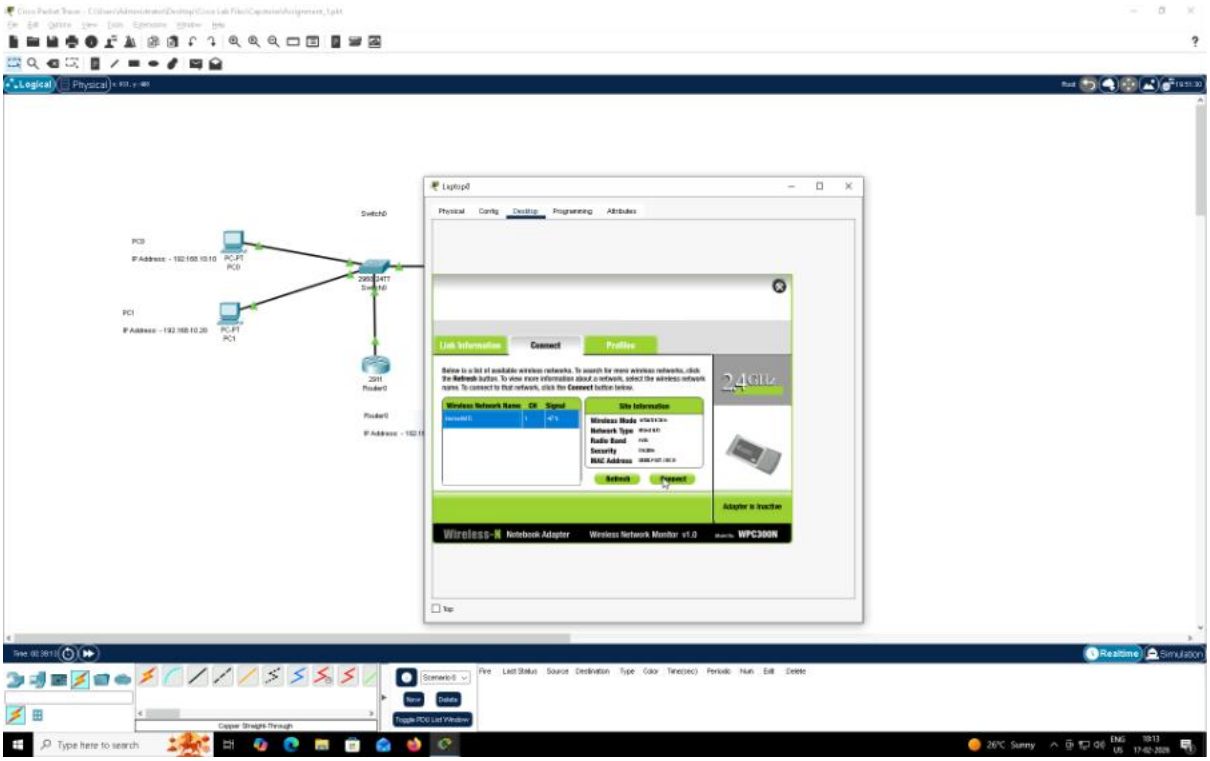


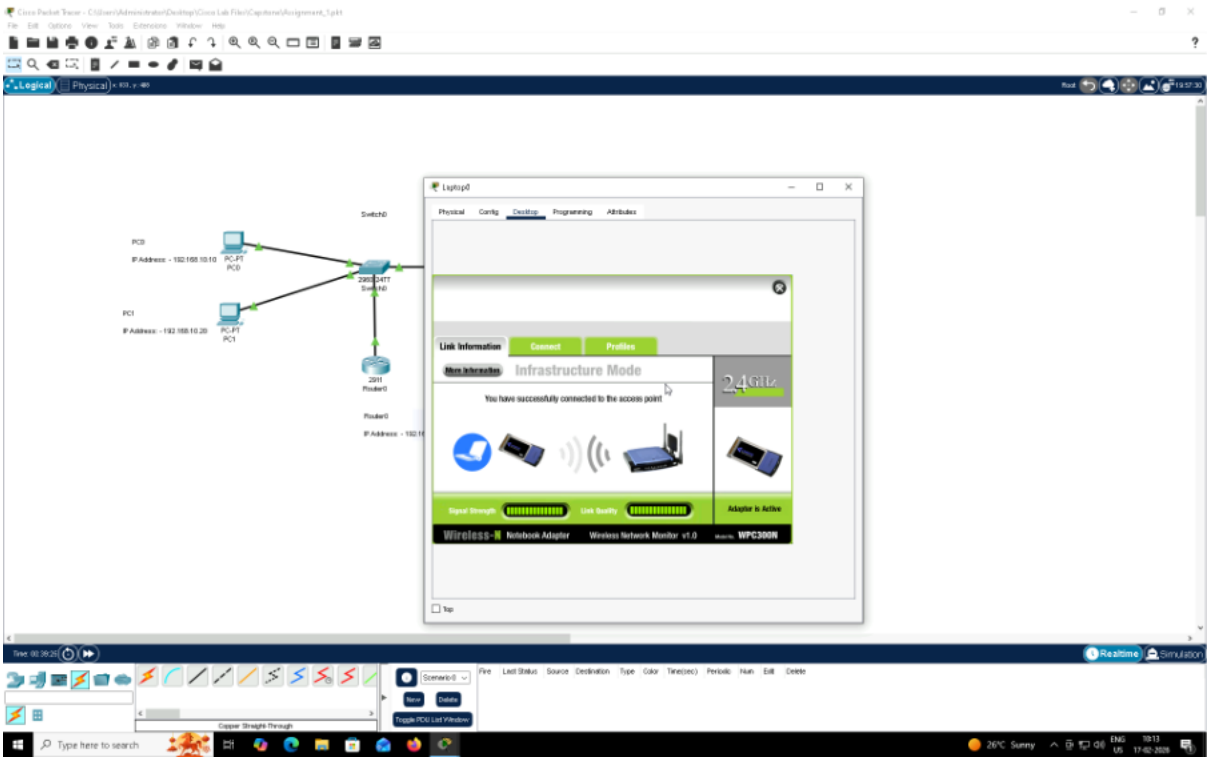
This error states that the Laptop0 has no wireless card installed and the laptop node in the Cicso Packet Tracer do not come with Wi-Fi by default so it should be added manually. We have added the wireless communication module.



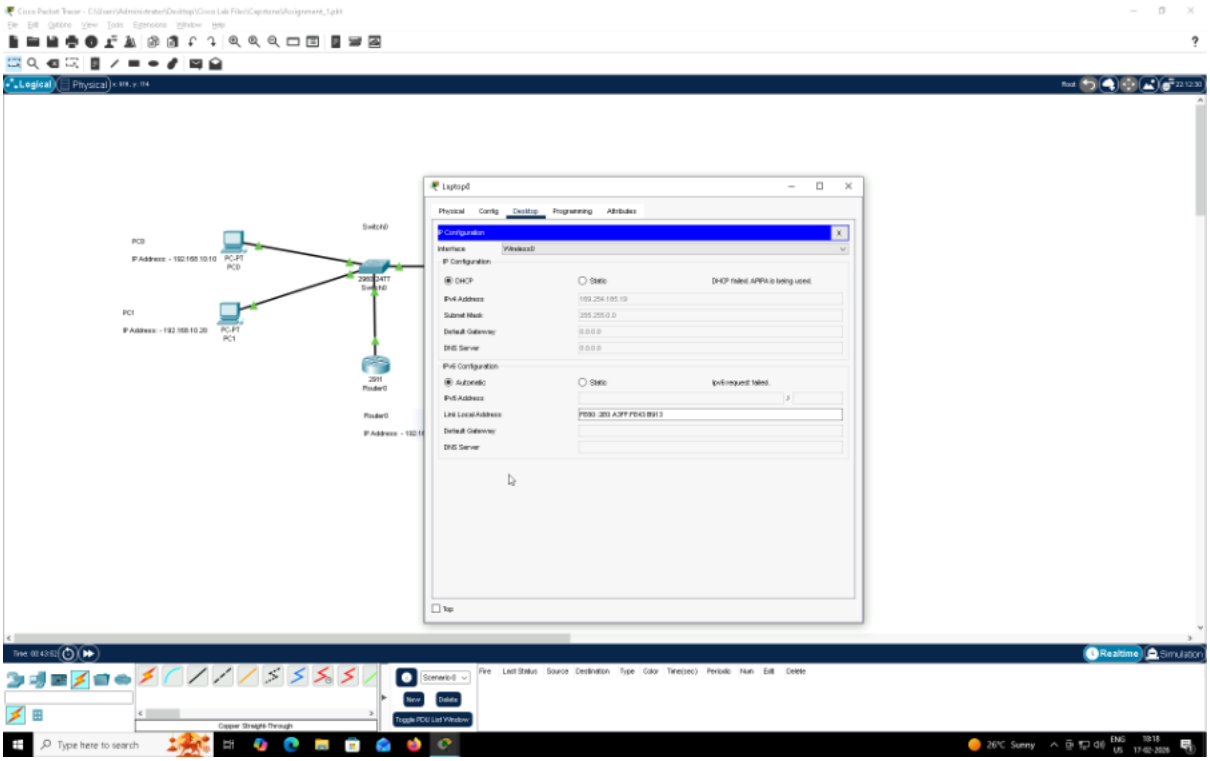


Assigning the Access Point Name (HomeWiFi) so that the Laptop0 can connect wirelessly

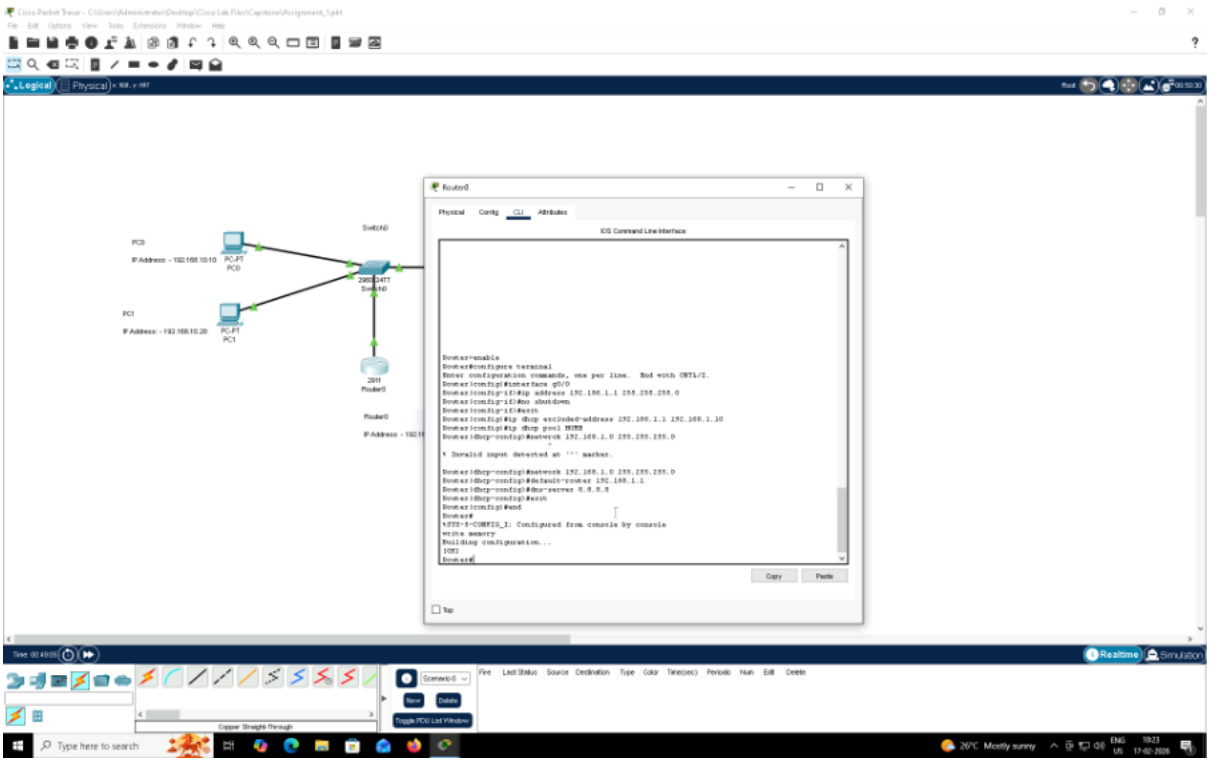




We have successfully added a Wireless network connection card and connected it to our HomeWiFi.

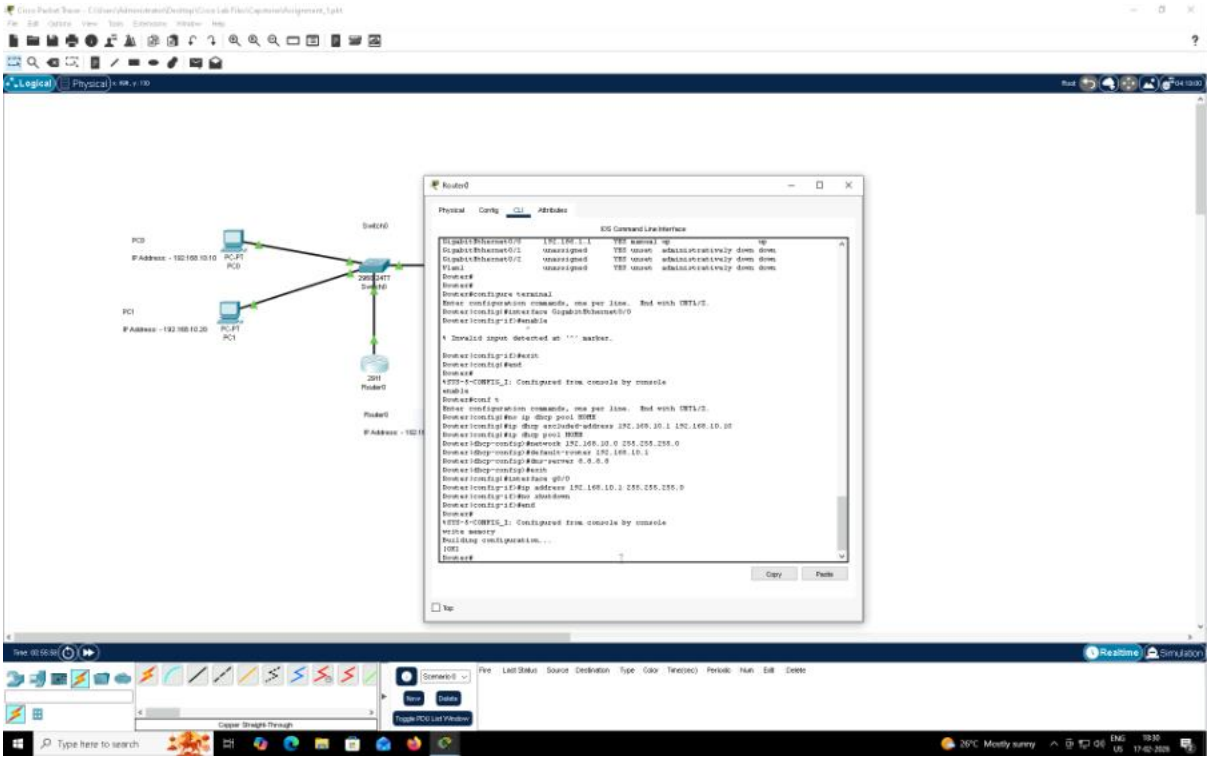


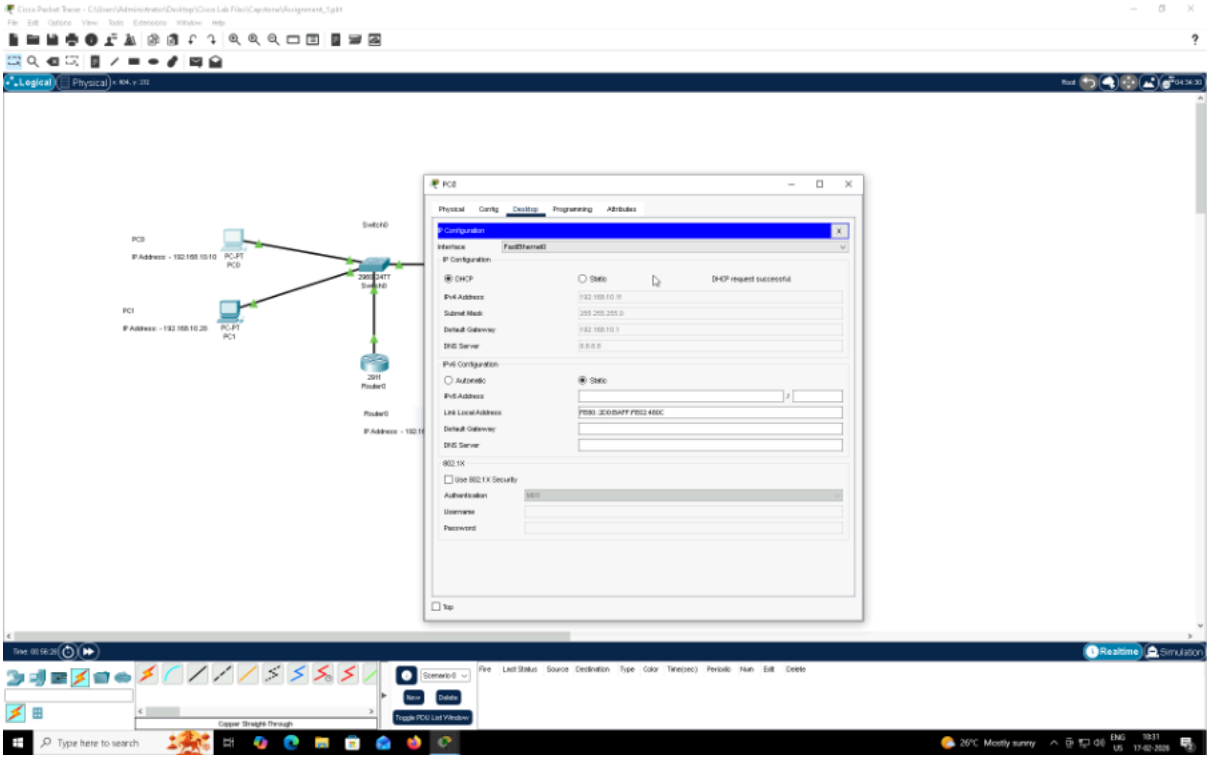
The DHCP fails and APIPA is being used because we haven't configured our Router yet as there are no devices that are providing IP Addresses.

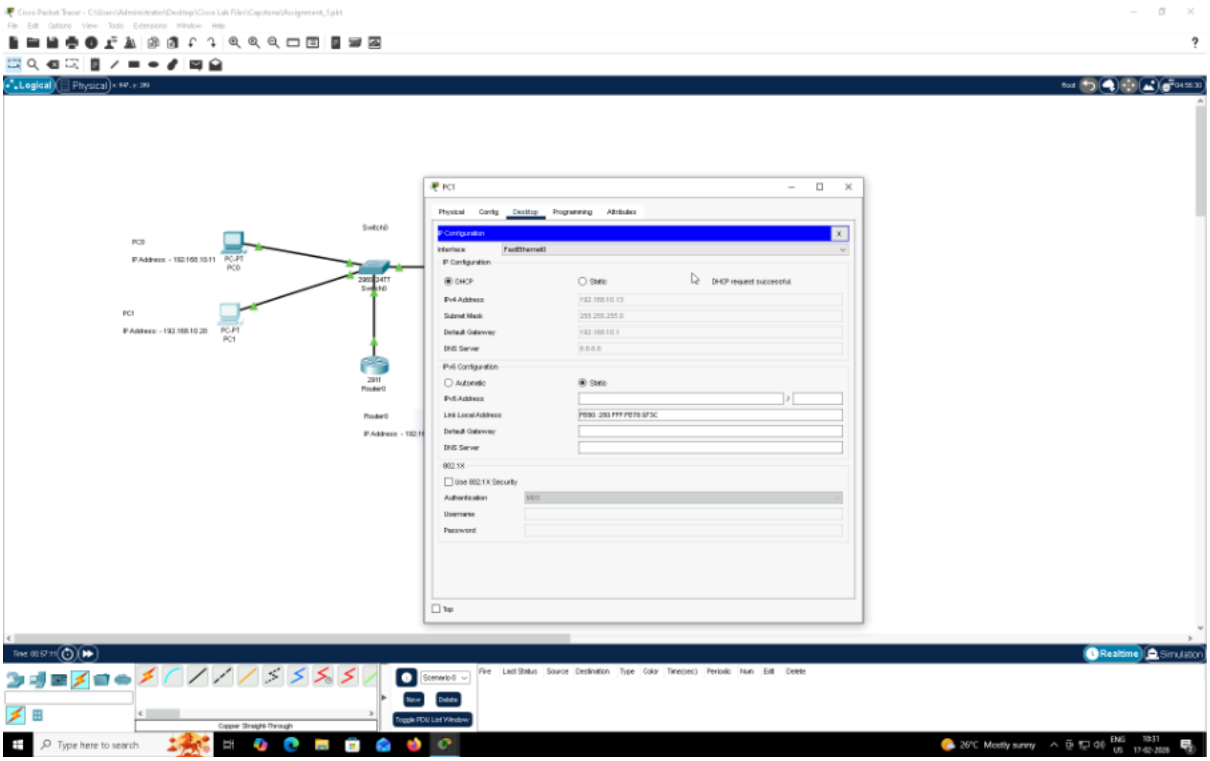


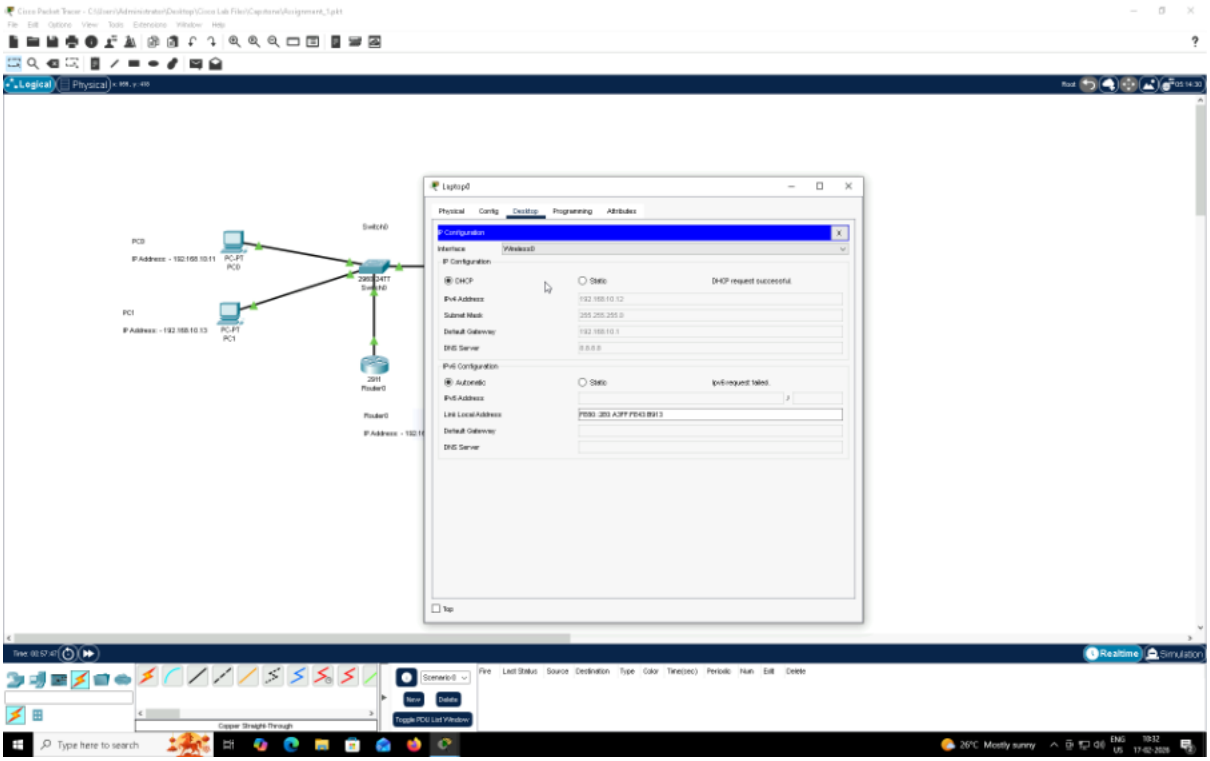
Here we have created a DHCP pool excluding the first 10 IP Addresses (192.168.1.1 – 192.168.1.10) using excluded-address command in the CLI of Router0 and now DHCP will

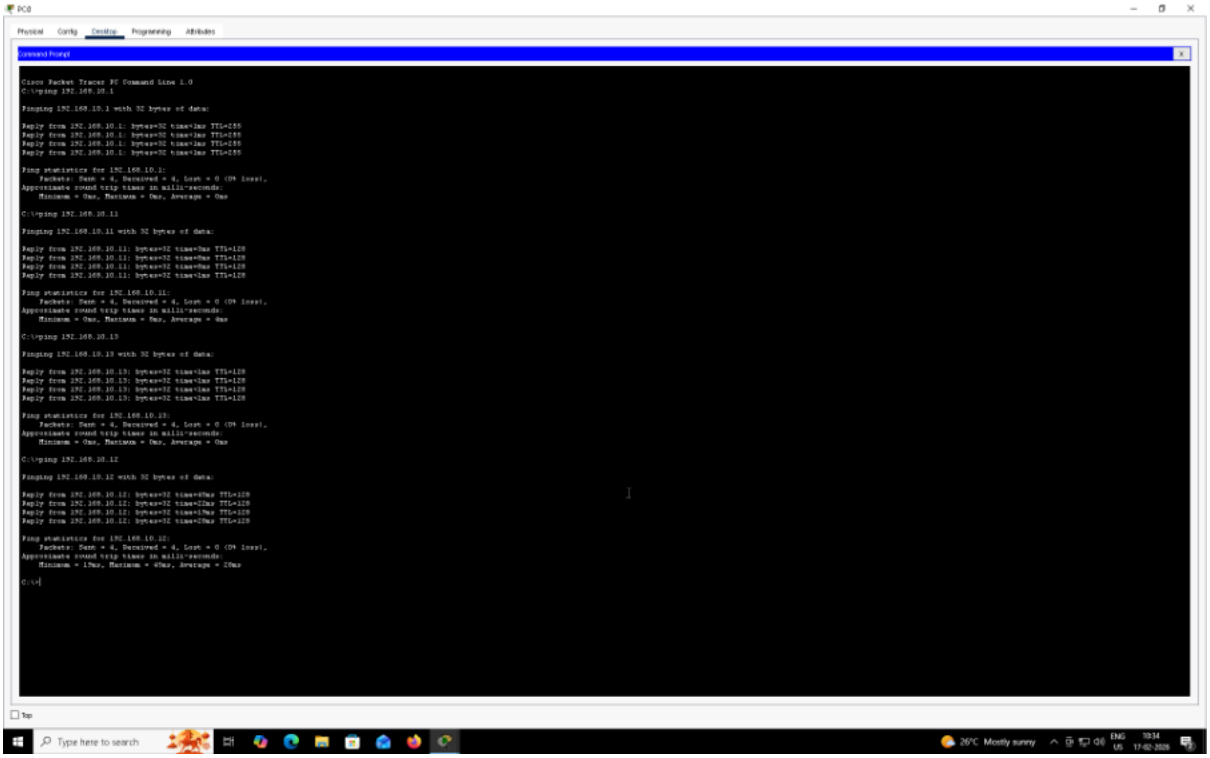
only assign 192.168.1.11 – 192.168.1.254.

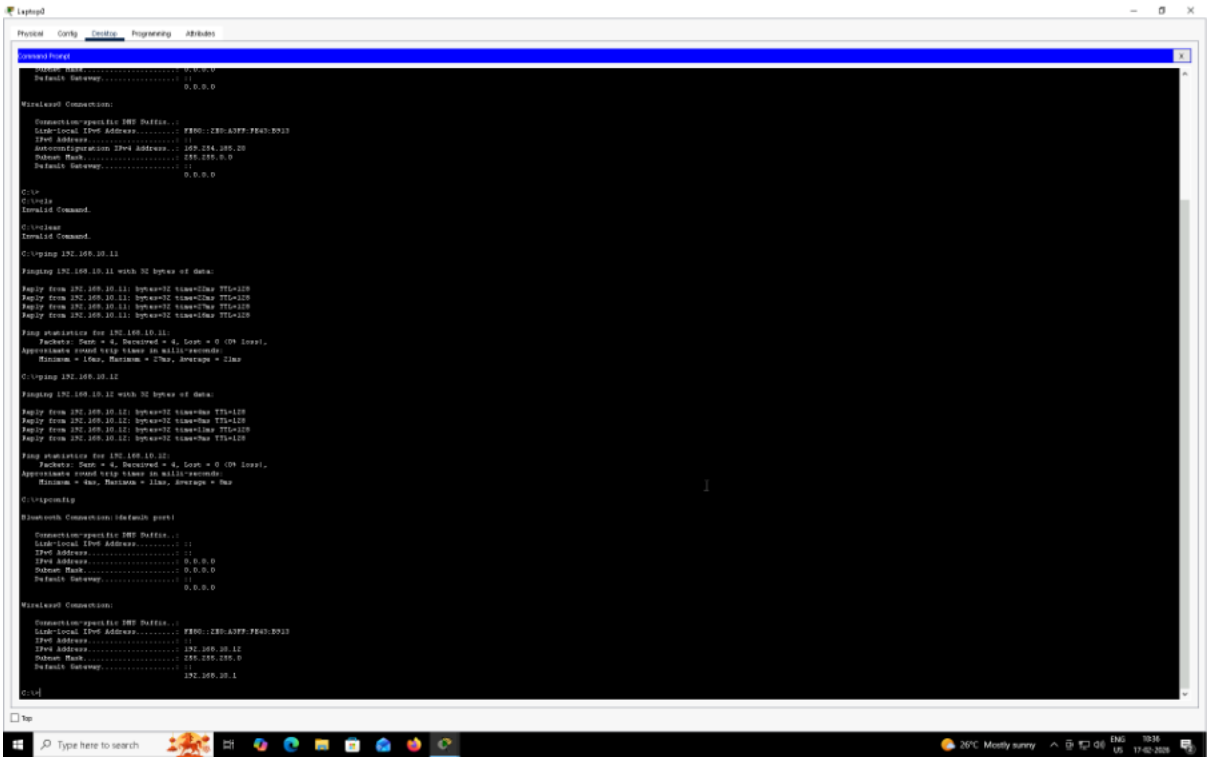










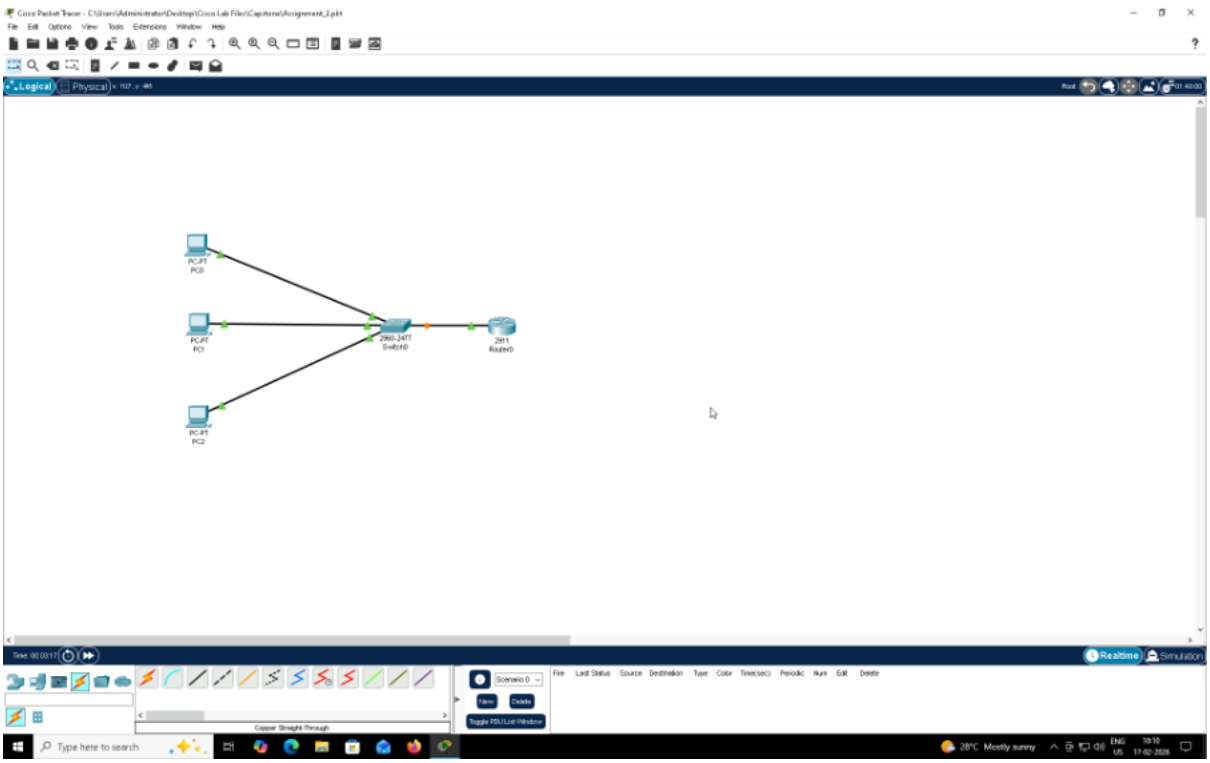


Now we can see that all the devices (PC0, PC1 and Laptop0) have been successfully assigned we have verified it using command prompt in PC0 and Laptop0.

Assignment 2: VLAN & Wireless Network Configuration

As part of this assignment, you need to complete below tasks:

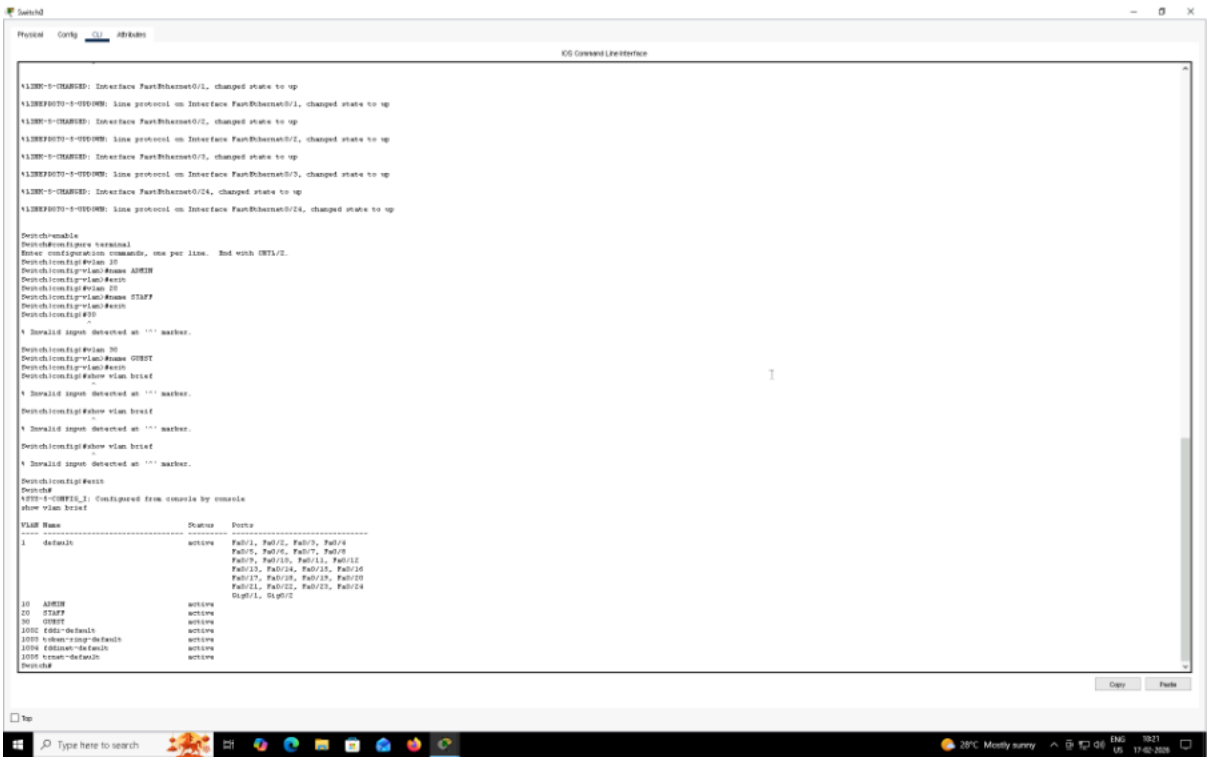
1. Create multiple VLANs on a managed switch
2. Configure inter-VLAN routing
3. Implement basic VLAN security policies
4. Configure a wireless access point with WPA3 encryption
5. Set up guest and private network SSIDs
6. Implement basic wireless security settings



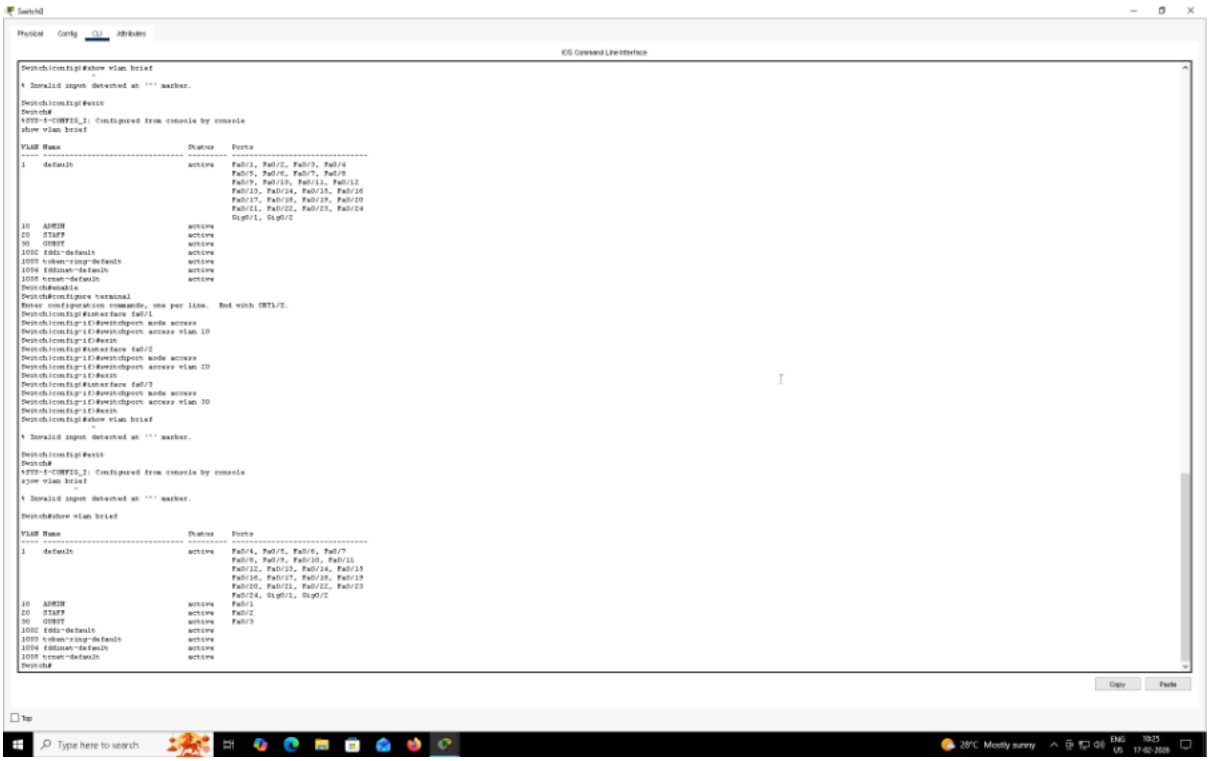
A new network topology was created consisting of a router, a Layer-2 Switch, and three PCs (Devices). The purpose of this topology is to implement VLAN segmentation and inter-VLAN routing in a controlled environment.

We will assume that: -

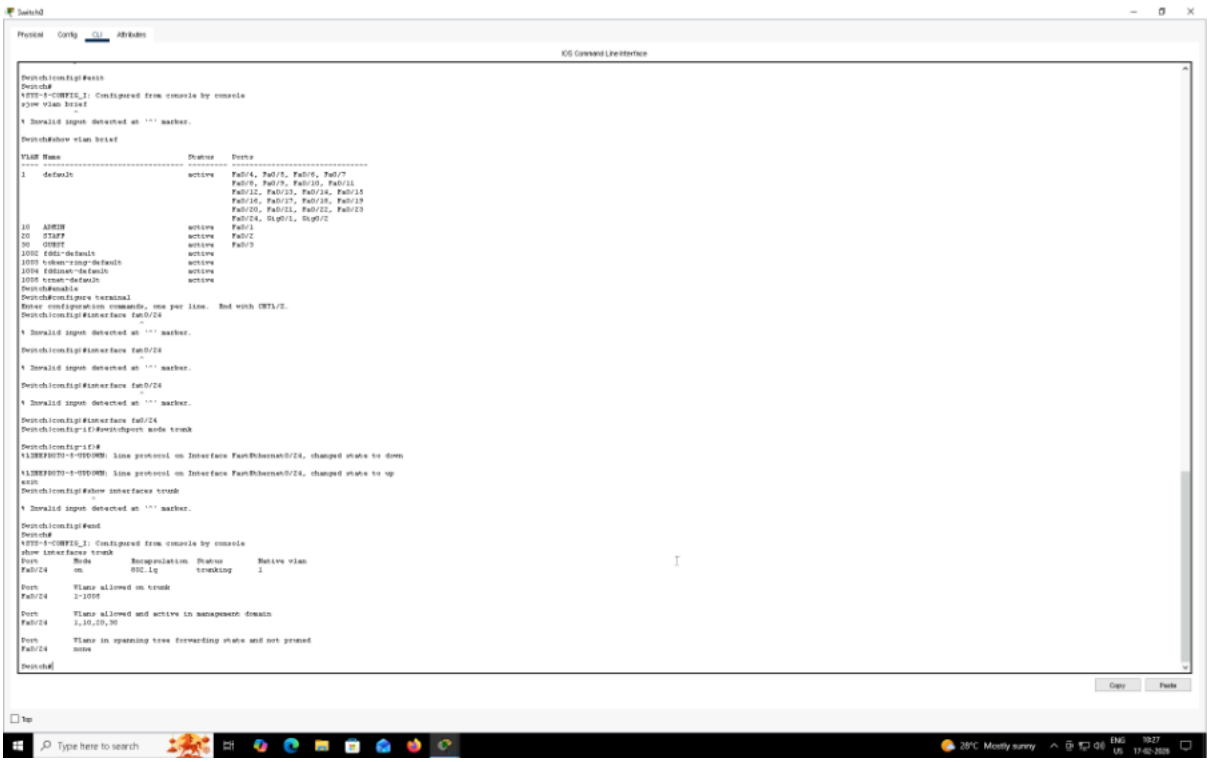
VLAN	Name	Device
10	ADMIN	PC0
20	STAFF	PC1
30	GUEST	PC2



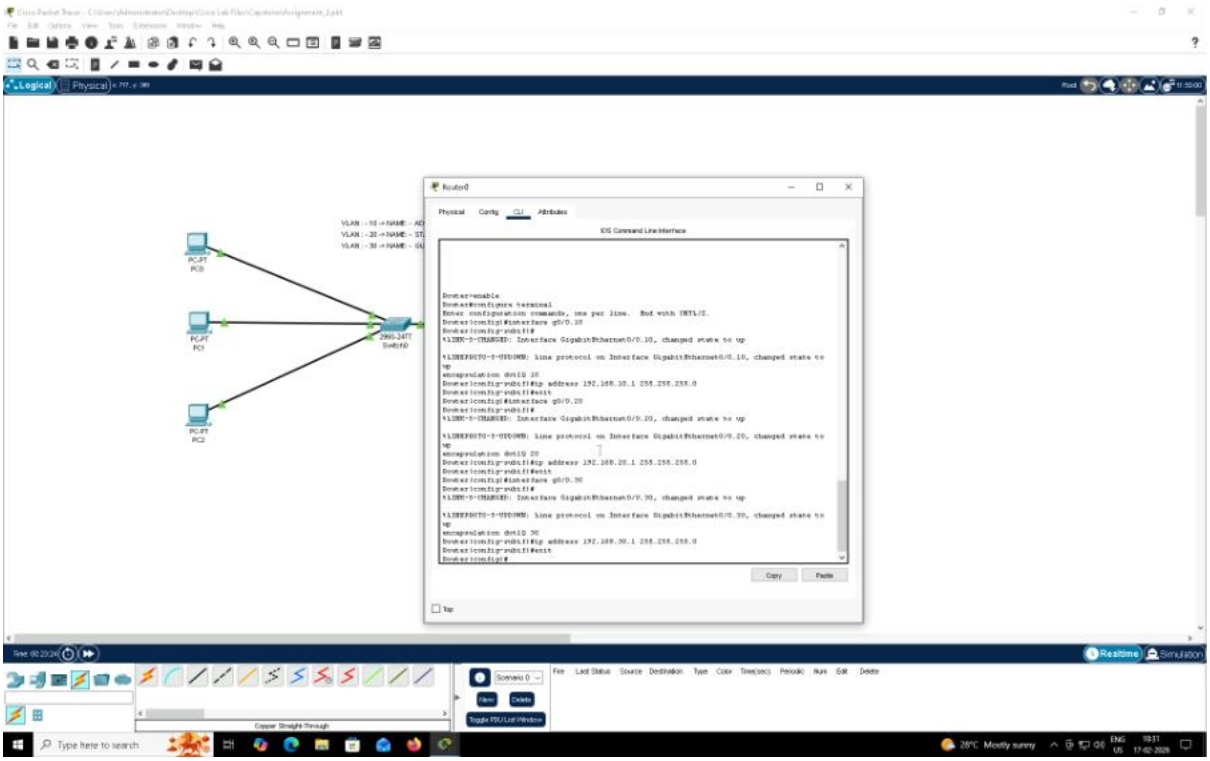
Three VLAN's were created on the switch to logically separate the network into different departments. This reduces broadcast traffic and improves network security by isolating devices within their own broadcast domain.



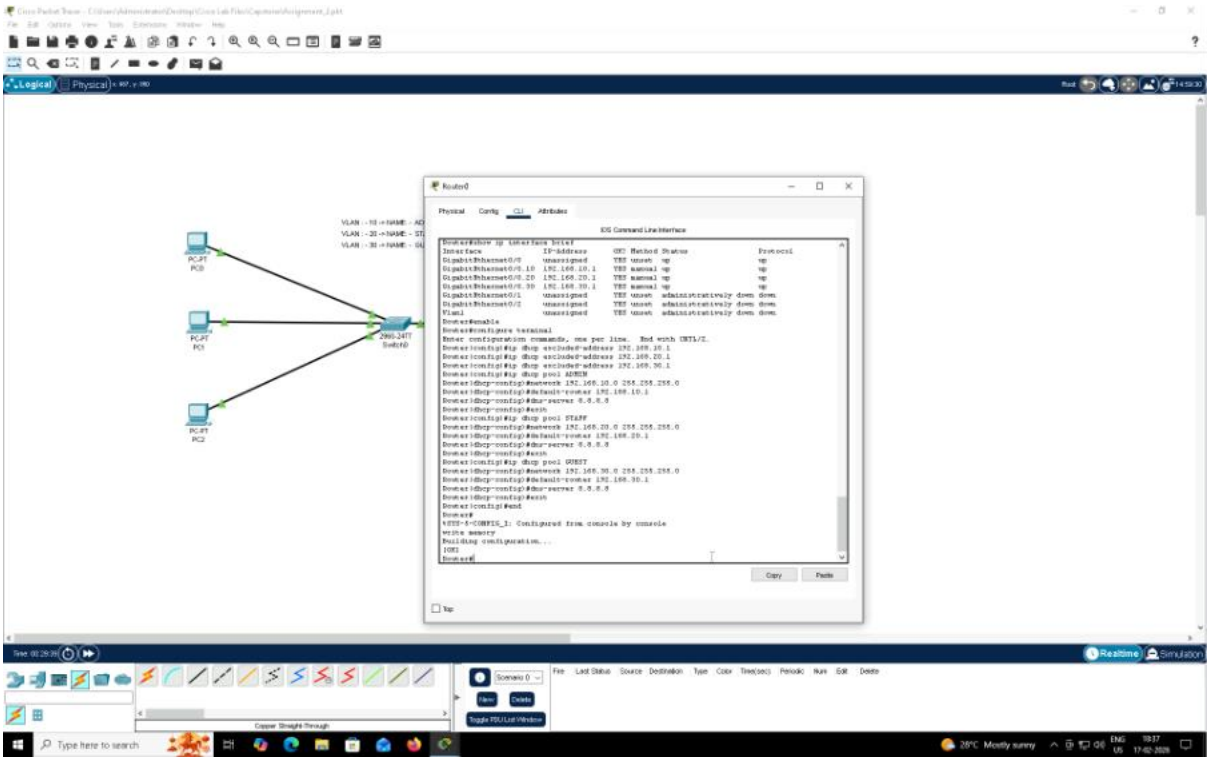
Switch access ports were assigned to respective VLANs so that each device belongs to a separate logical network segment.



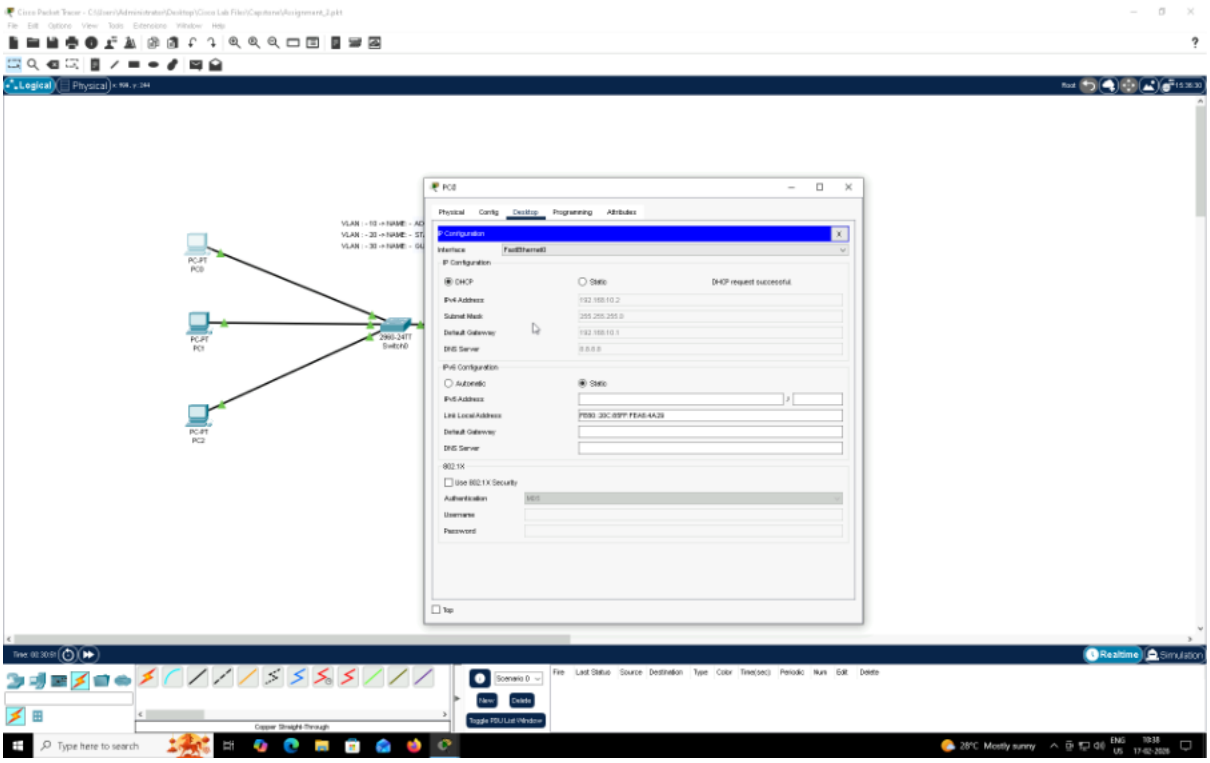
The switch uplink port was configured as a trunk port to allow multiple VLAN traffic to pass between the switch and router using IEEE 802.1Q tagging.



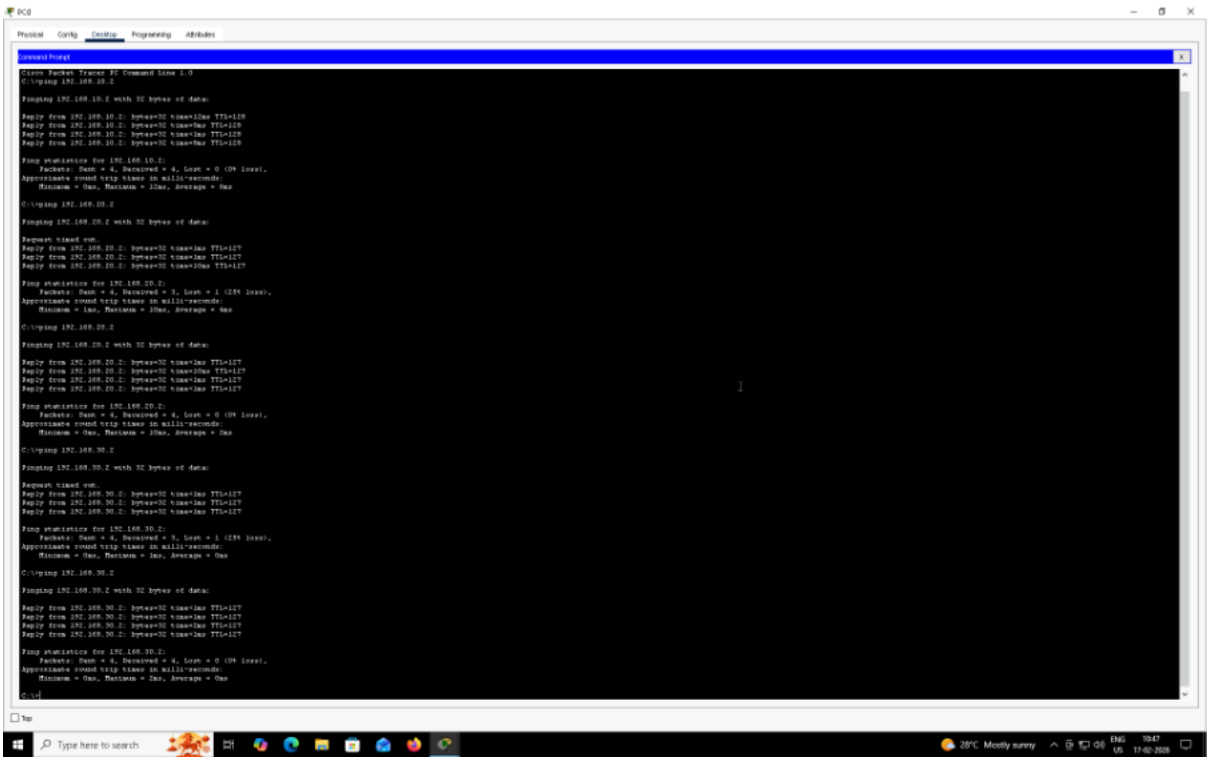
Router sub interfaces were created and tagged with VLAN IDs using the router-on-a-stick method to enable communication between multiple VLAN networks.



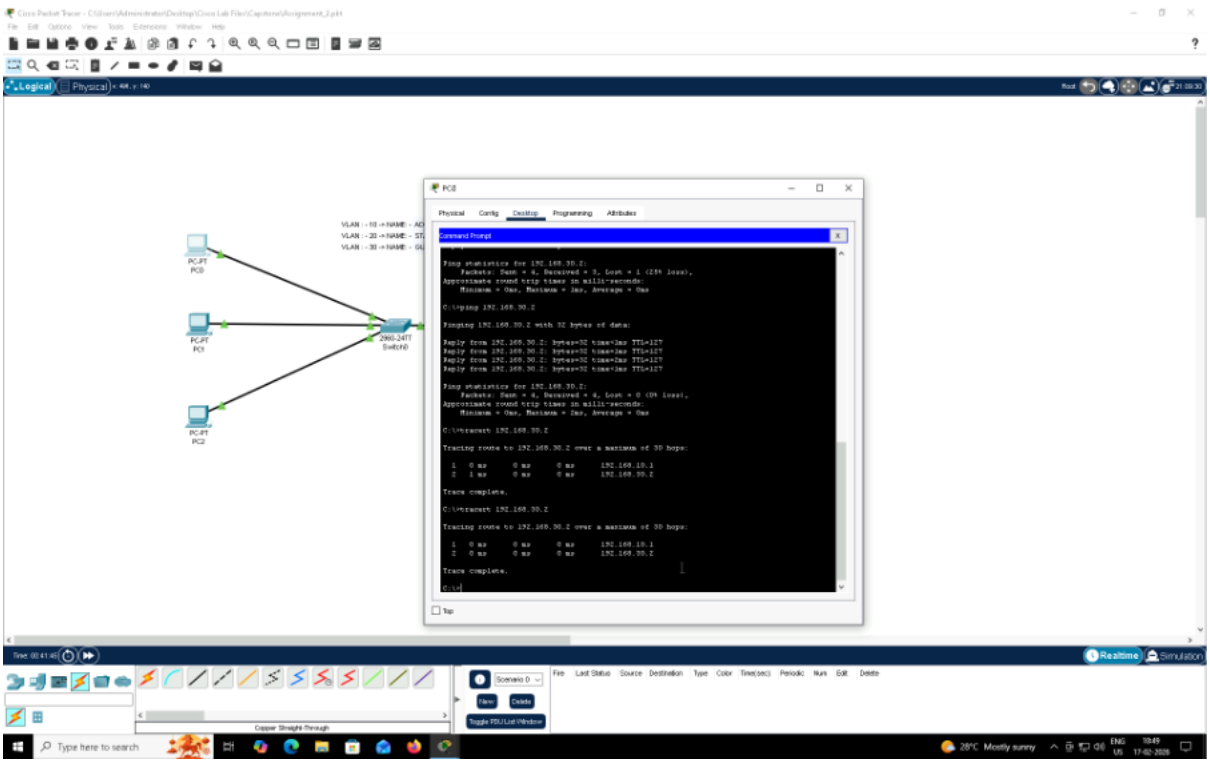
Separate DHCP protocols were configured on the router for each VLAN network to automatically assign IP addresses and gateway information to devices within their respective VLAN.



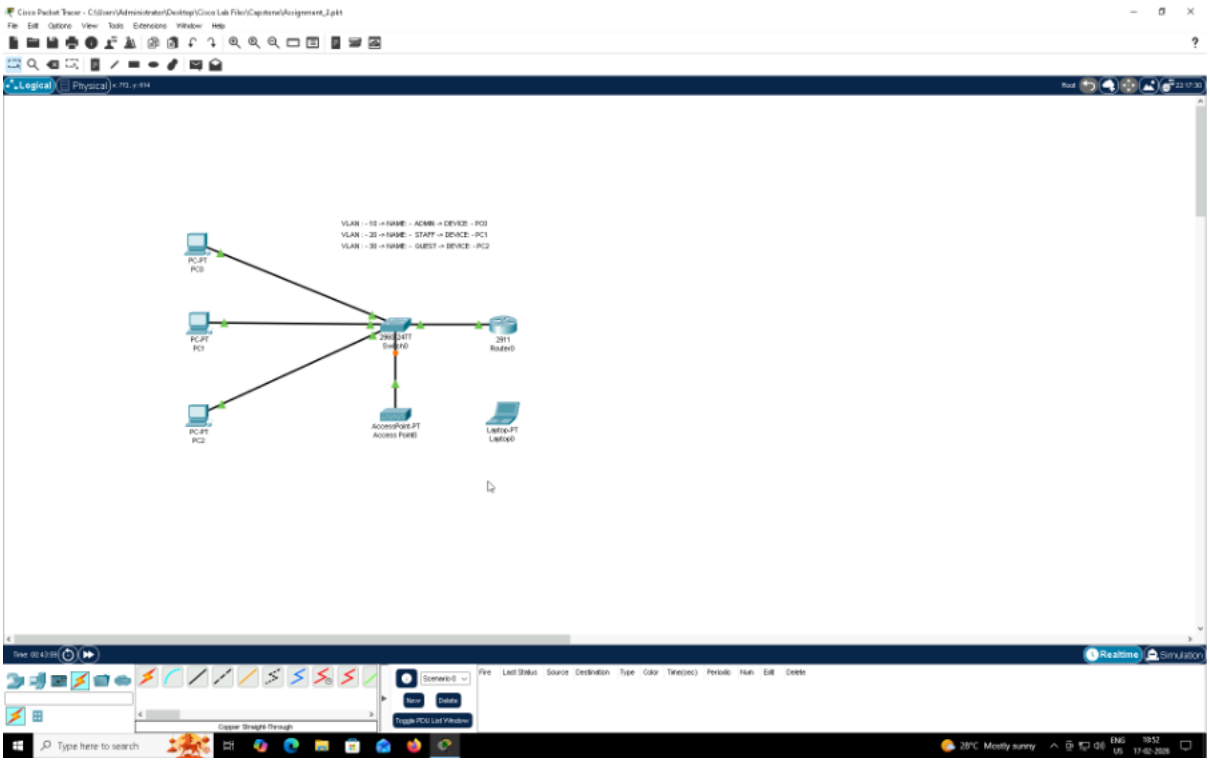
End devices successfully obtained IP addresses from their respective VLAN DHCP pools, confirming proper segmentation and DHCP Operation.



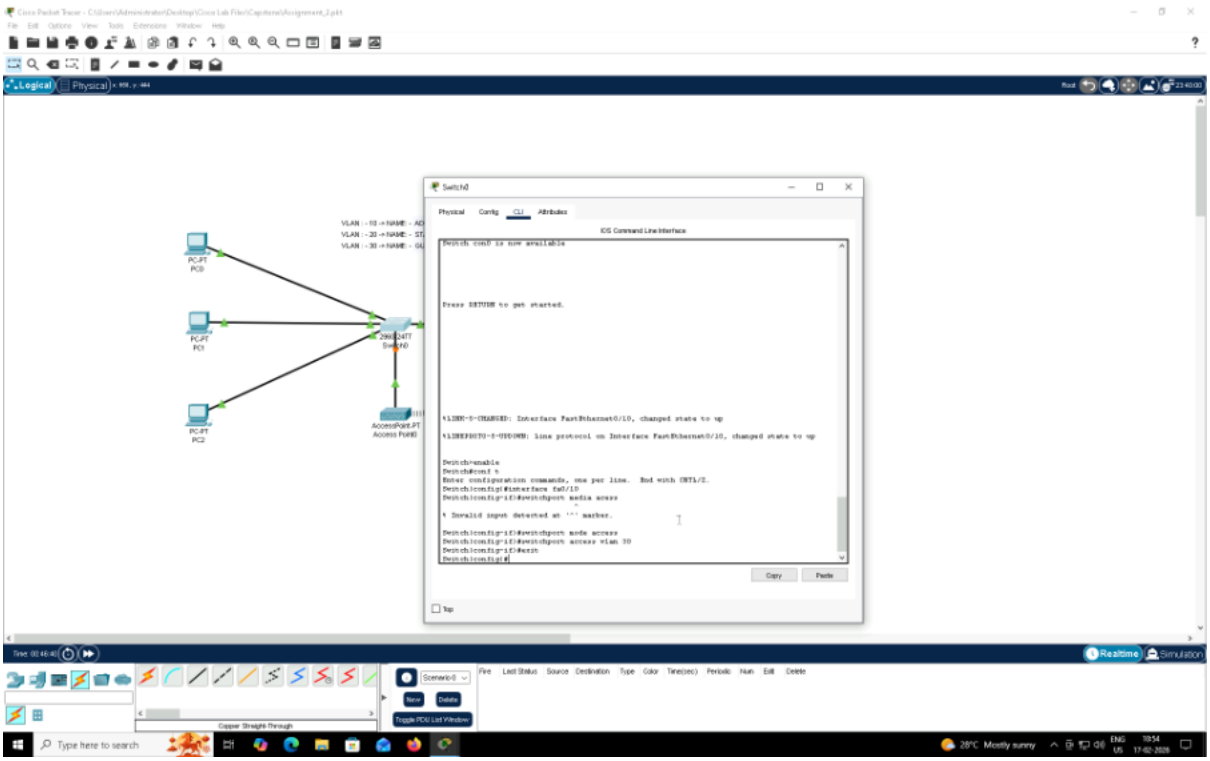
Devices located in different VLAN networks successfully communicated through the router. This confirmed proper router-on-a-stick configuration and Layer-3 routing between VLANs.



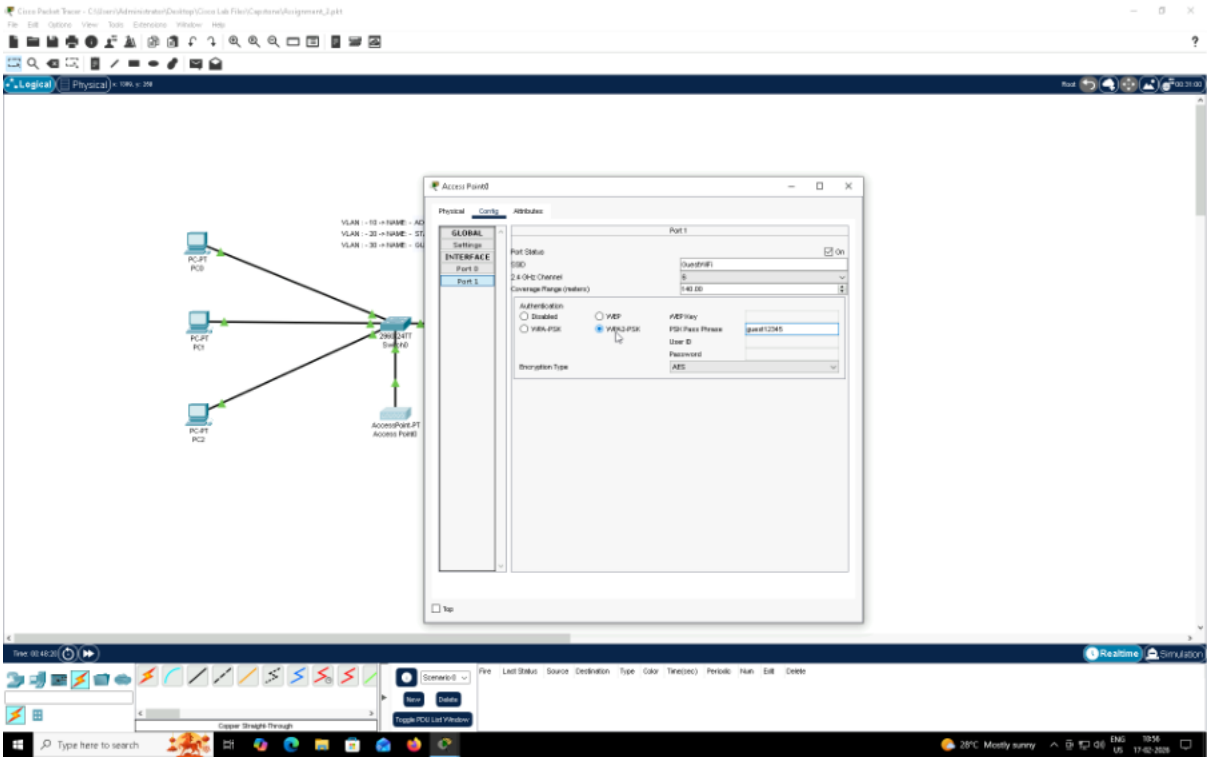
Traceroute confirmed that traffic between VLANs was routed through the router gateway, verifying correct Layer-3 forwarding behaviour.



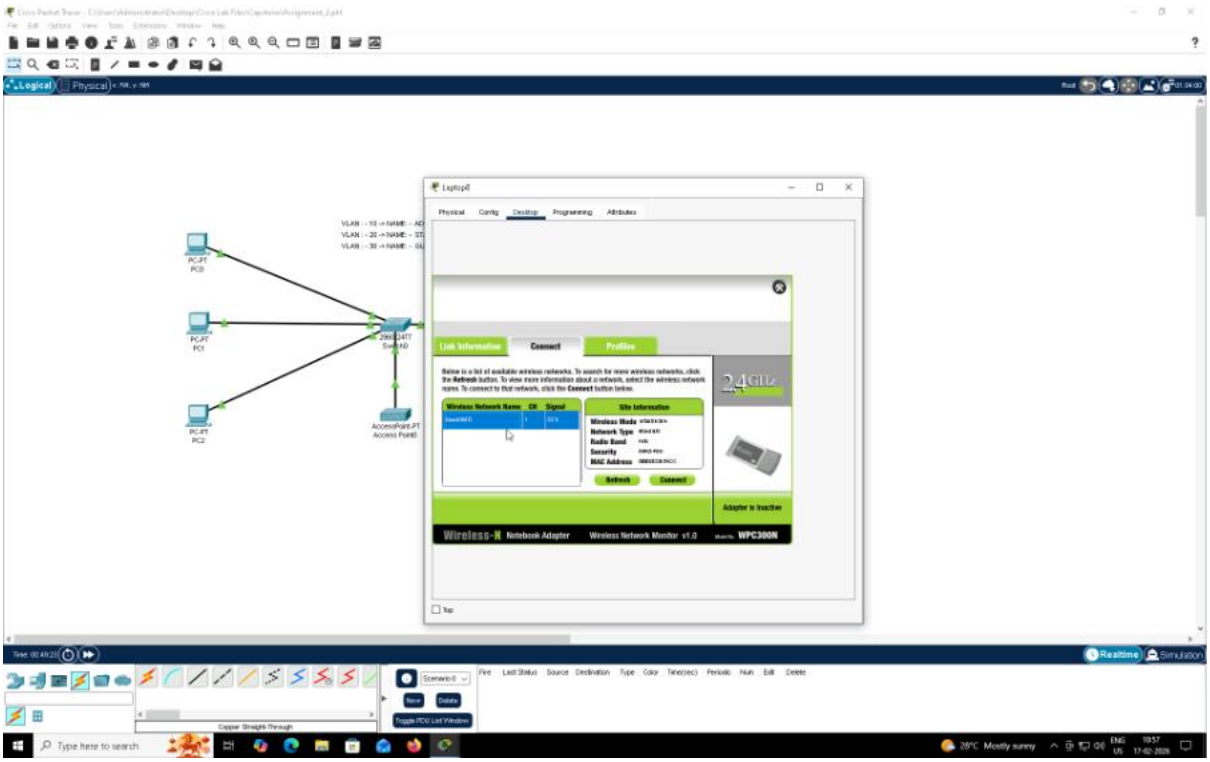
A wireless access point and laptop were added to extend network connectivity to wireless users within the VLAN environment.

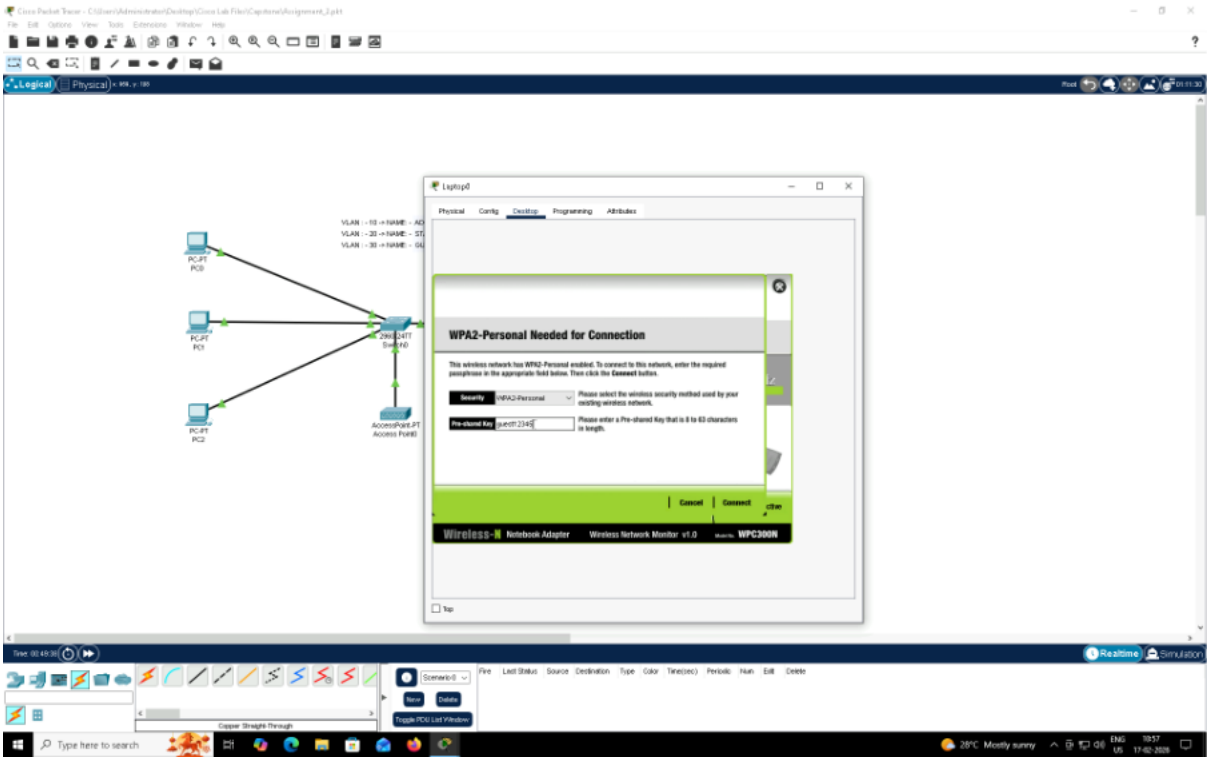


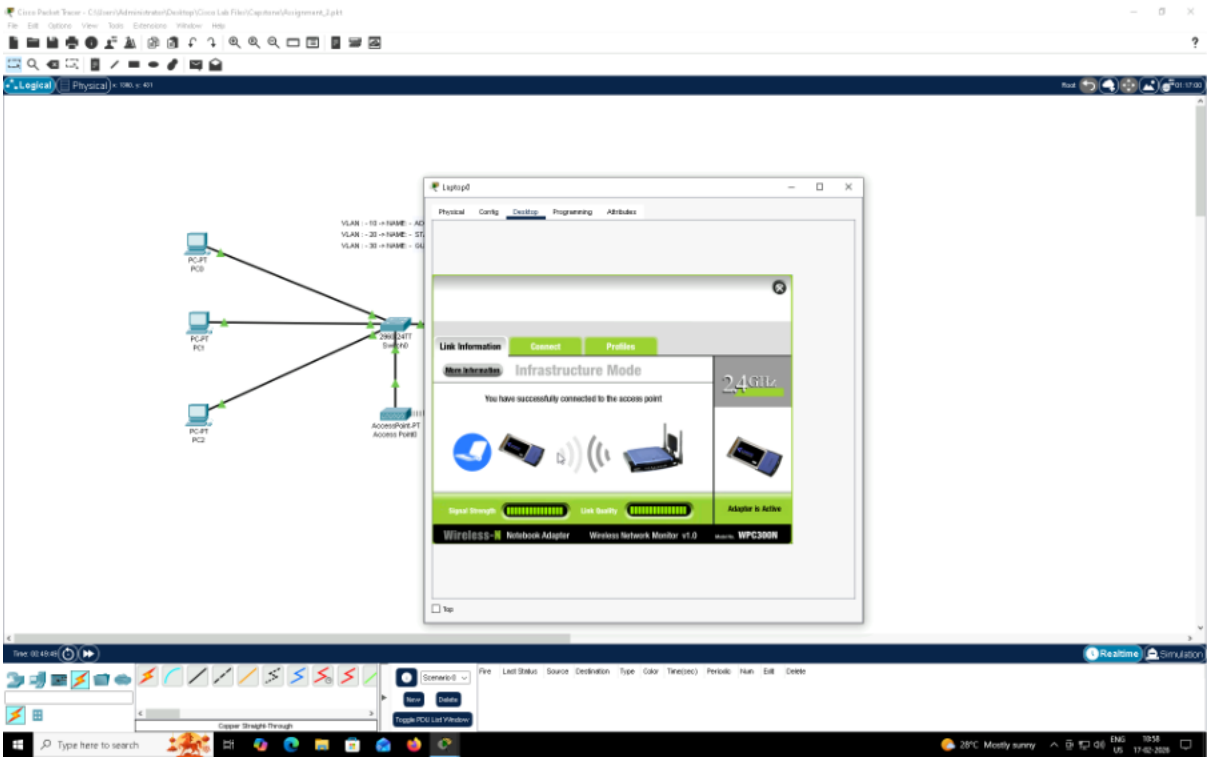
The access point port was assigned to the Guest VLAN to ensure wireless users are isolated from internal wired networks.

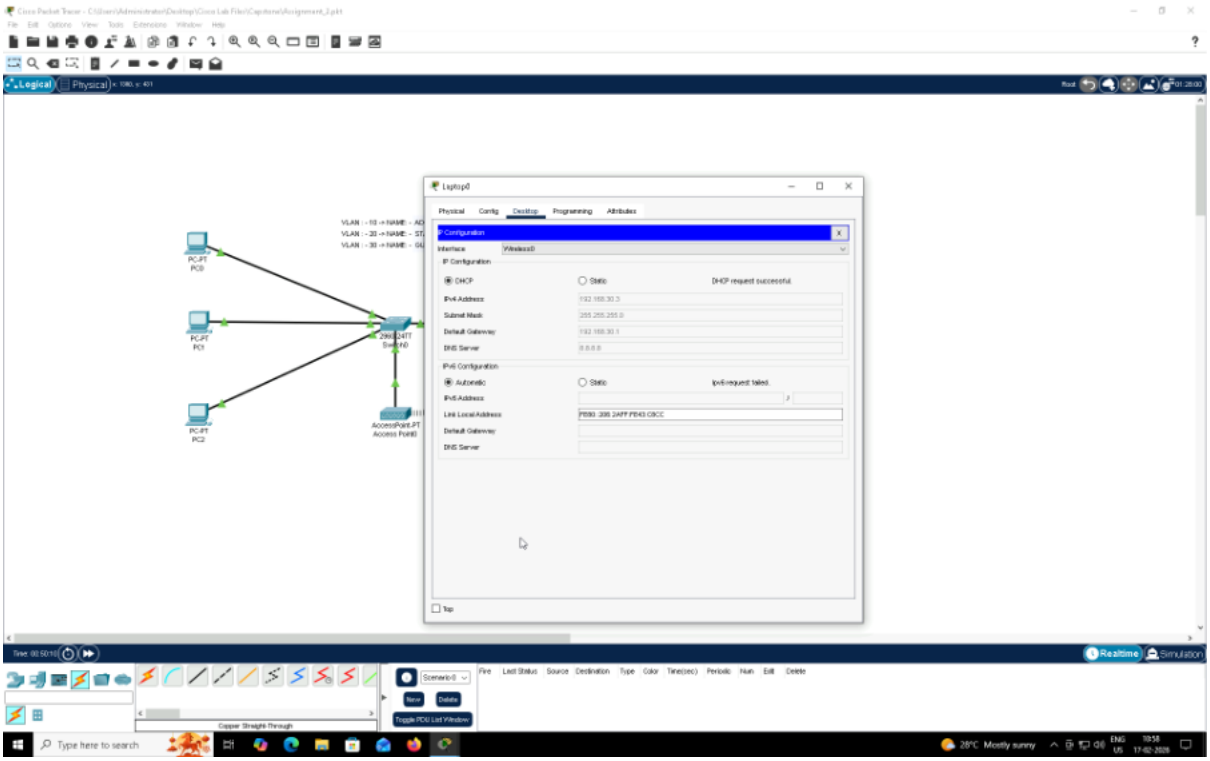


WPA2 encryption was configured on the wireless network to protect communication and prevent unauthorized access.

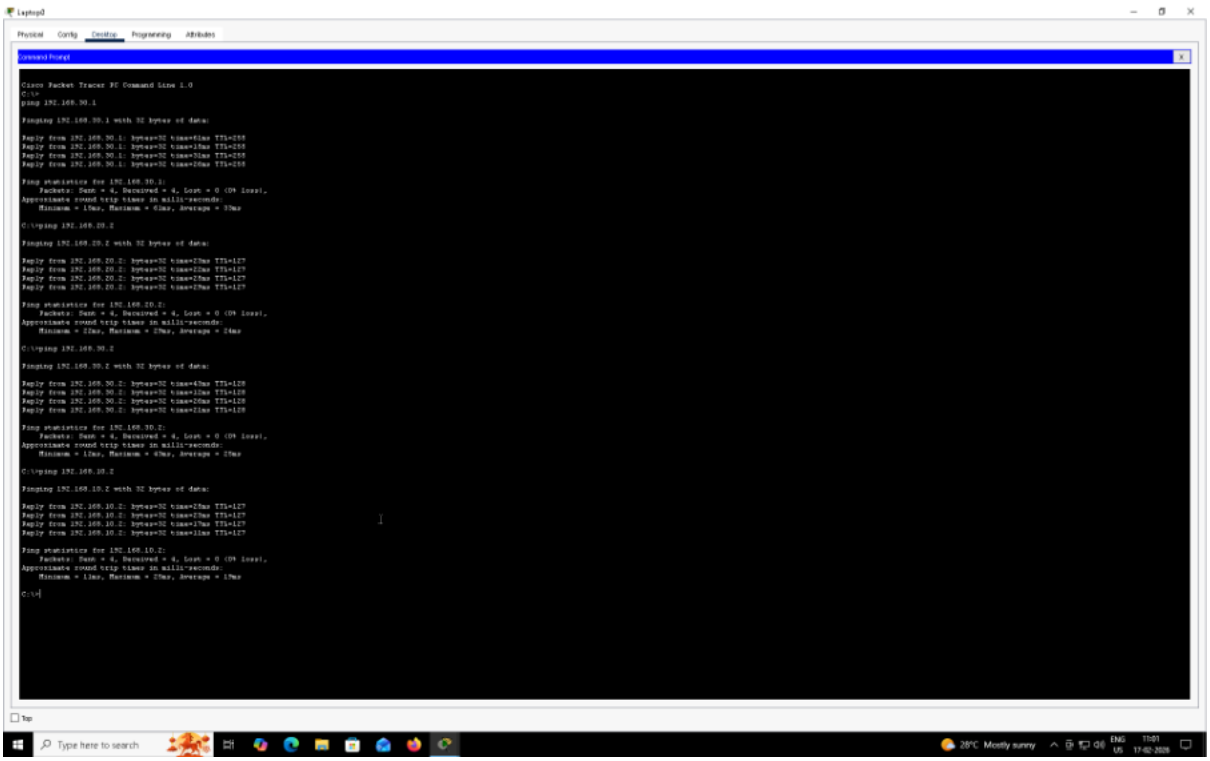








The wireless client successfully authenticated and received an IP address from the Guest VLAN DHCP pool, confirming secure wireless access.



The wireless device communicated successfully with other VLAN networks, confirming proper VLAN mapping and secure wireless configuration.

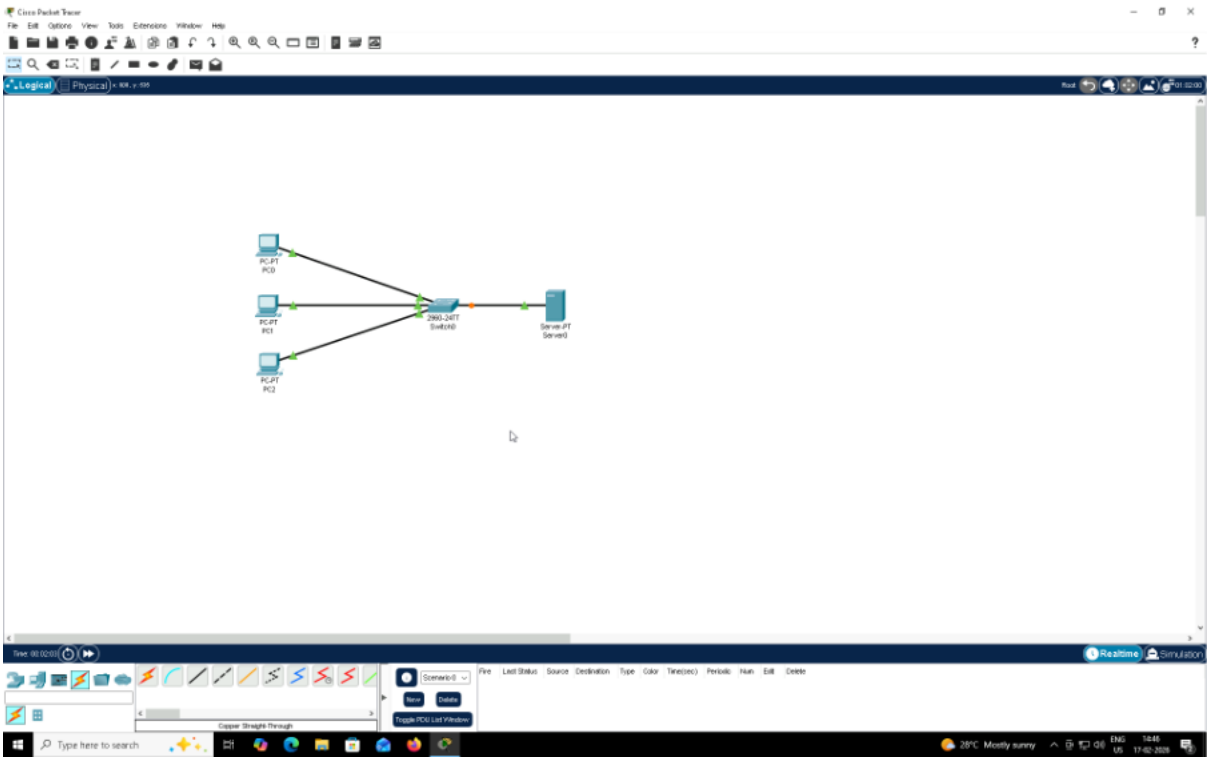
Assignment 3: DHCP and DNS Configuration

As part of this assignment, you need to complete below tasks:

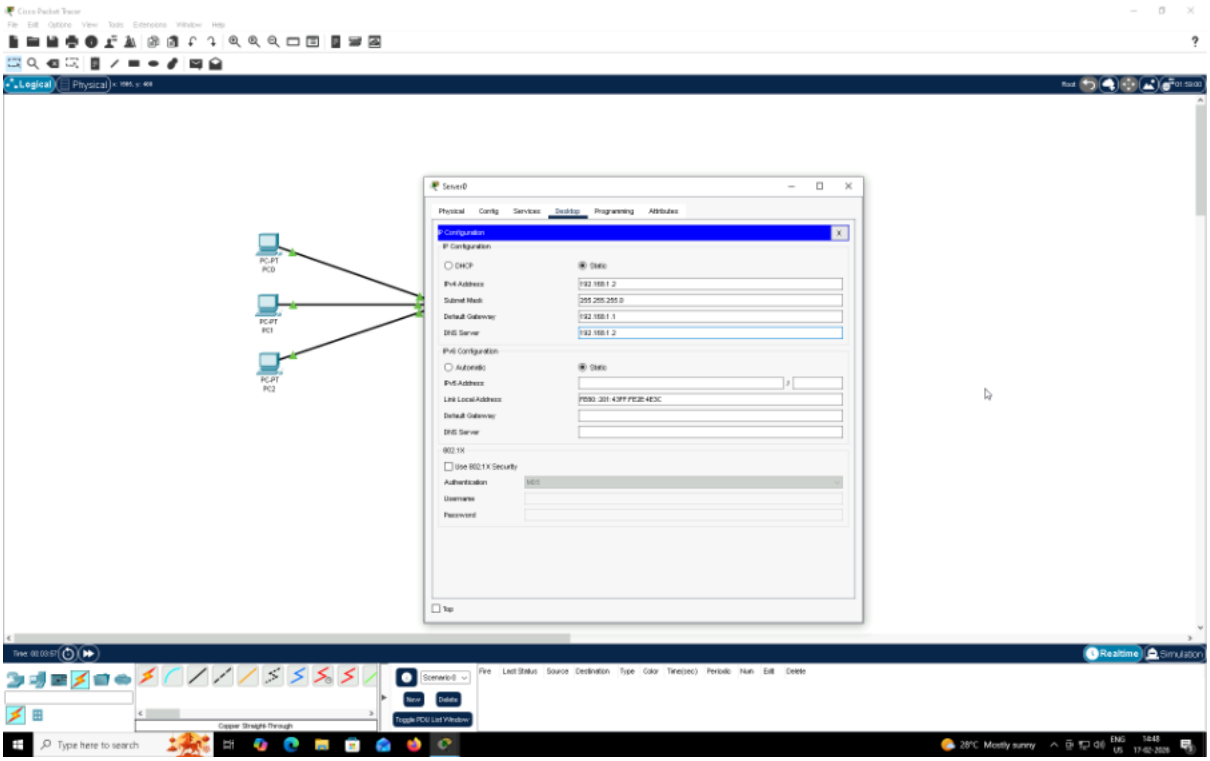
1. Set up a DHCP server with IP address pools

2. Configure DNS resolution

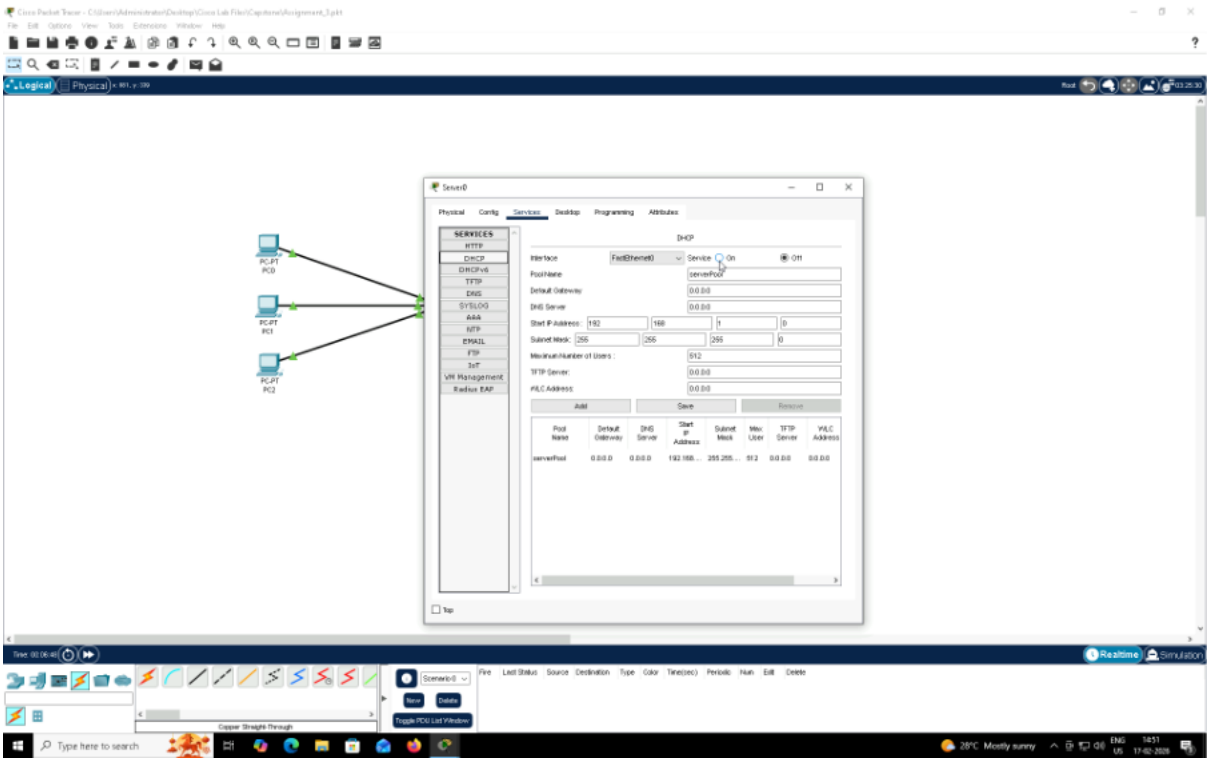
3. Implement DHCP reservations for specific device



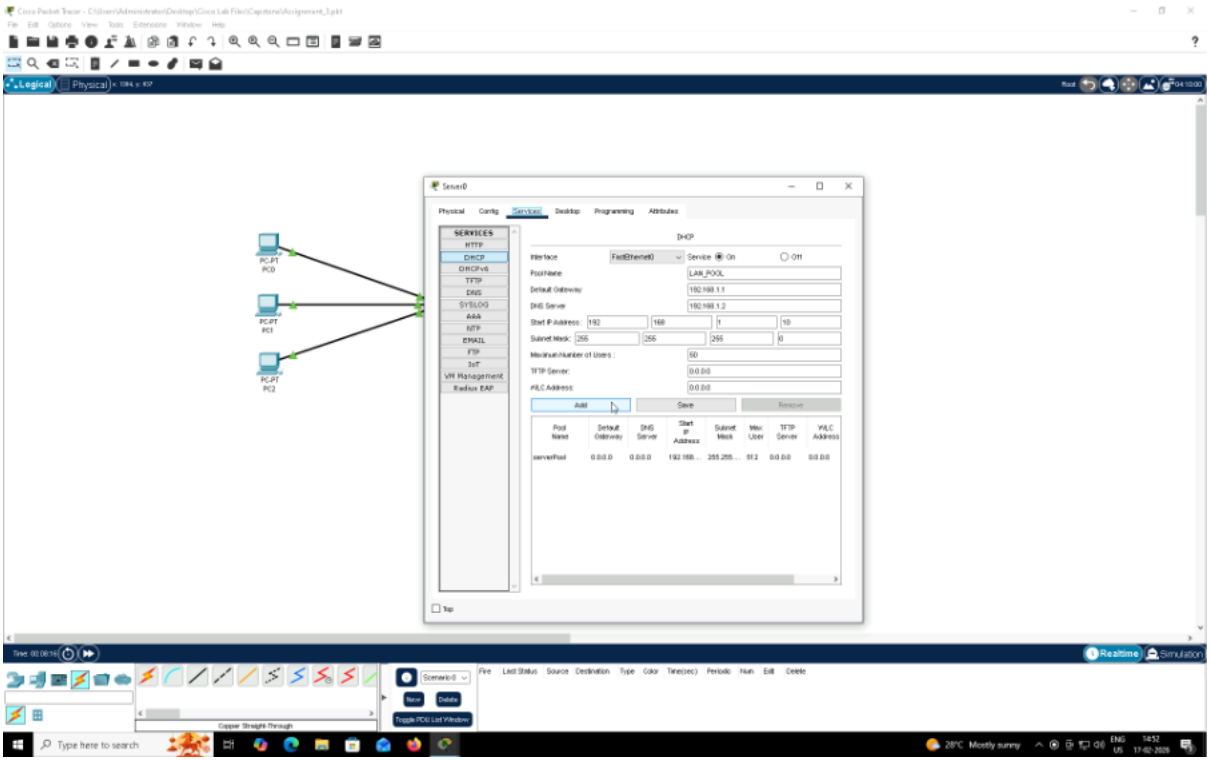
All end devices were connected to a central switch to a central switch to allow communication within the LAN and access the network services hosted on the server.

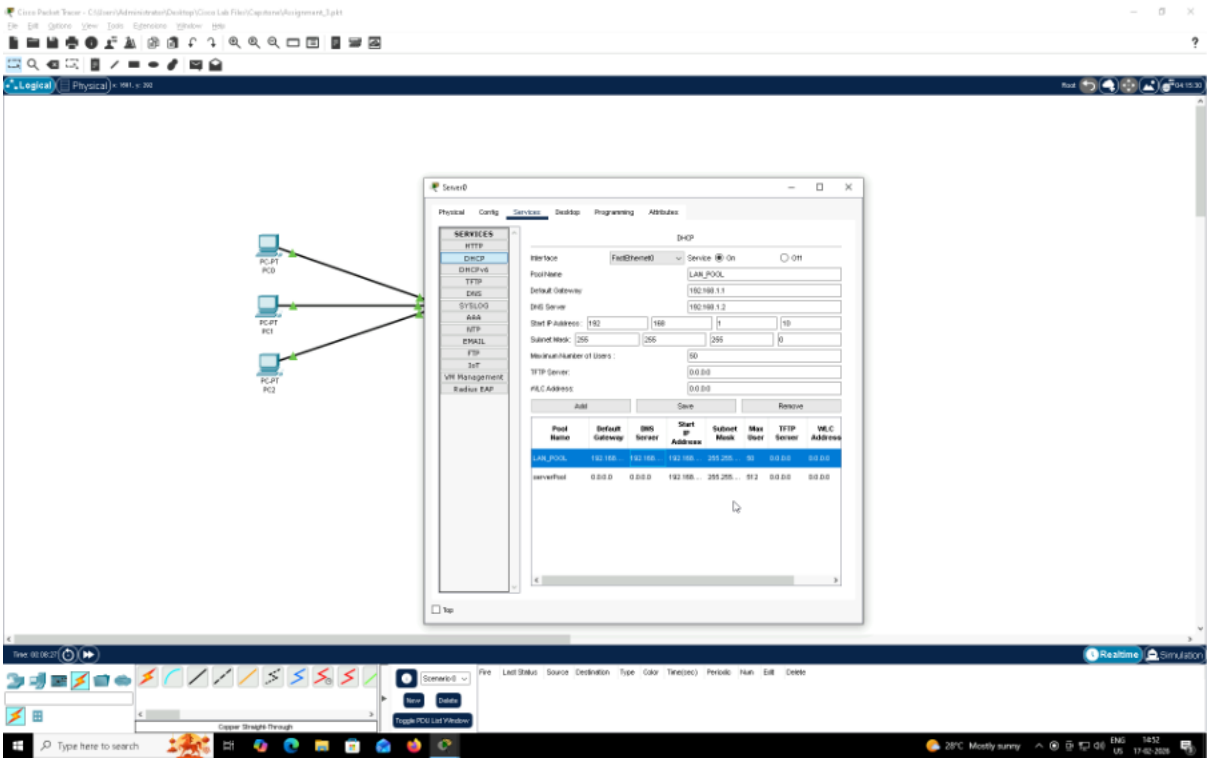


A static IP Address was assigned to the server so that clients can reliably access DHCP and DNS Services.

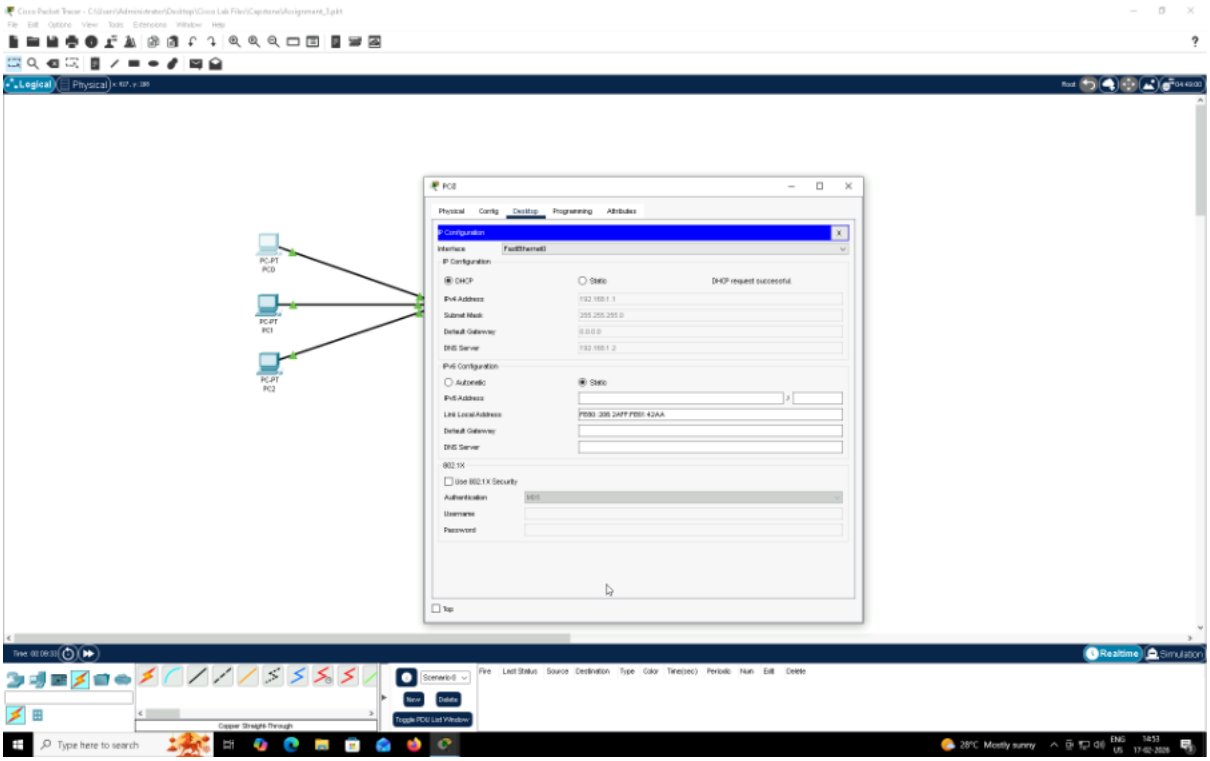


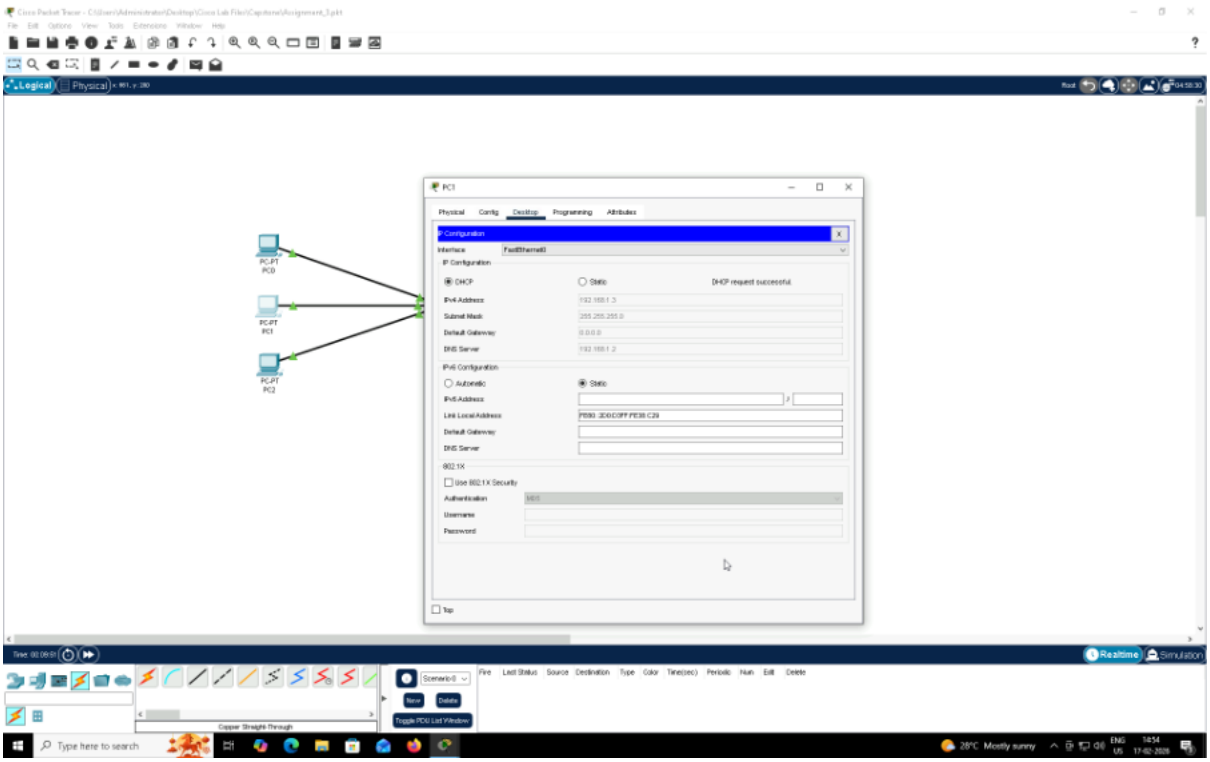
DHCP services were enabled.

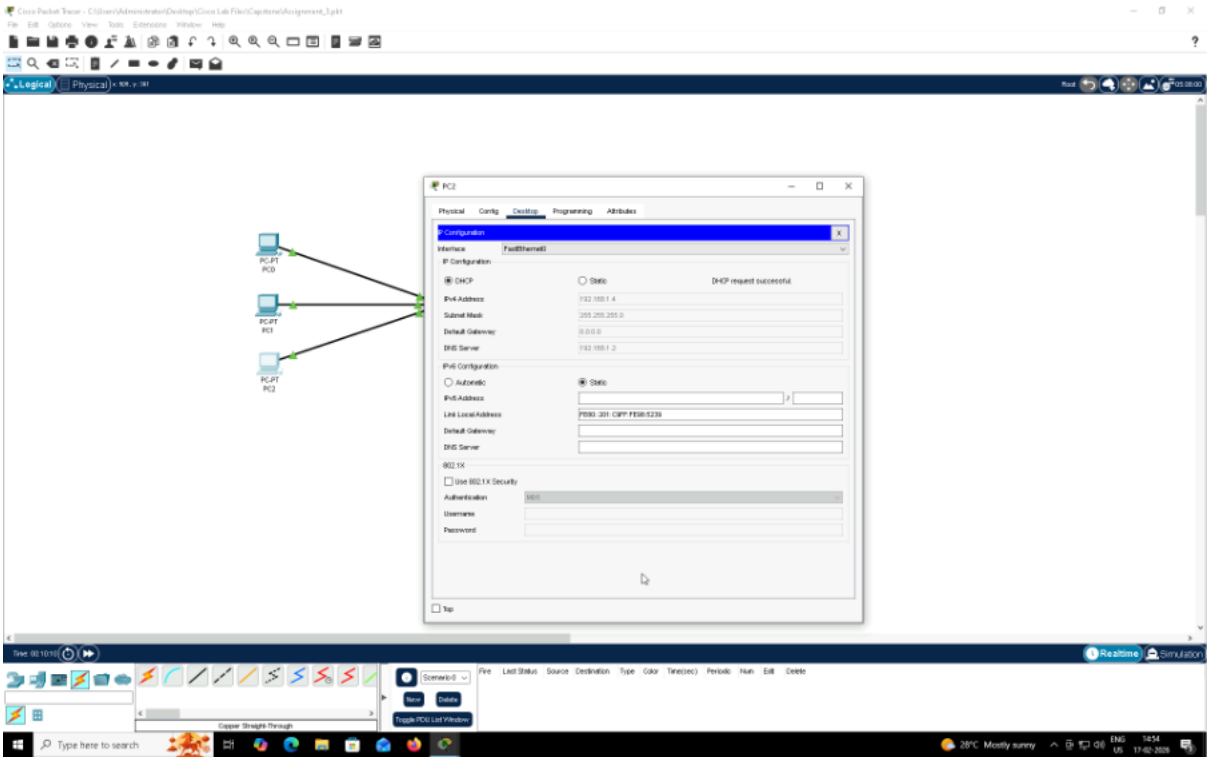




The DHCP service was enabled on the server and configured to automatically assign IP addresses, gateway and DNS information to client systems within the network.

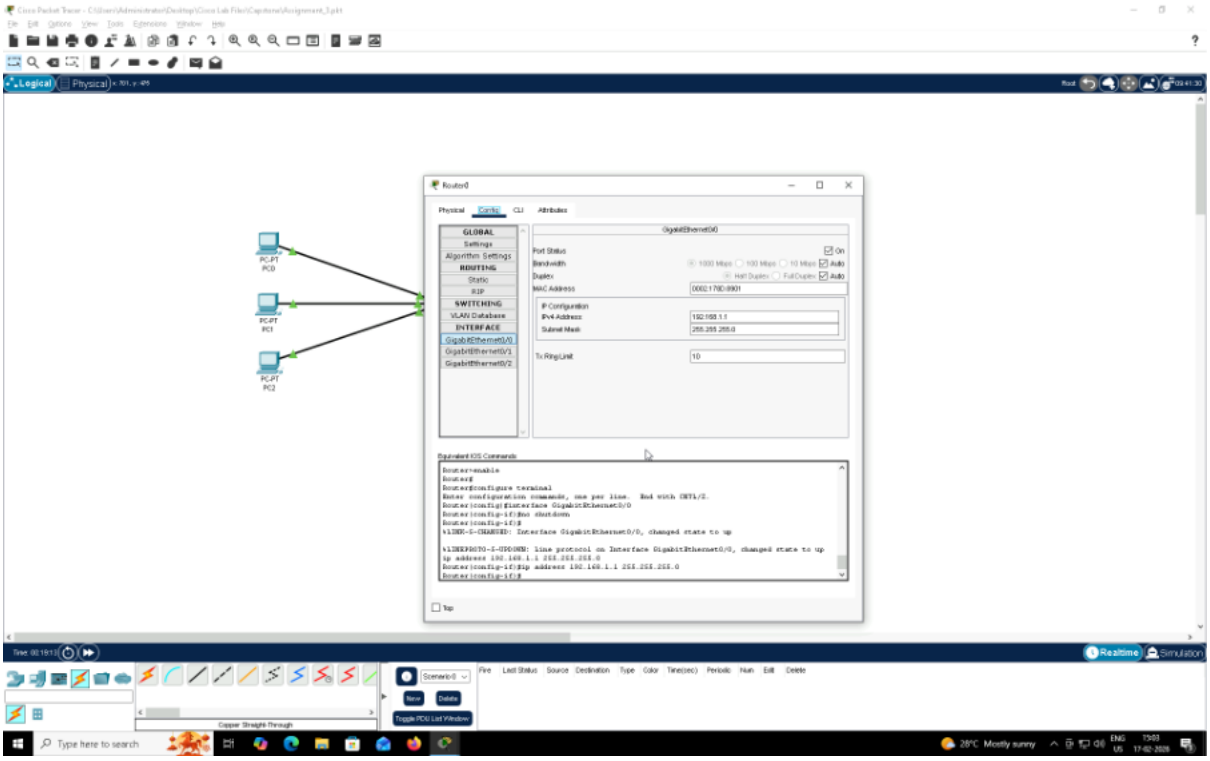






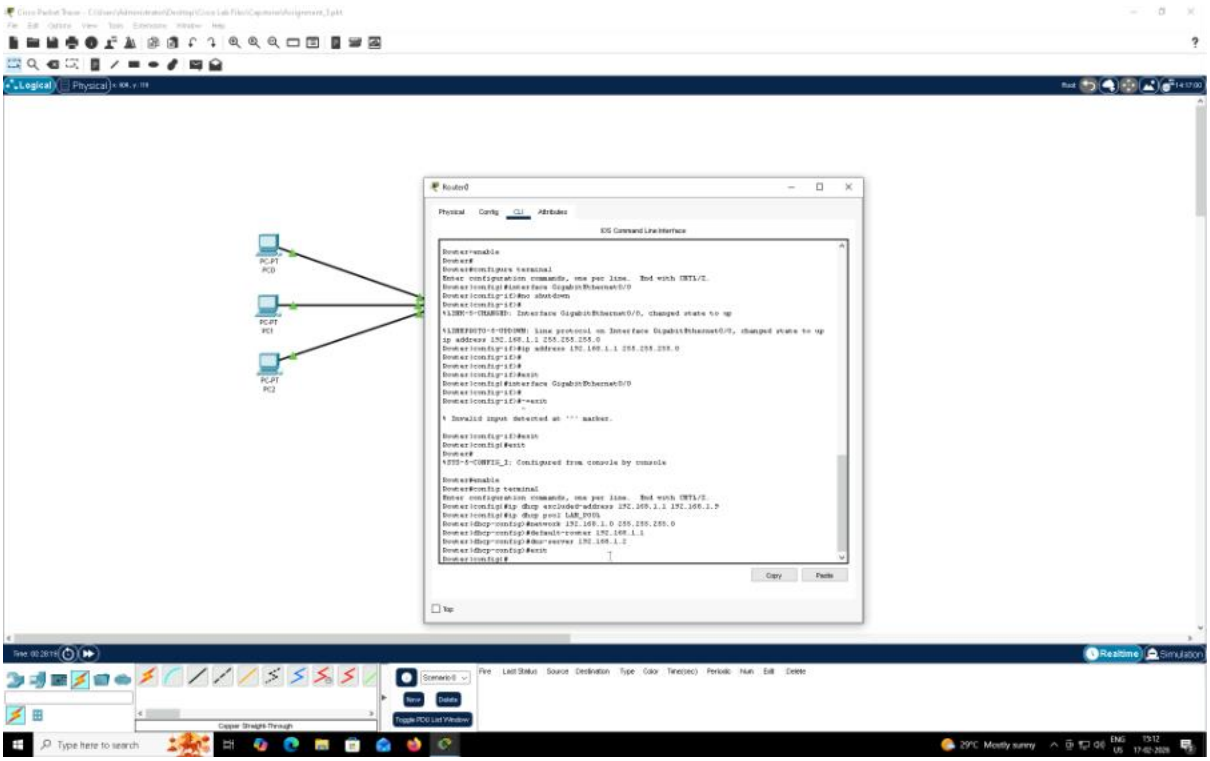
Client systems successfully obtained network configuration automatically from the DHCP server confirming proper operation of the DHCP service.

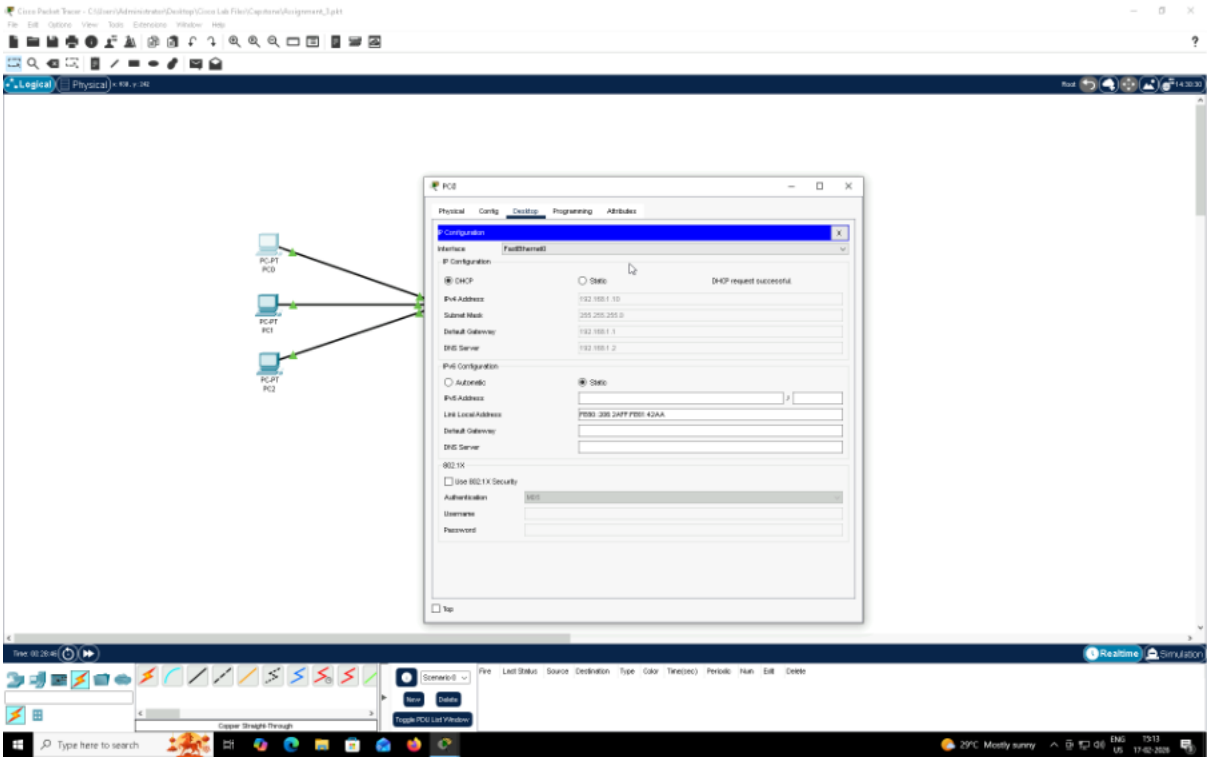
During DHCP configuration, clients received incorrect gateway addresses because no Layer-3 device existed in the network. To resolve this, a router was added to act as the default gateway for all hosts.

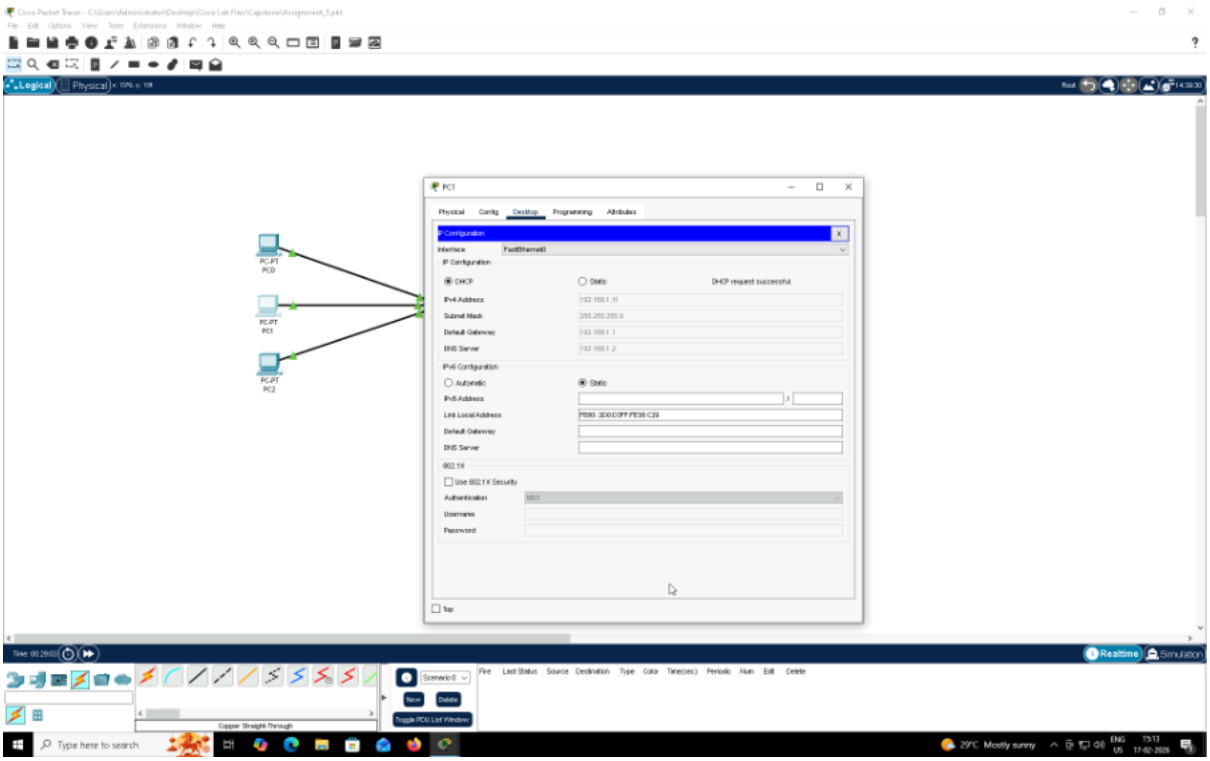


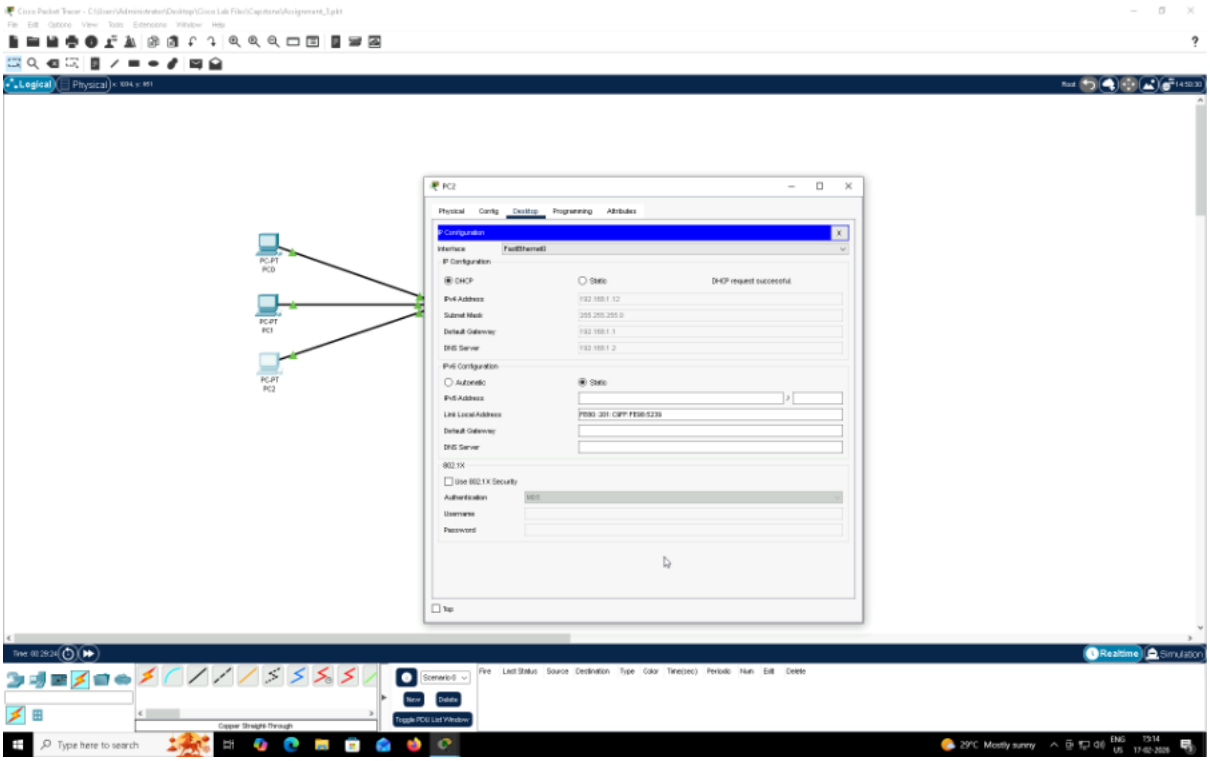
The router interface has been configured with IP address 192.168.1.1/24, and all clients will now use it as gateway.

DHCP Server migrated from server to router because the router acts as the default gateway and handles broadcast requests efficiently in a LAN environment.

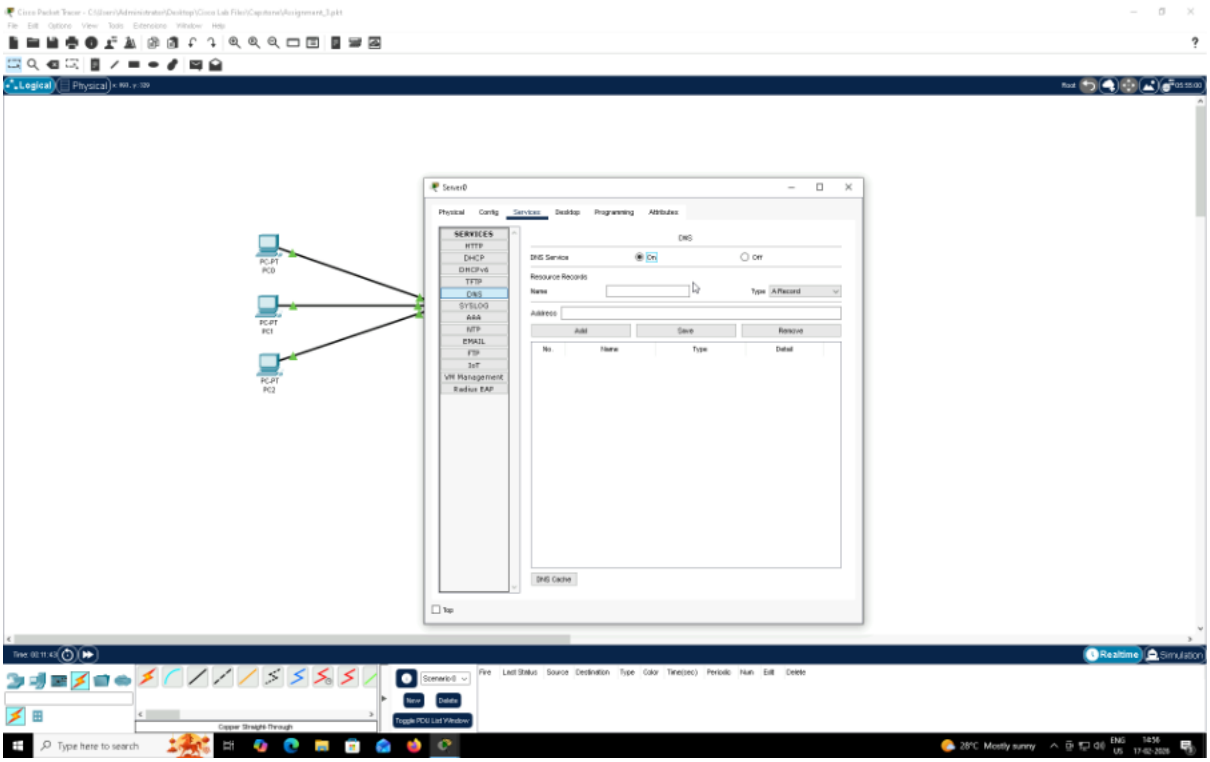

Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shut
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#--exit
^
% Invalid input detected at '^' marker.
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router>enable
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
Router(config)#ip dhcp pool LAN_POOL
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 192.168.1.2
Router(dhcp-config)#exit
Router(config)#
Copy
Paste
Top
PC-PT PC0
PC-PT PC1
PC-PT PC2
Realtime
Simulation
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Scenario 0
New
Delete
Toggle PDU List Window
Copper Straight-Through
Type here to search
29°C Mostly sunny
ENG
US
17-02-2026


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful
IPv4 Address
Subnet Mask
Default Gateway
DNS Server
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::205:2AFF:FE61:42AA
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
PC-PT PC0
PC-PT PC1
PC-PT PC2
Realtime
Simulation
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Scenario 0
New
Delete
Toggle PDU List Window
Copper Straight-Through
Type here to search
29°C Mostly sunny
ENG
US
17-02-2026

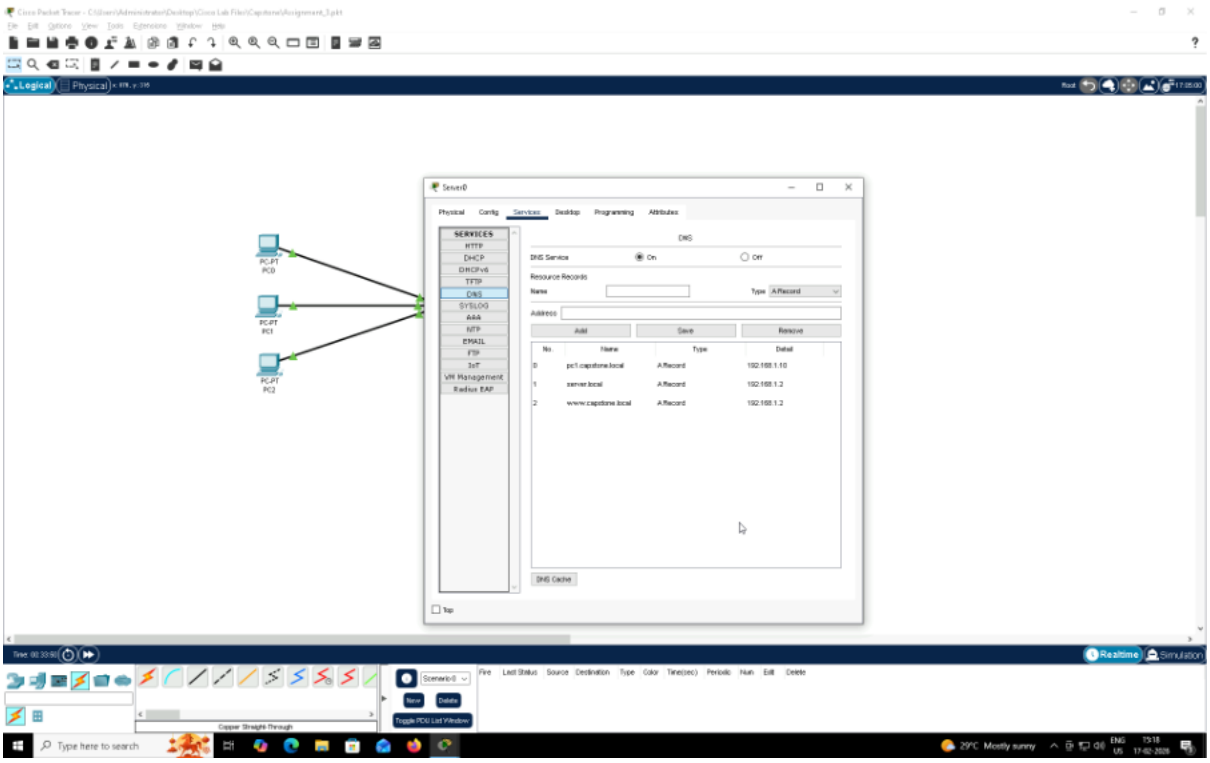




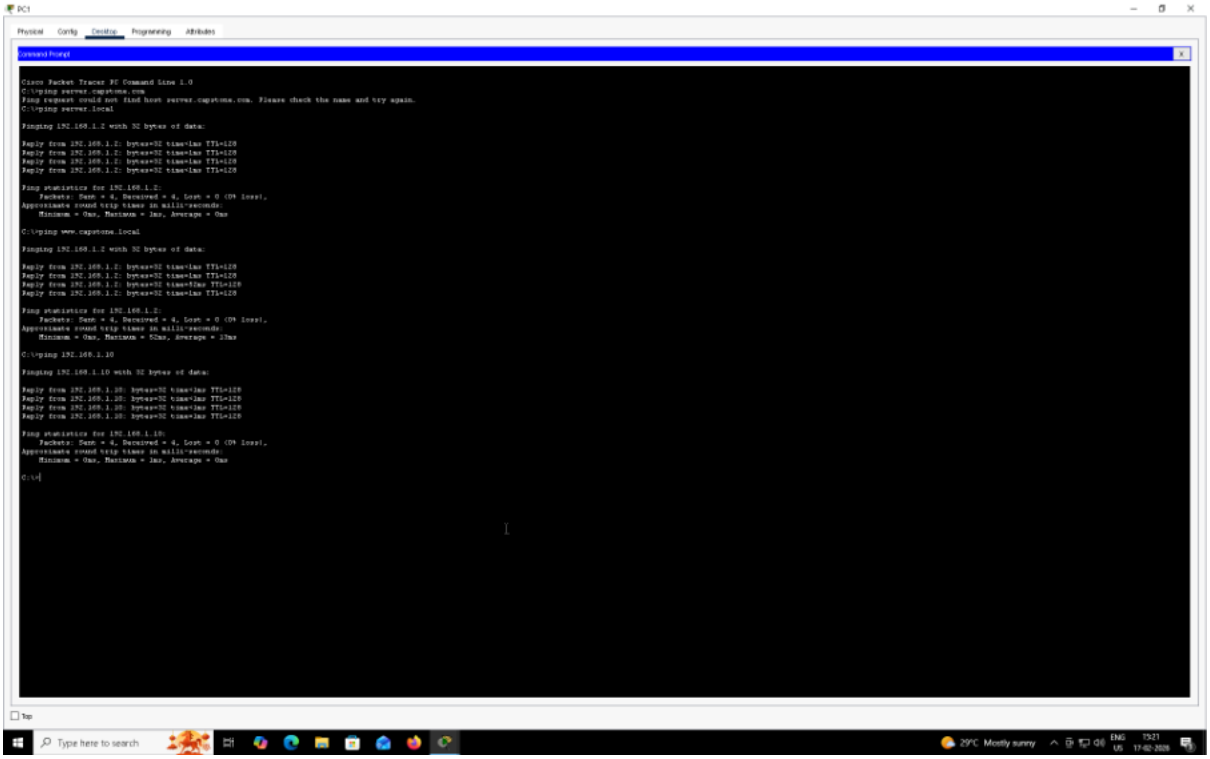
We can now see that the IP address have been allotted starting from 192.168.1.10 and the reserved ones were not assigned.



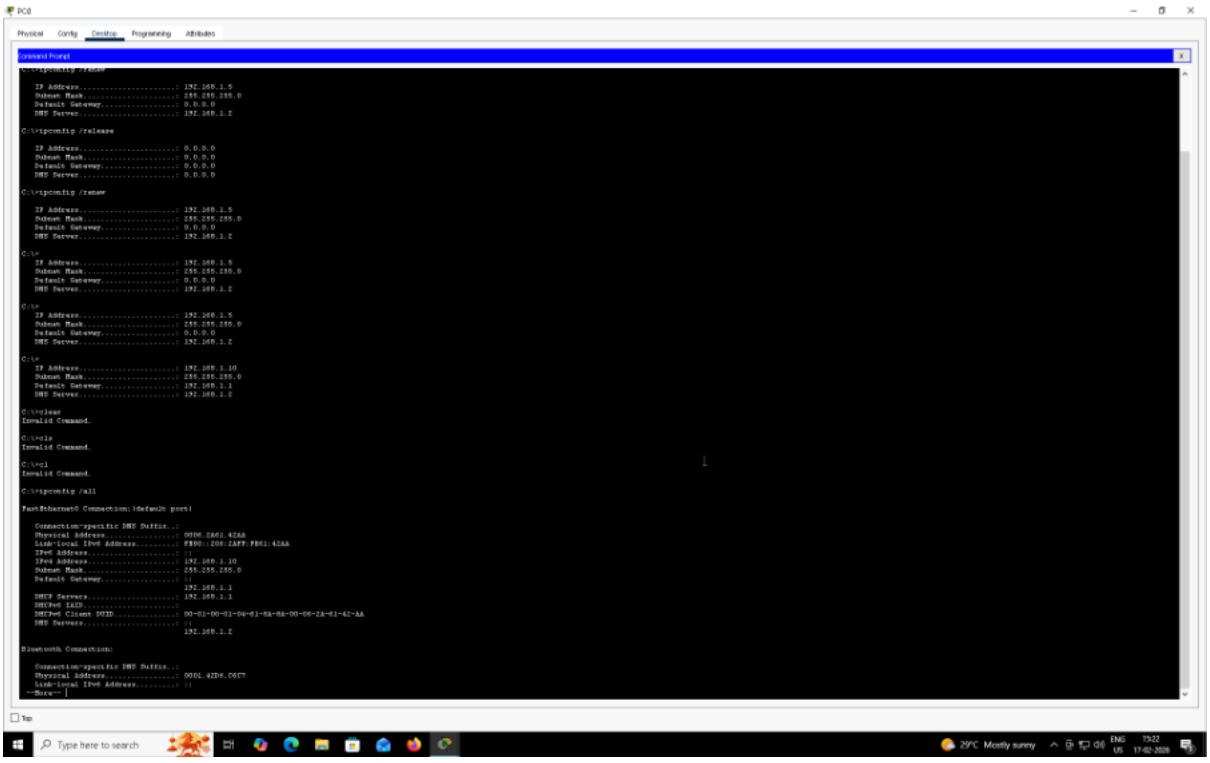
DNS has been enabled.



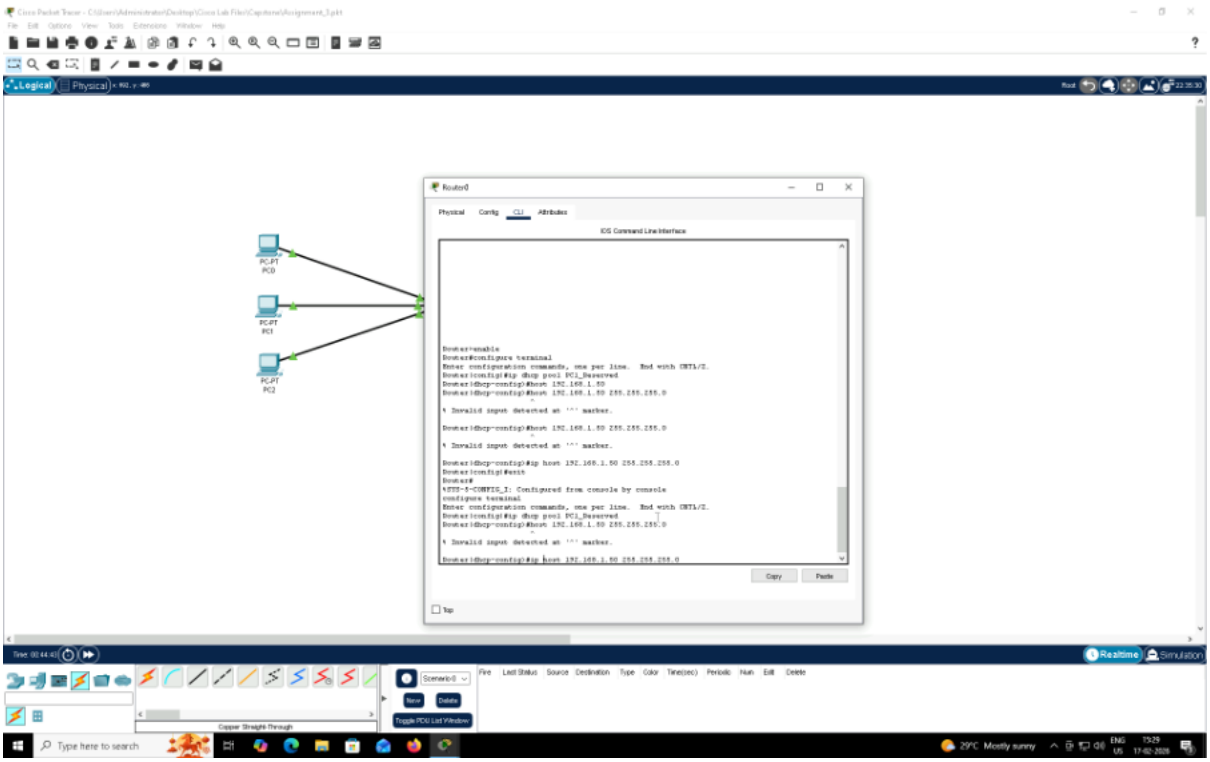
DNS server was configured to translate domain names into IP addresses. We have added an link (www.capstone.local) for simulation



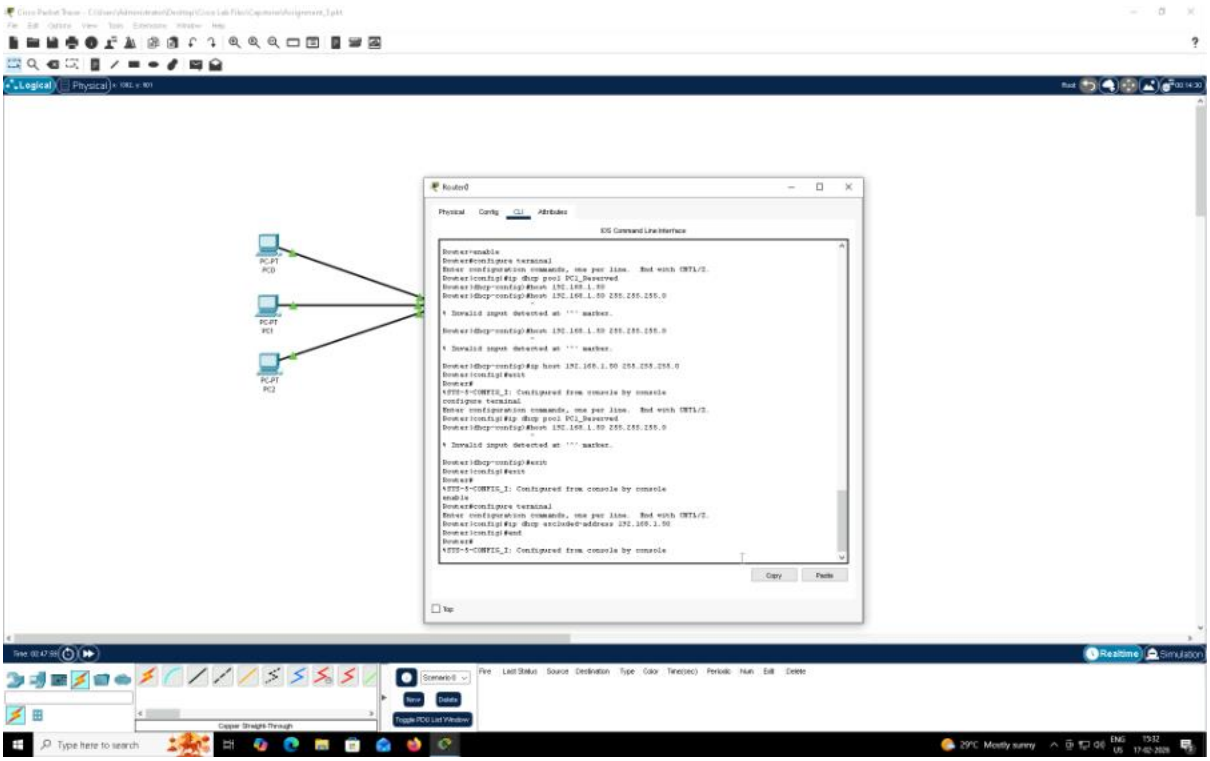
Hostname resolution was verified using ping command from client machine.

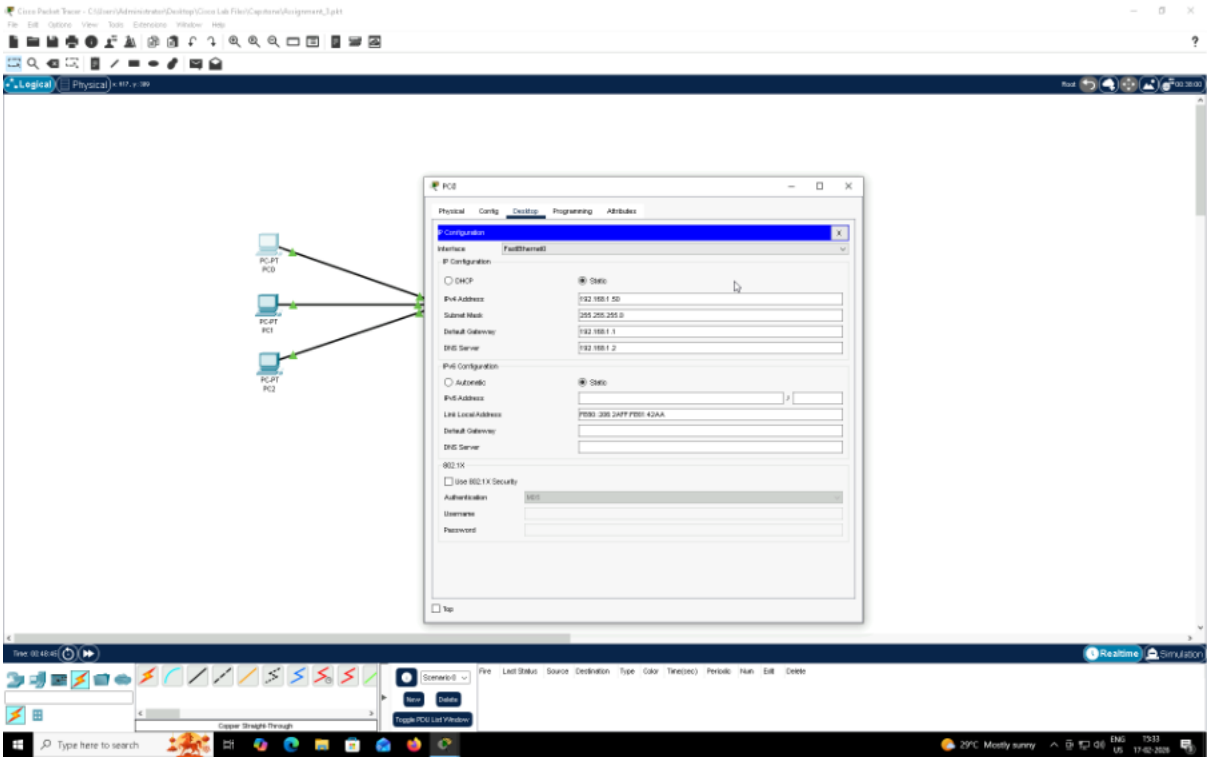


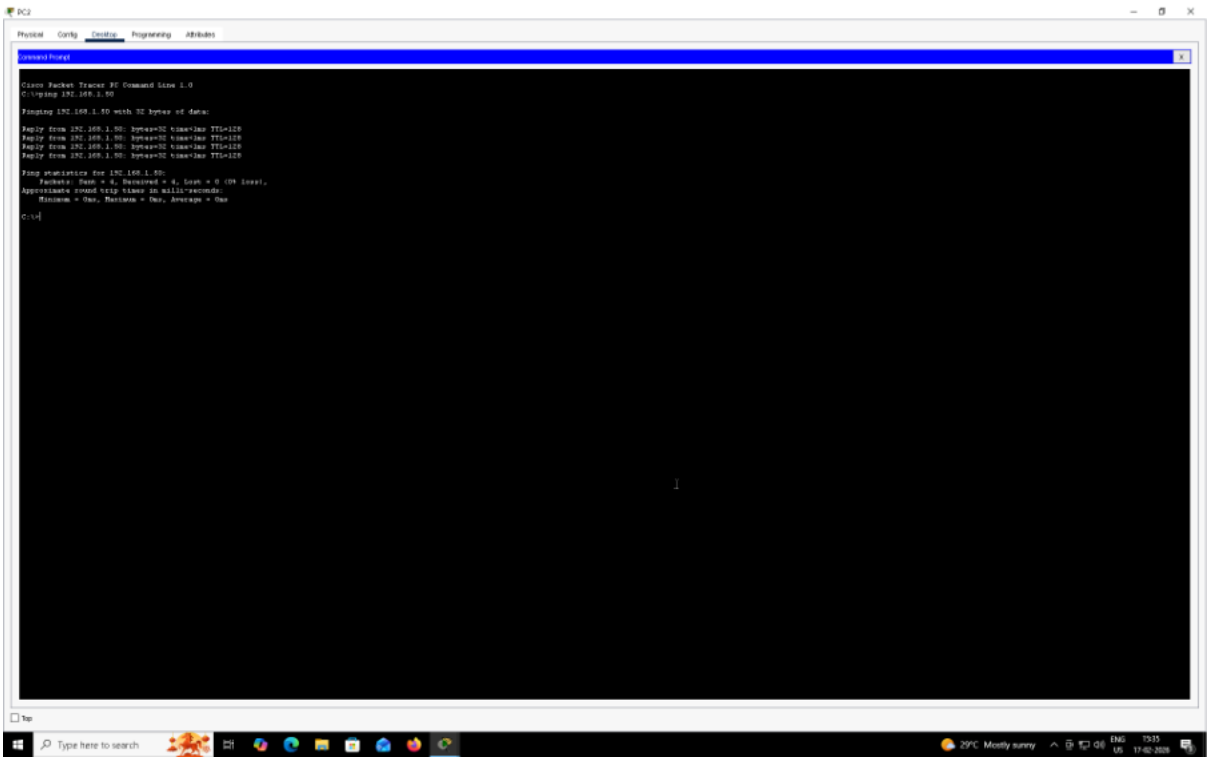
For implementing DHCP Reservation, we will need the MAC Address, we will use PC0's MAC Address (0001.42D6.C6C7).



Since the MAC-based DHCP binding is not supported yet by Packet Tracer so we will do using excluding the IP from the DHCP pool and manually assign it to the host.


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool PC1_Reserved
Router(dhcp-config)#host 192.168.1.50
Router(dhcp-config)#host 192.168.1.50 255.255.255.0
% Invalid input detected at '^' marker.
Router(dhcp-config)#host 192.168.1.50 255.255.255.0
% Invalid input detected at '^' marker.
Router(dhcp-config)#ip host 192.168.1.50 255.255.255.0
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool PC1_Reserved
Router(dhcp-config)#host 192.168.1.50 255.255.255.0
% Invalid input detected at '^' marker.
Router(dhcp-config)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.1.50
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top
PC-PT PC0
PC-PT PC1
PC-PT PC2
Realtime
Simulation
Copper Straight-Through
Type here to search
29°C Mostly sunny


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
192.168.1.50
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
192.168.1.2
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
Username
Password
Top
PC-PT PC0
PC-PT PC1
PC-PT PC2
Realtime
Simulation
Copper Straight-Through
Type here to search
29°C Mostly sunny

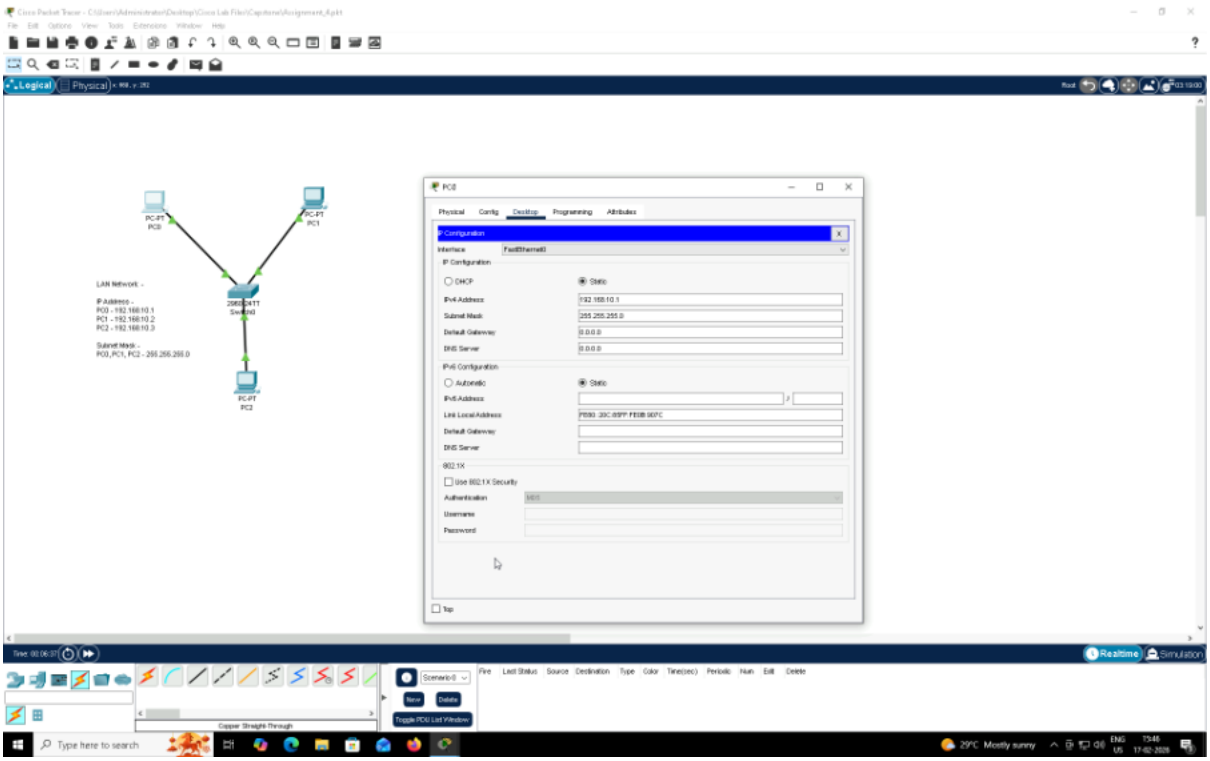


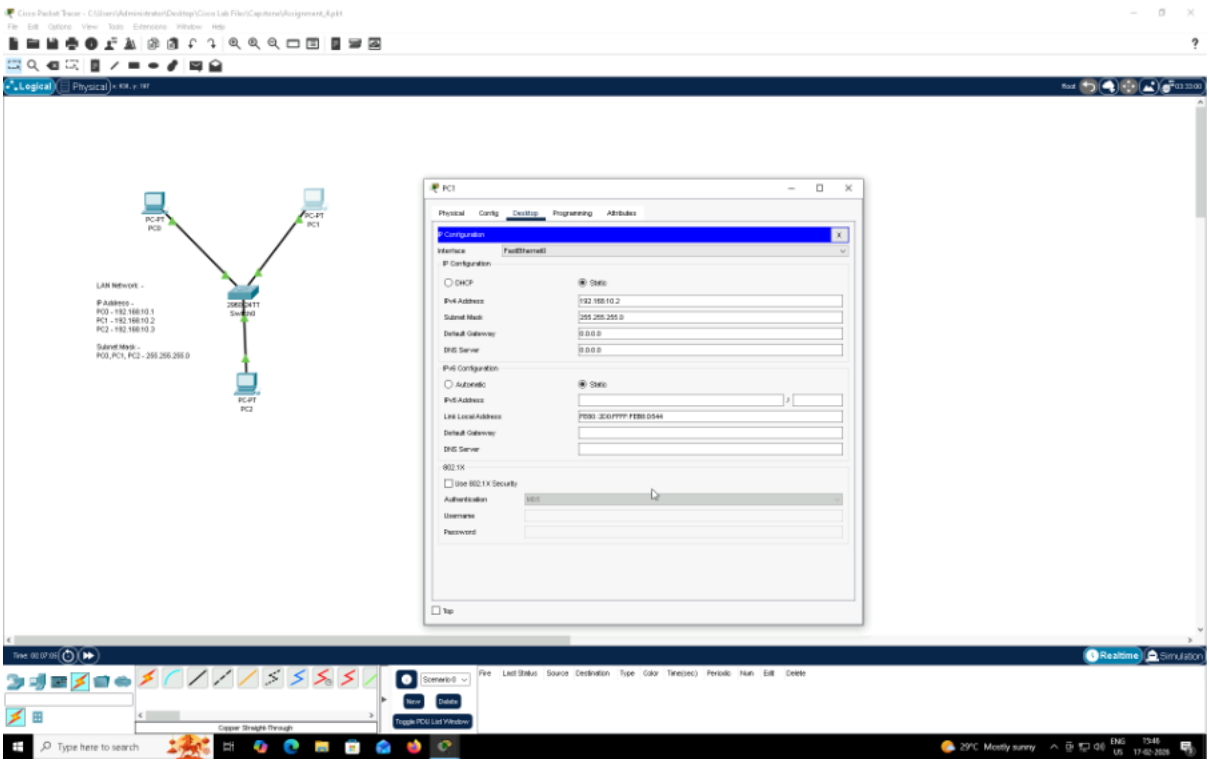
The reserved host was configured with a manually assigned IP address that was excluded from the DHCP pool. Connectivity was verified using ping, confirming successful communication within the network.

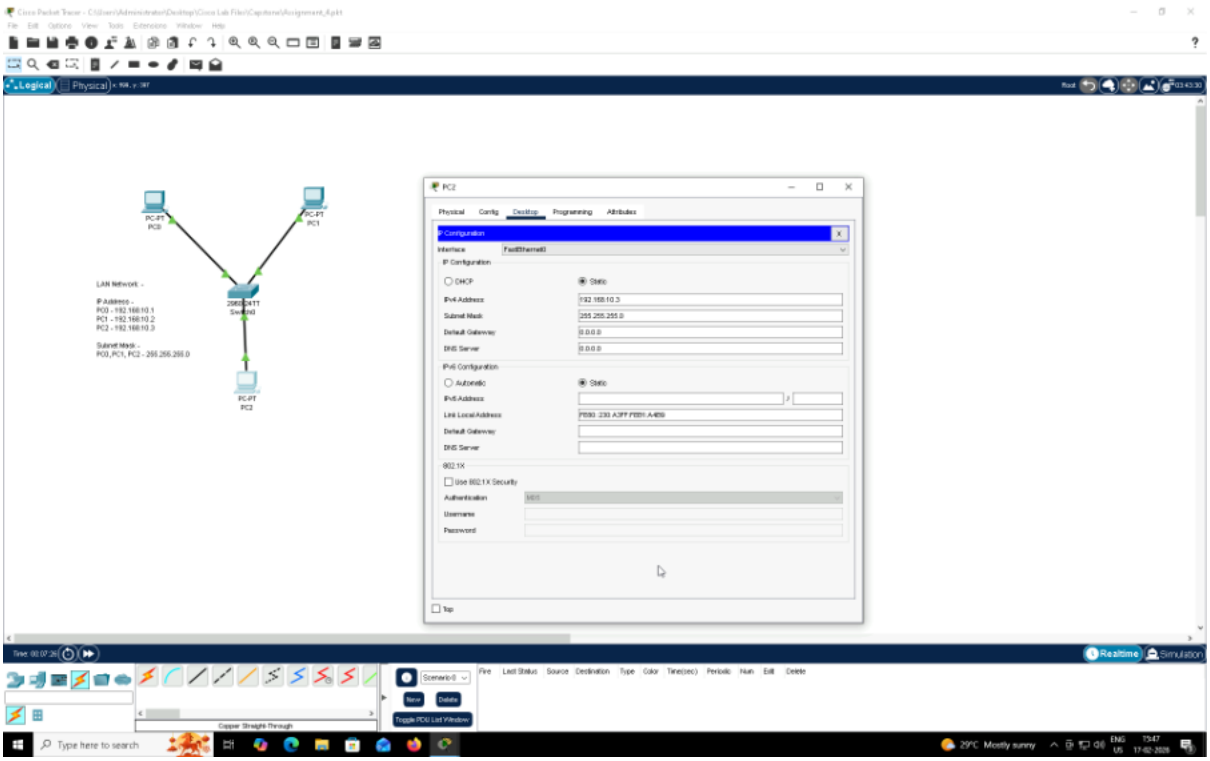
Assignment 4: Network Type Classification

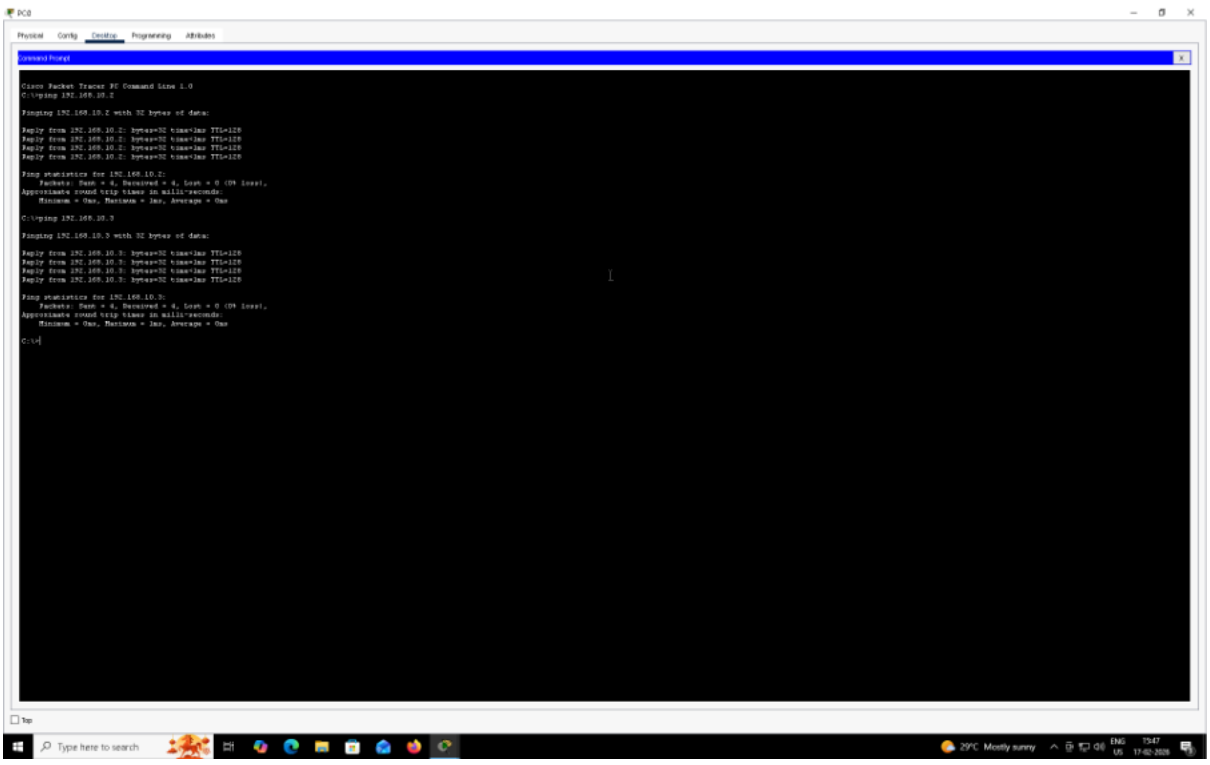
As part of this assignment, you need to complete below tasks:

- LAN network in a single building
- WAN network across multiple locations
- WLAN for wireless connectivity
- SOHO network setup

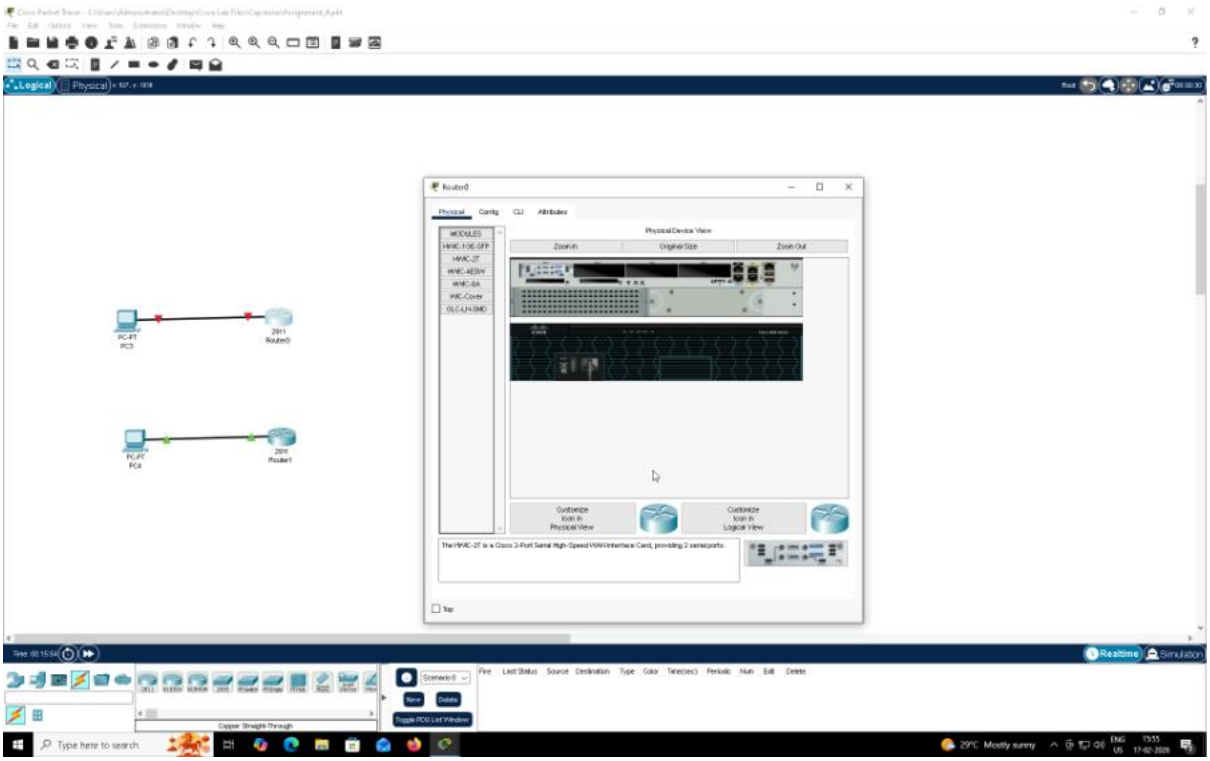

PC-PT
PC0
PC-PT
PC1
LAN Network: -
IP Address -
PC0 - 192.168.10.1
PC1 - 192.168.10.2
PC2 - 192.168.10.3
Subnet Mask -
PC0,PC1, PC2 - 255.255.255.0
PC-PT
PC2
PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
192.168.10.1
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
802.1X
Use 802.1X Security
Authentication
Username
Password
Top
Realtime
Simulation
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Scenario 0
New
Delete
Toggle PDU List Window
Copper Straight-Through
Type here to search
29°C Mostly sunny
ENG
US
17-02-2026


PC-PT
PC0
PC-PT
PC1
LAN Network: -
IP Address -
PC0 - 192.168.10.1
PC1 - 192.168.10.2
PC2 - 192.168.10.3
Subnet Mask -
PC0,PC1, PC2 - 255.255.255.0
PC-PT
PC2
PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
802.1X
Use 802.1X Security
Authentication
Username
Password
Top
Realtime
Simulation
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Scenario 0
New
Delete
Toggle PDU List Window
Copper Straight-Through
Type here to search
29°C Mostly sunny
ENG
US
17-02-2026

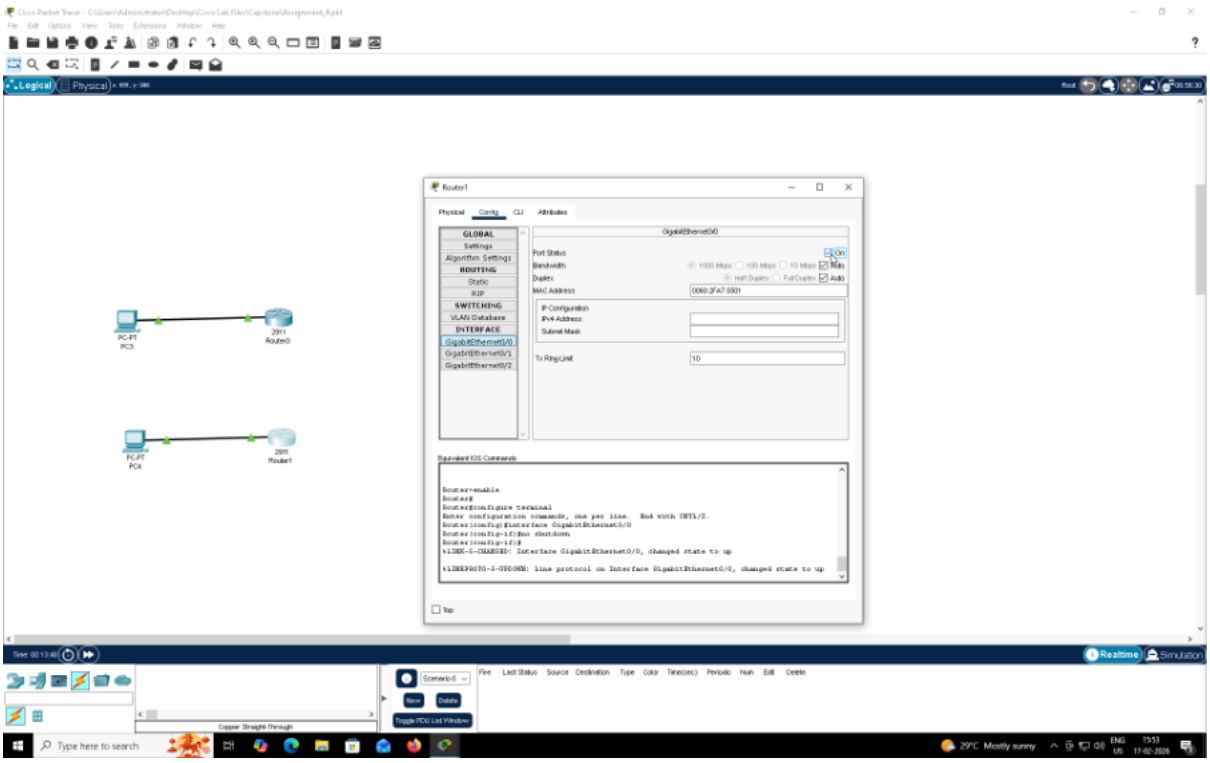


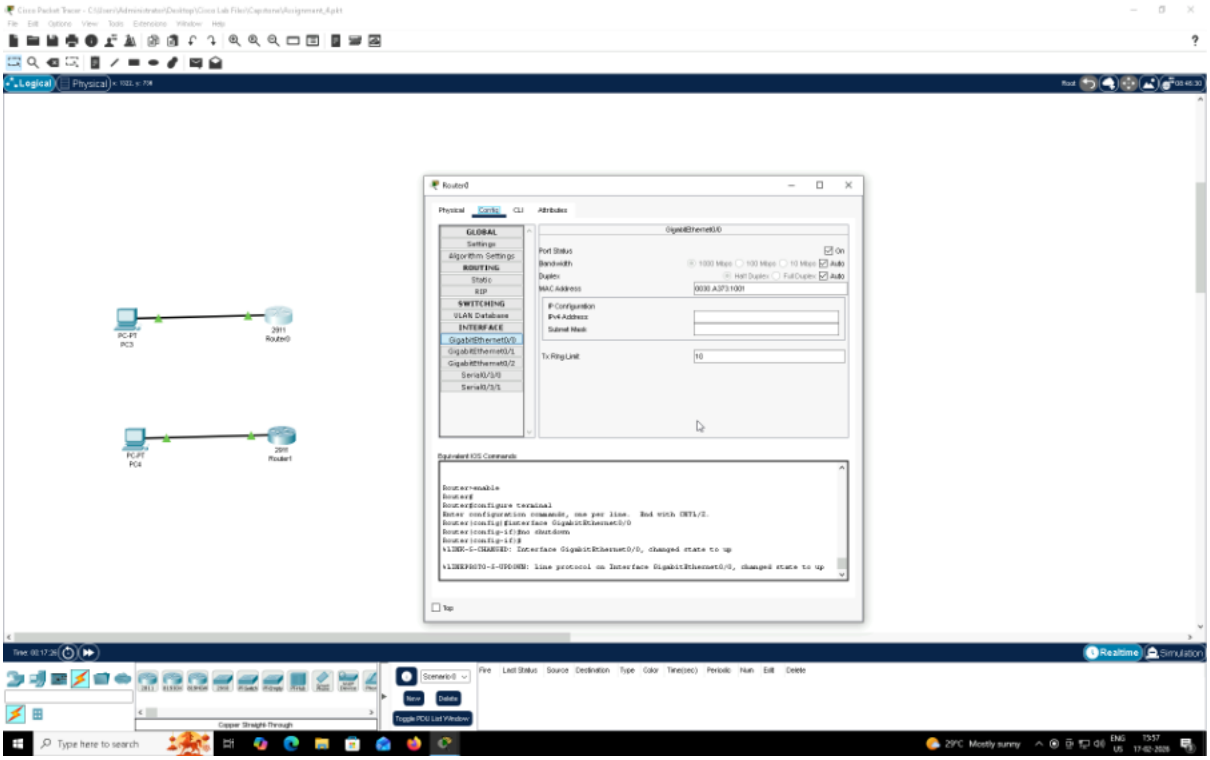


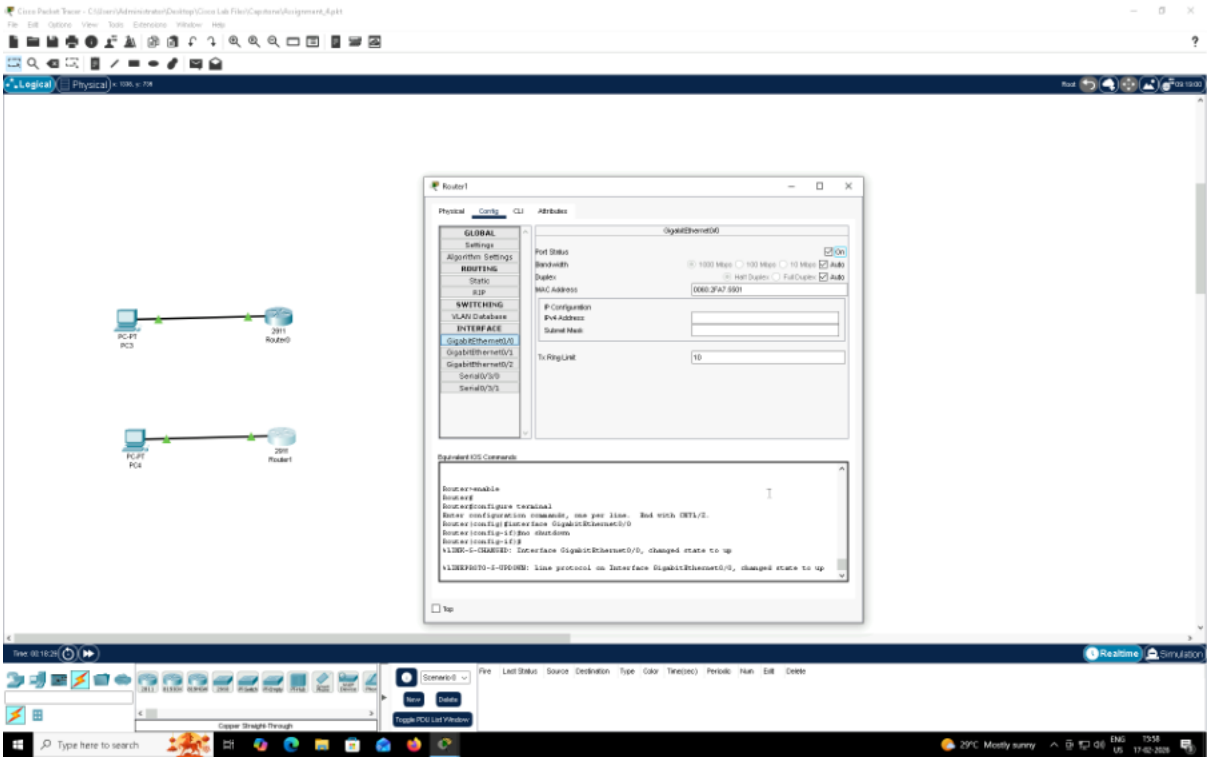
A LAN Topology was created using a switch and multiple PCs connected in the same subnet. Devices communicated directly without a router since they belonged to the same network.



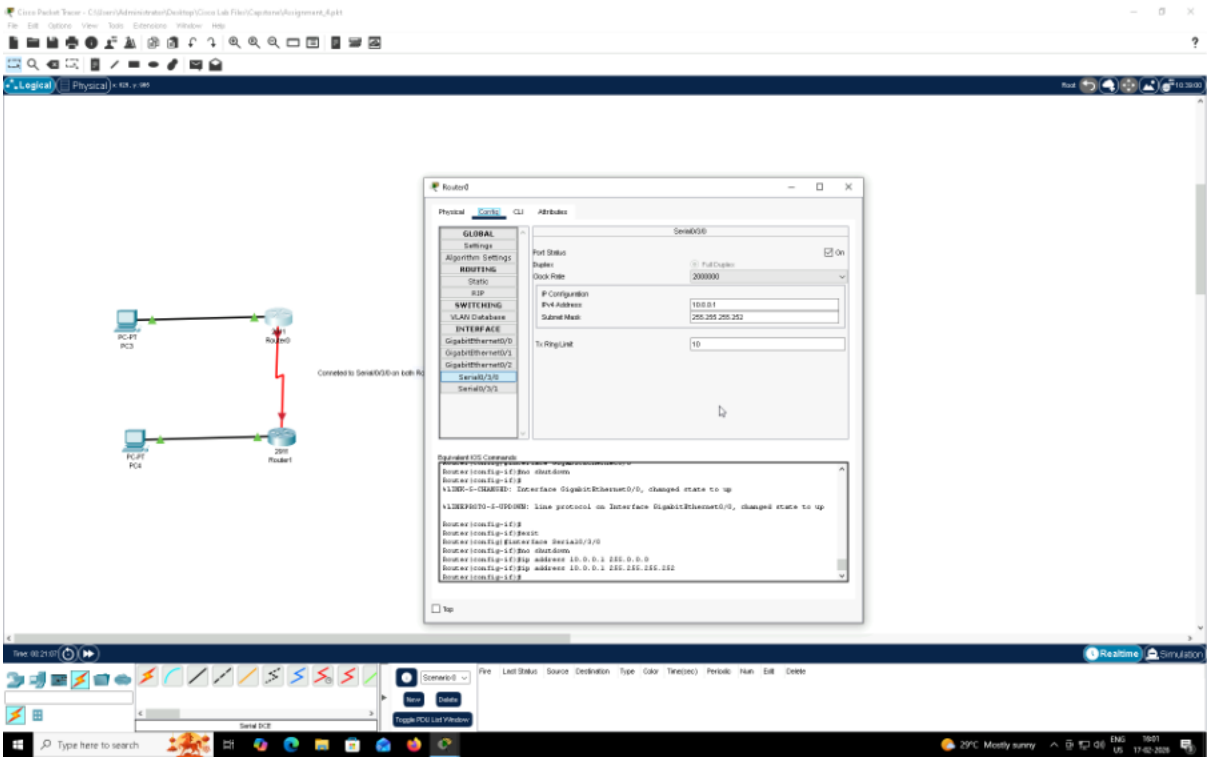
Power off the router and add HWIC-2T module in both routers.

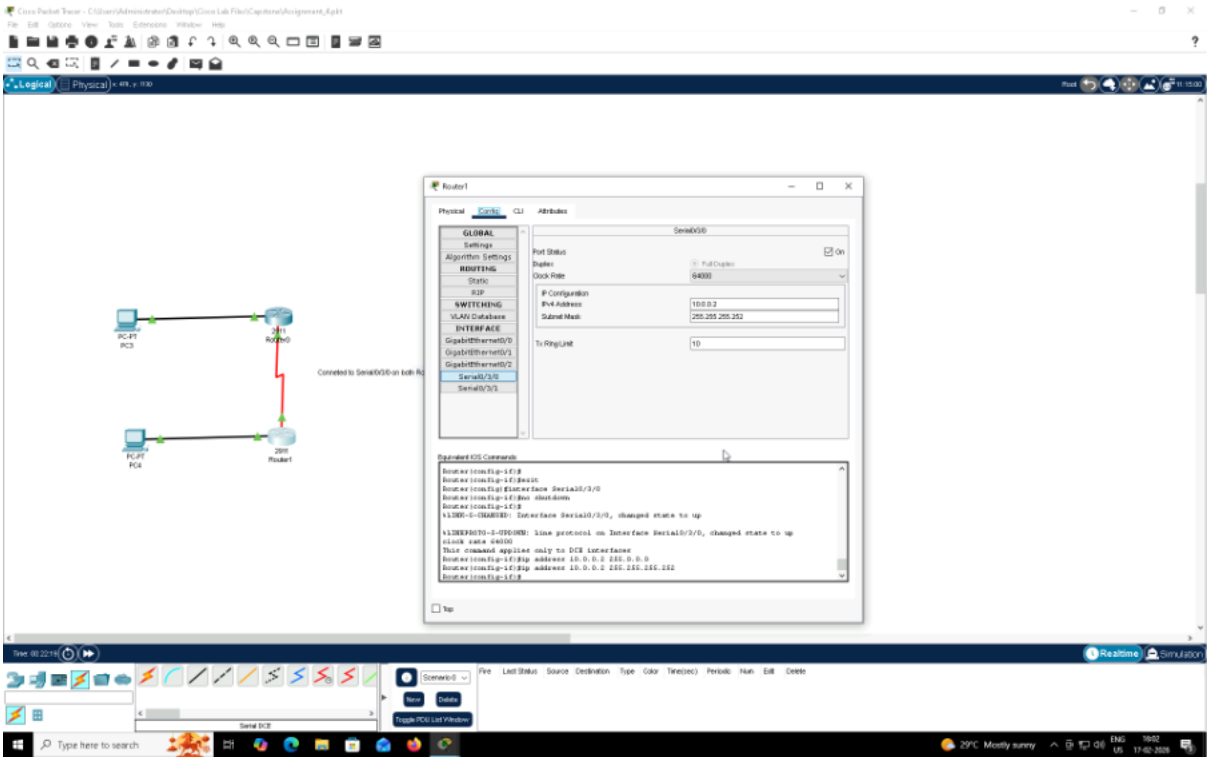


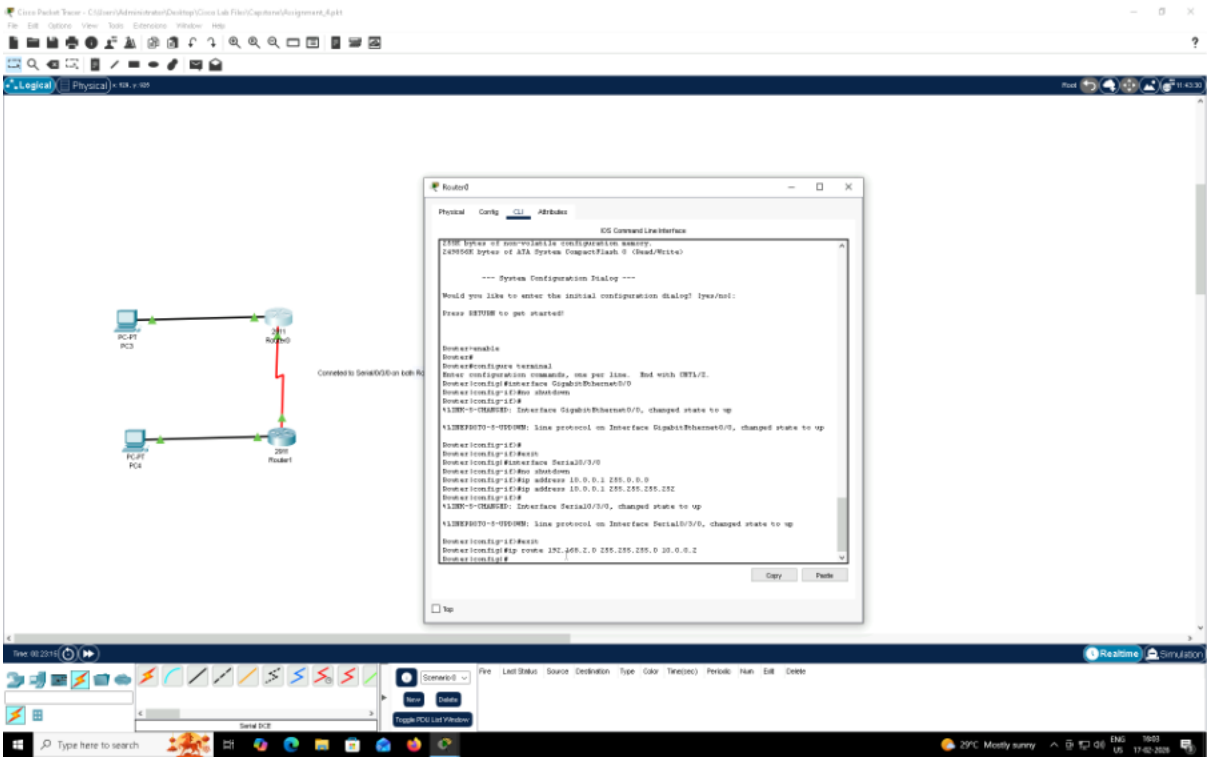


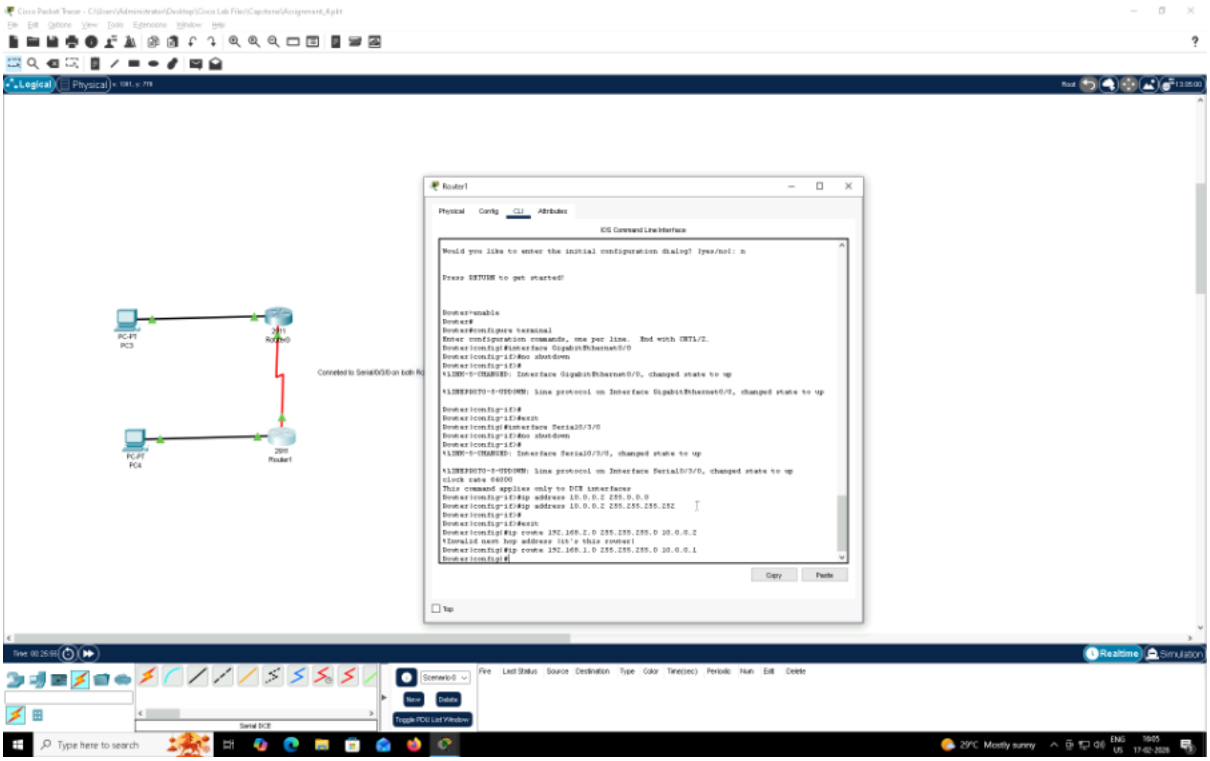


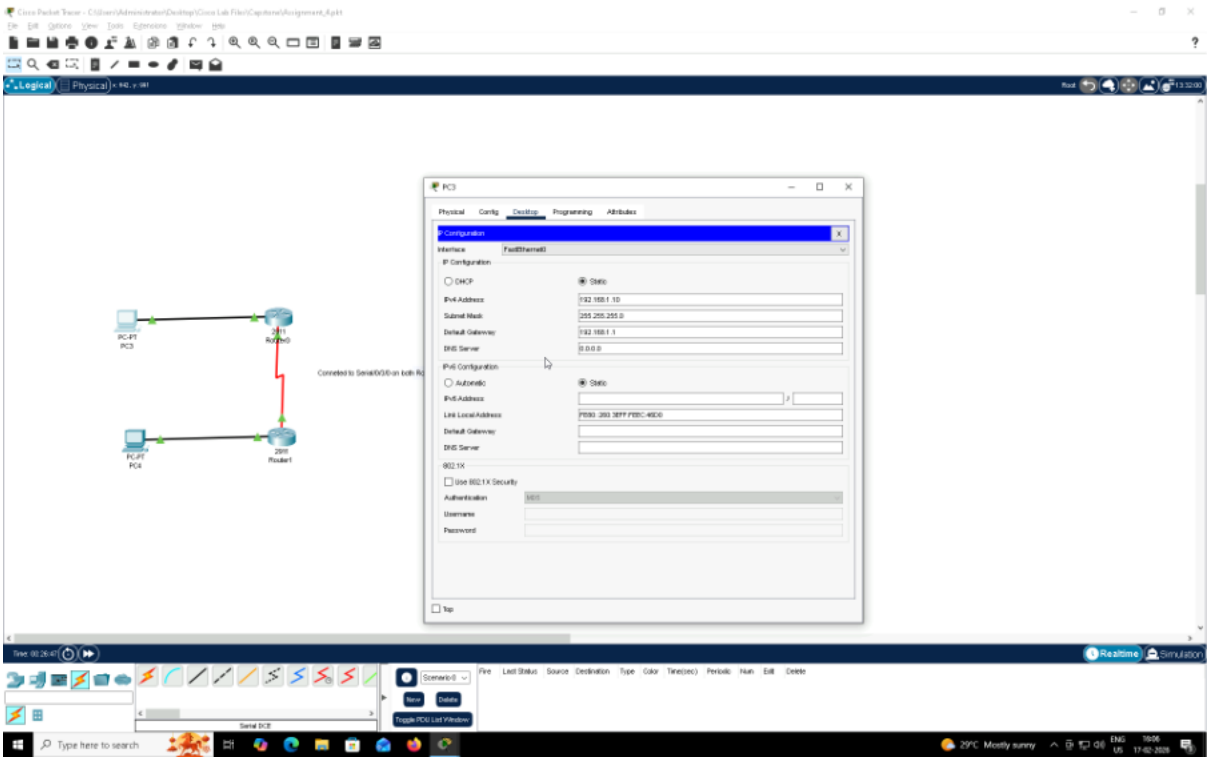
A serial DCE cable was use to simulate WAN communication between routers. The DCE router provides clock synchronization required for data transmission.

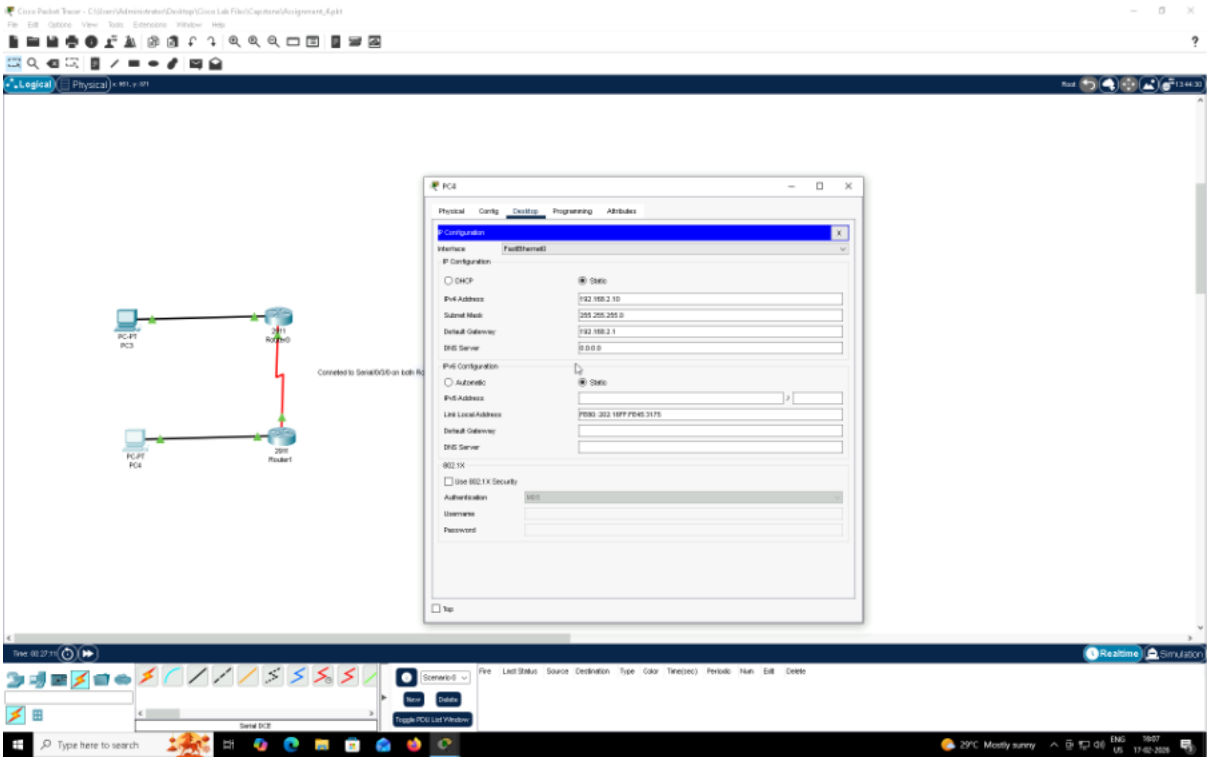

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
Serial0/3/0
Serial0/3/1
Port Status
On
Clock Rate
2000000
IP Configuration
IPv4 Address
10.0.0.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10
Equivalent IOS Commands
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#no shutdown
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#
Top
PC-PT
PC3
PC-PT
PC4
2911
Router0
2911
Router1
Connected to Serial0/3/0 on both Ro
Realtime
Simulation
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Scenario 0
New
Delete
Toggle PDU List Window
Serial DCE
Type here to search
29°C Mostly sunny
ENG
US
17-02-2026

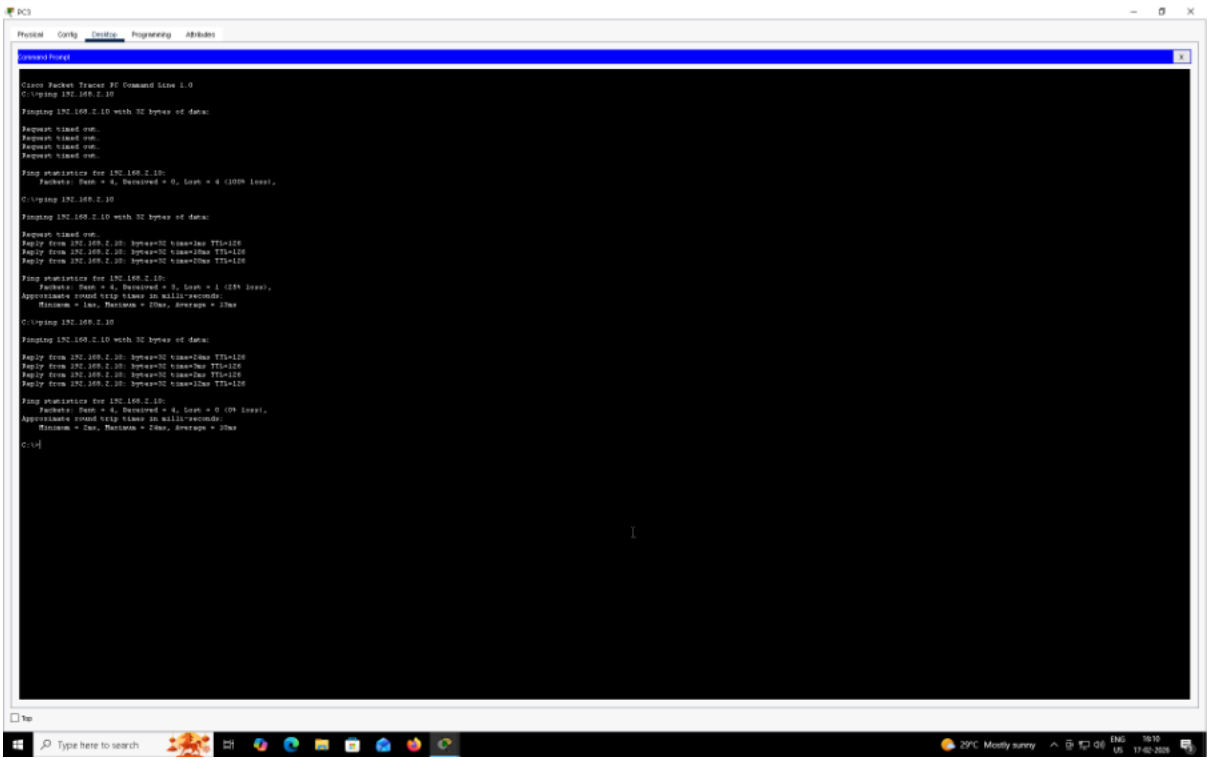

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
Serial0/3/0
Serial0/3/1
Port Status
On
Clock Rate
64000
IP Configuration
IPv4 Address
10.0.0.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10
Equivalent IOS Commands
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#
Top
PC-PT
PC3
PC-PT
PC4
2911
Router0
2911
Router1
Connected to Serial0/3/0 on both Ro
Realtime
Simulation
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Scenario 0
New
Delete
Toggle PDU List Window
Serial DCE
Type here to search
29°C Mostly sunny
ENG
US
17-02-2026


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/3/0
Router(config-if)#no shutdown
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/3/0, changed state to up
Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#
Copy
Paste
Top
PC-PT
PC3
PC-PT
PC4
2911
Router0
2911
Router1
Realtime
Simulation
Type here to search
29°C Mostly sunny

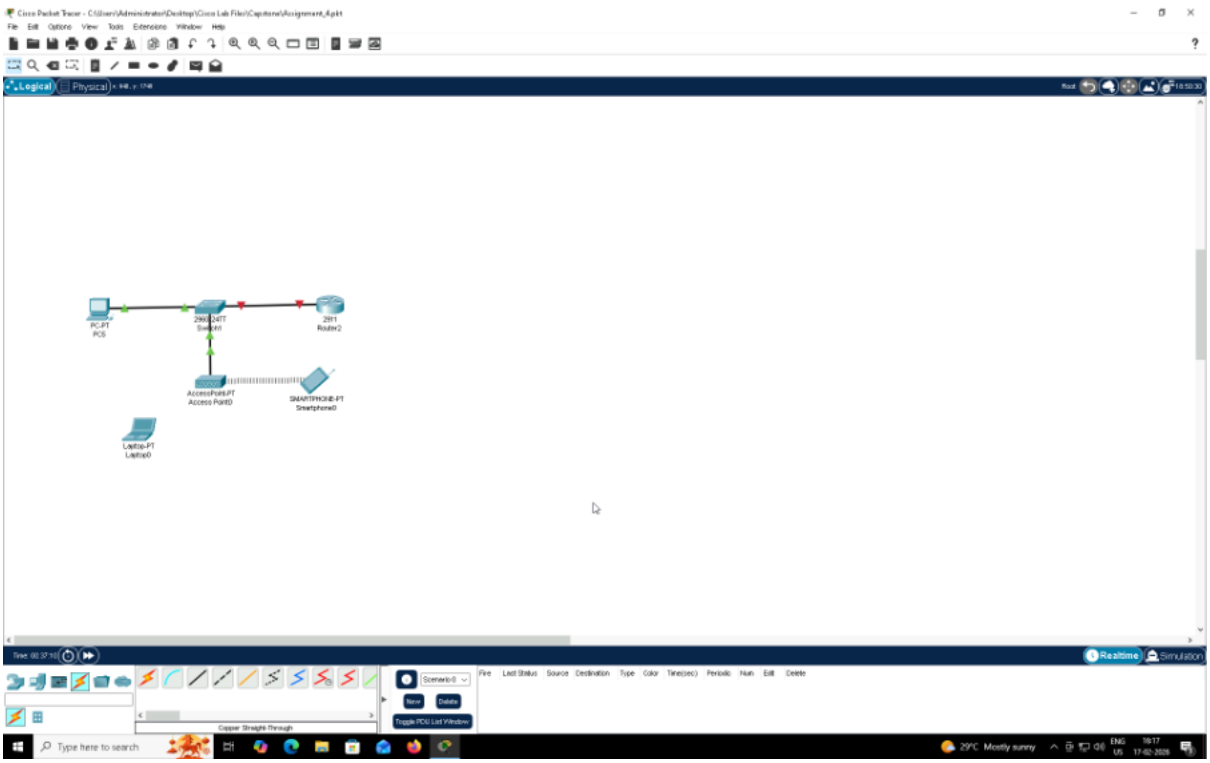

Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]: n
Press RETURN to get started!
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/3/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/3/0, changed state to up
clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#
Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
%Invalid next hop address (it's this router)
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#
Copy
Paste
Top
PC-PT
PC3
PC-PT
PC4
2911
Router0
2911
Router1
Realtime
Simulation
Type here to search
29°C Mostly sunny

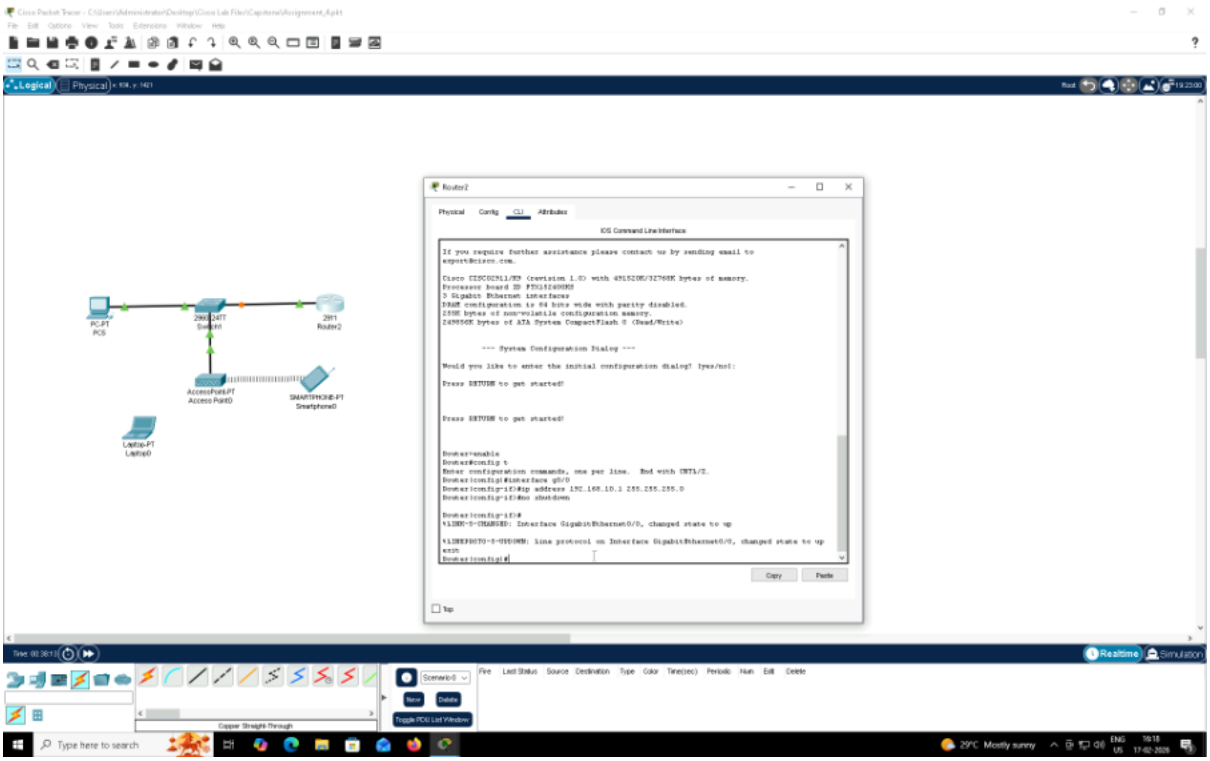

PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
DNS Server
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
Username
Password
Top
PC-PT
PC3
PC-PT
PC4
Router0
Router1
Realtime
Simulation


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
DNS Server
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
Username
Password
Top
PC-PT
PC3
PC-PT
PC4
Router0
Router1
Realtime
Simulation

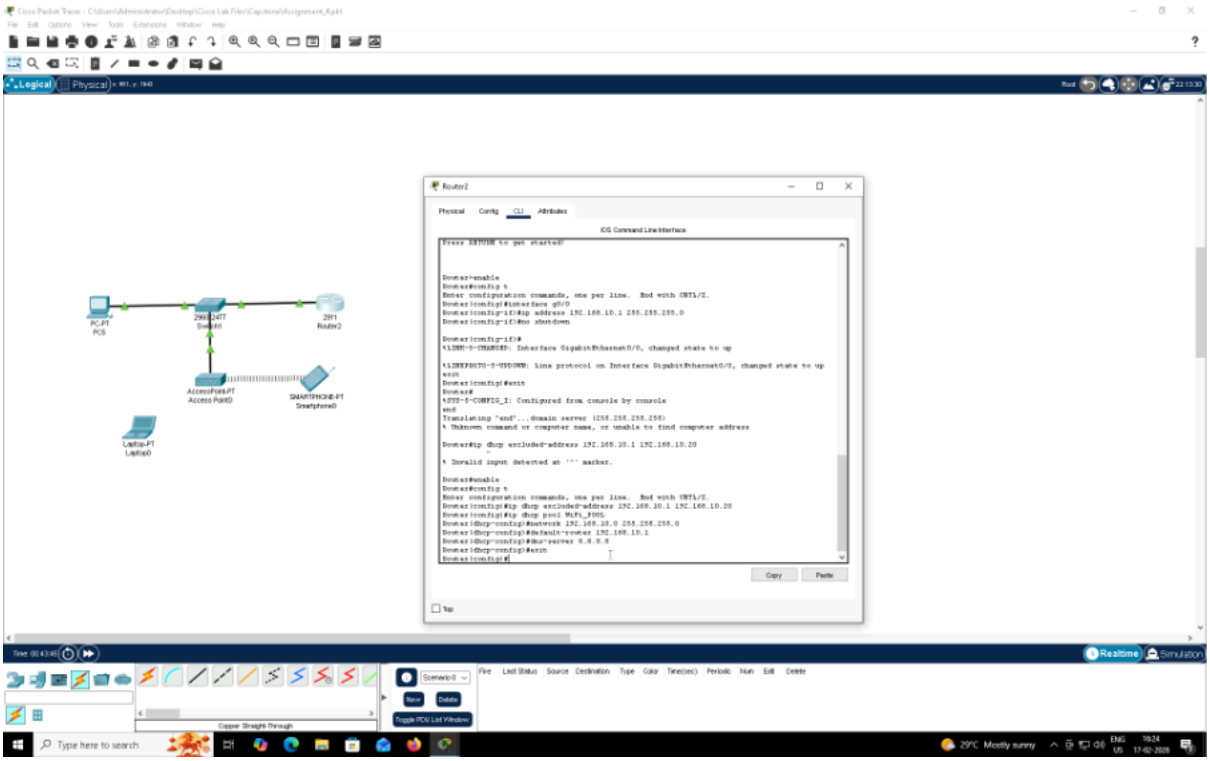


A WAN Topology was created using two routers connected via serial link. Static routing was configured to allow communication between two different networks.

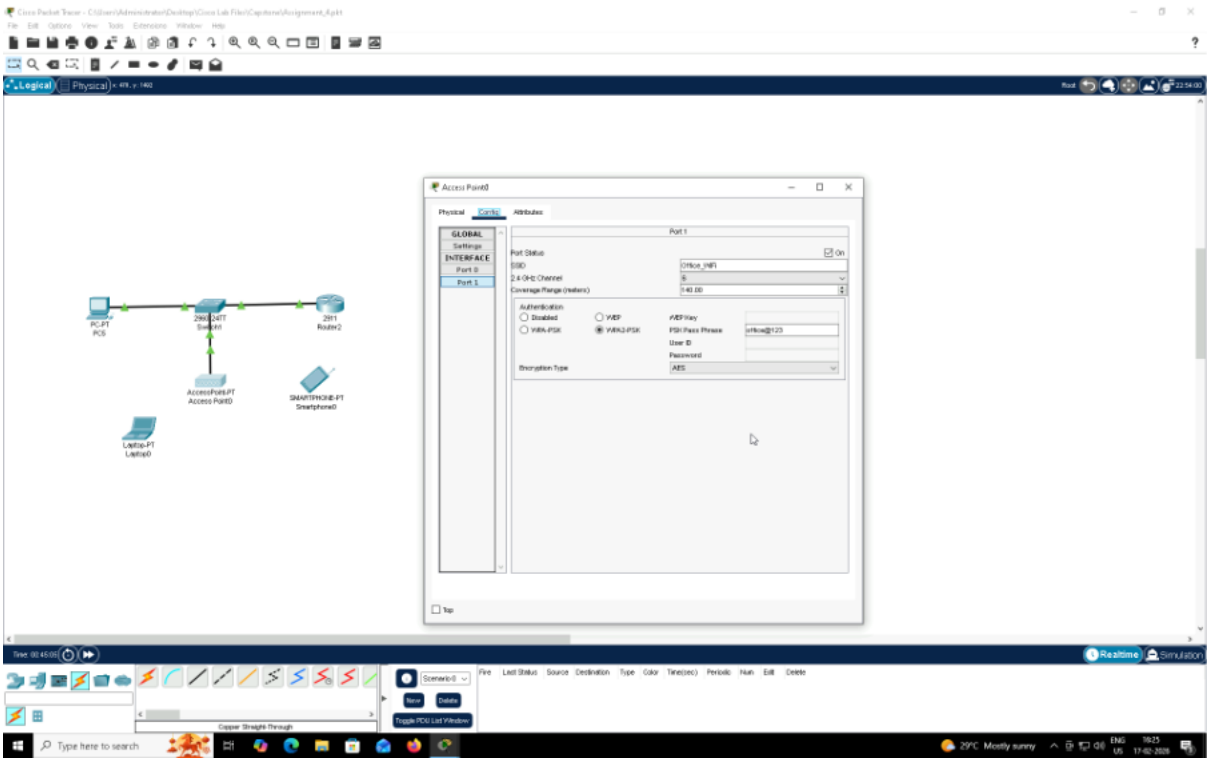




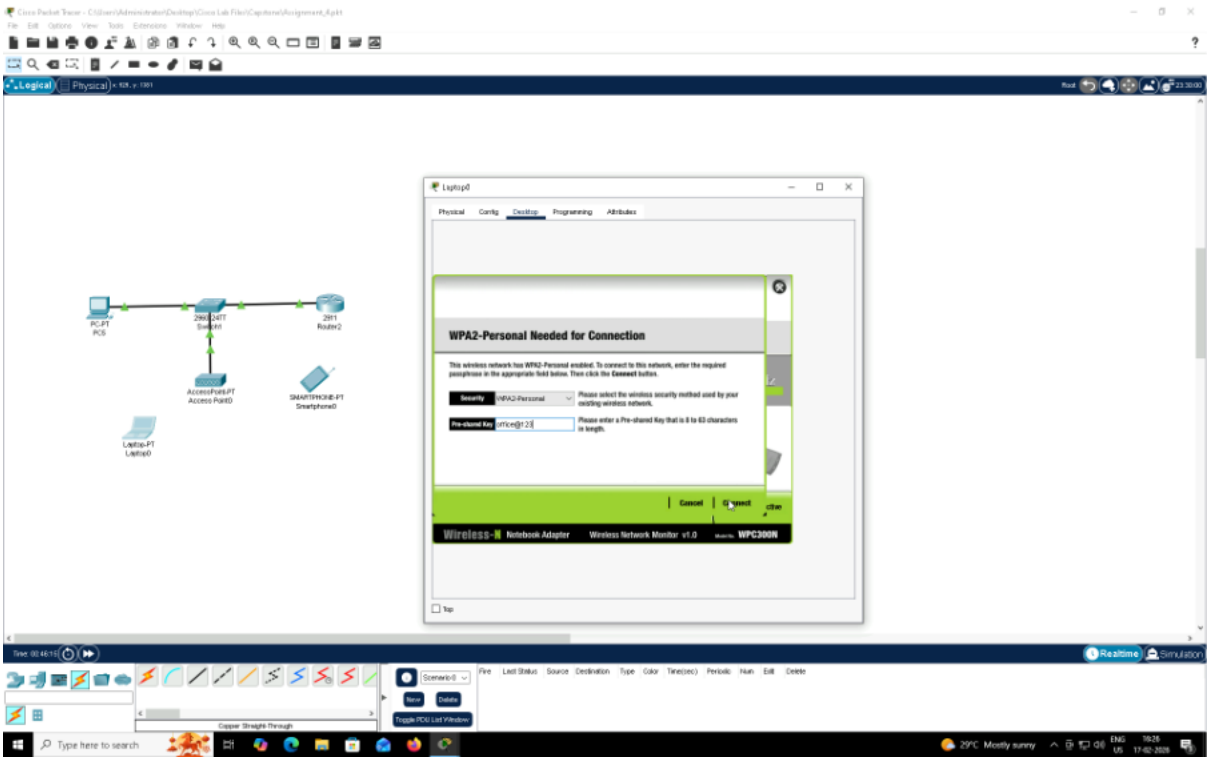
The router interface was configured as the default gateway to enable communication between wireless clients and the network.

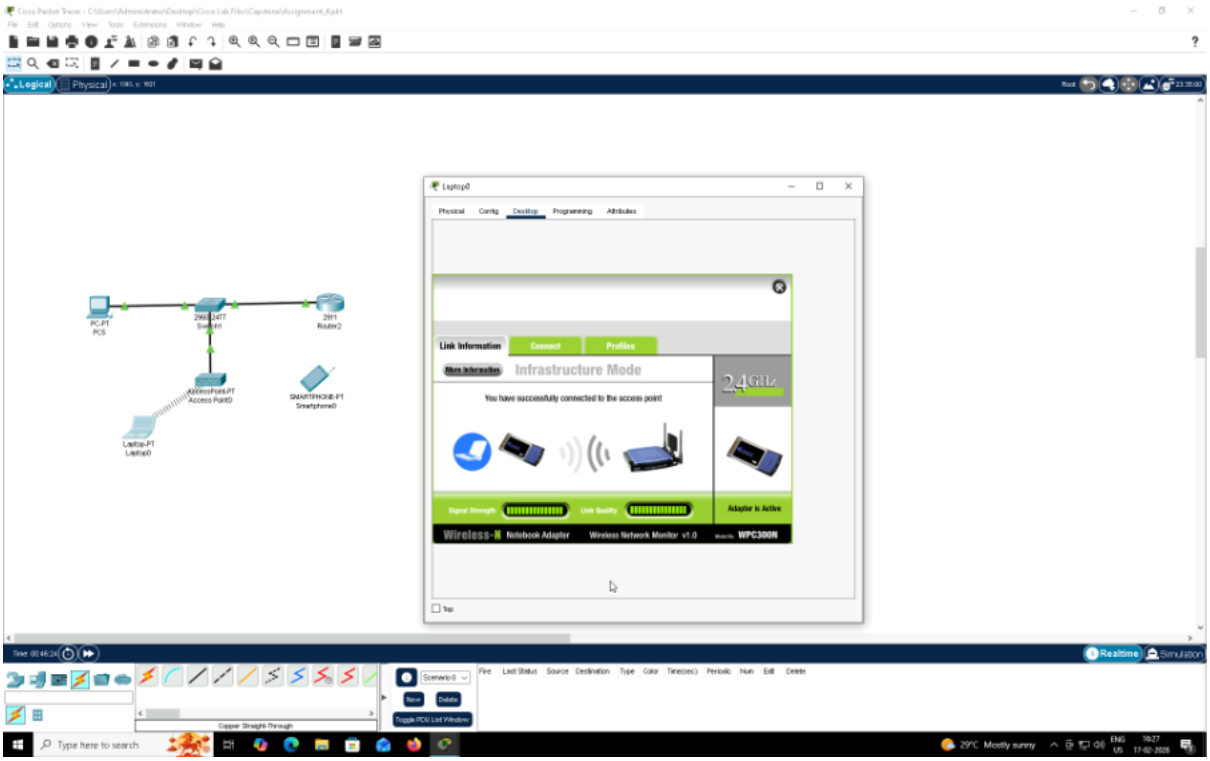


A DHCP pool was configured to automatically assign IP addresses to wireless clients connecting to the network.

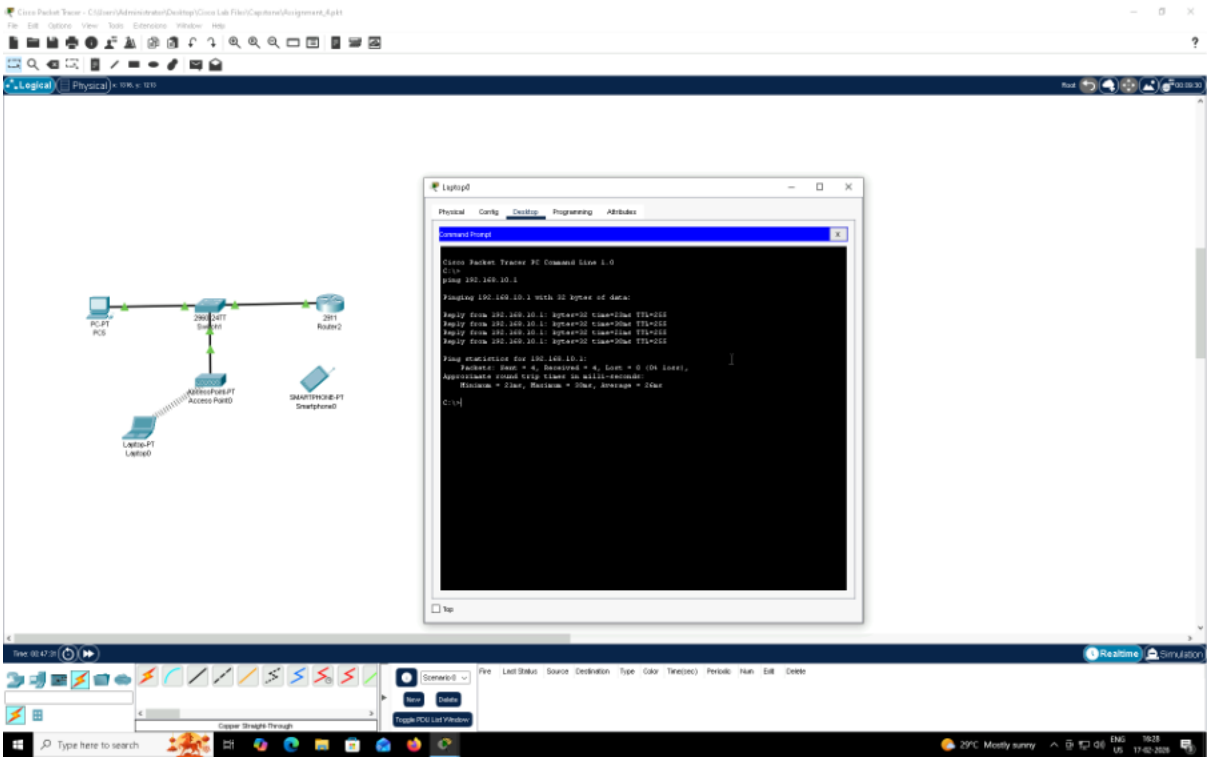


The access point was configured with WPAS2 security to provide secure wireless connectivity for users.

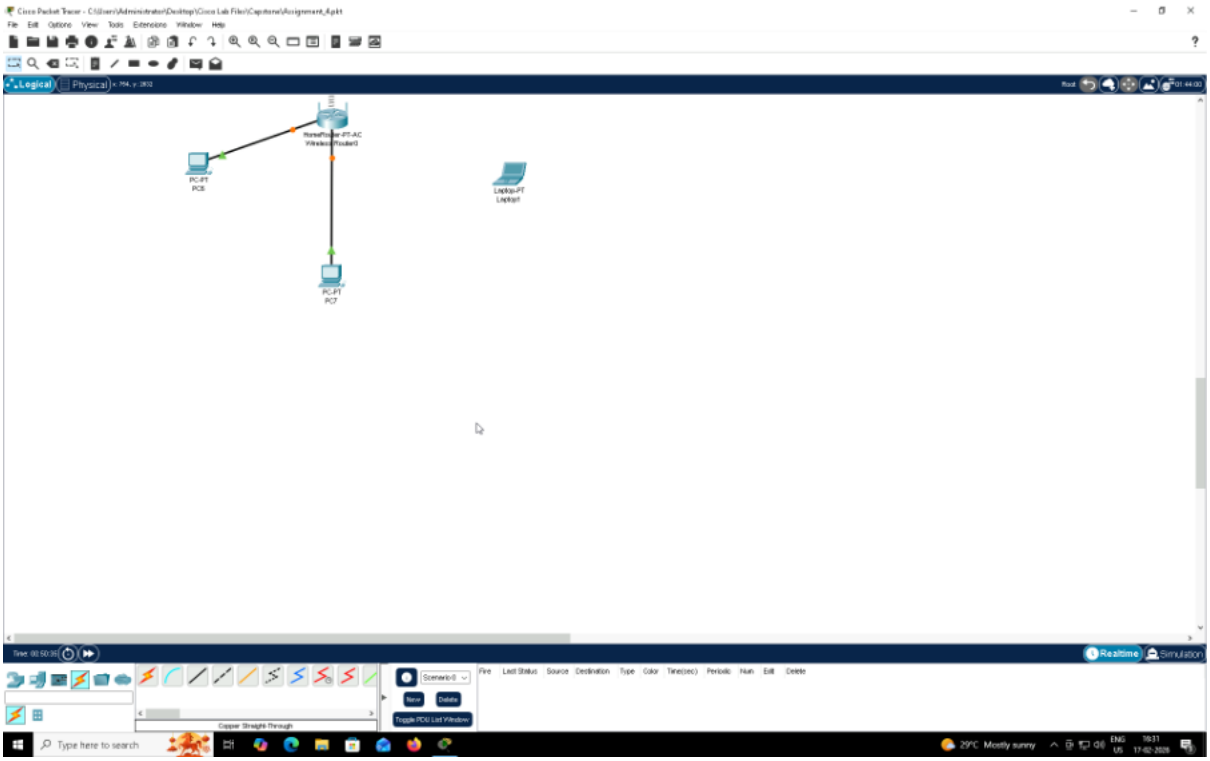


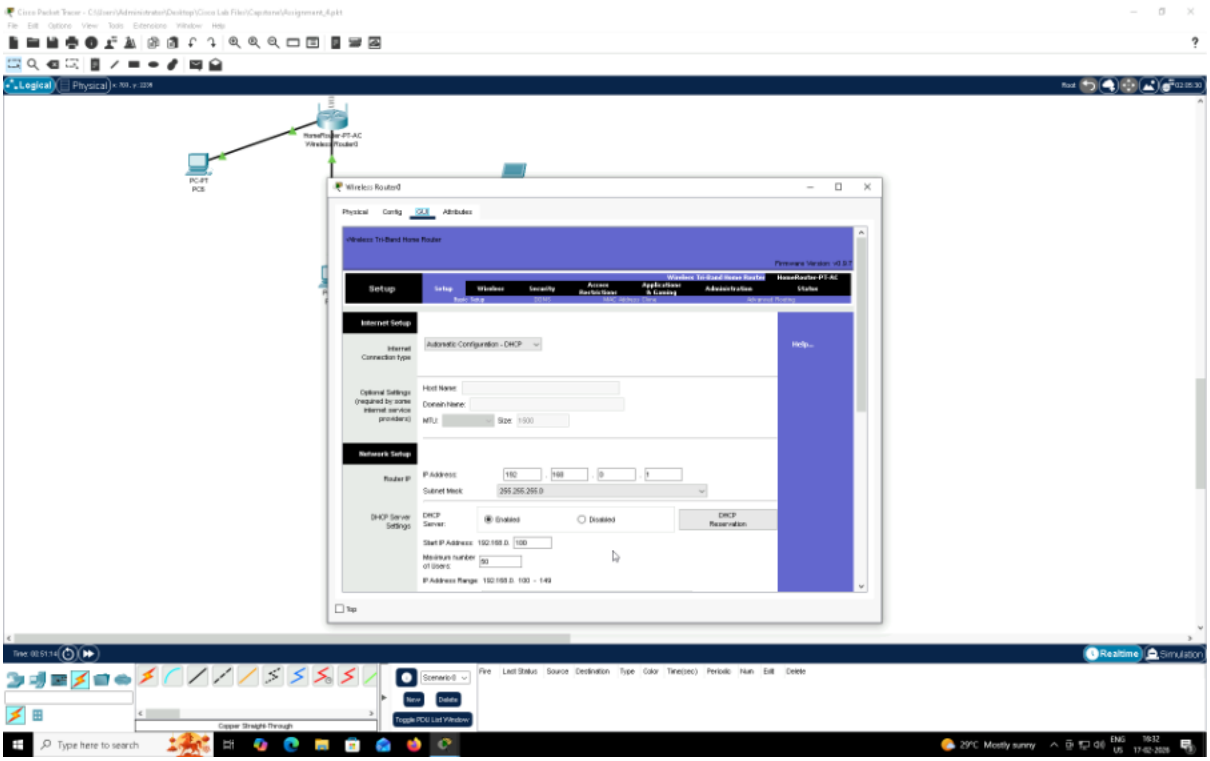


The wireless client successfully authenticated and joined the wireless network using WPA2 encryption.

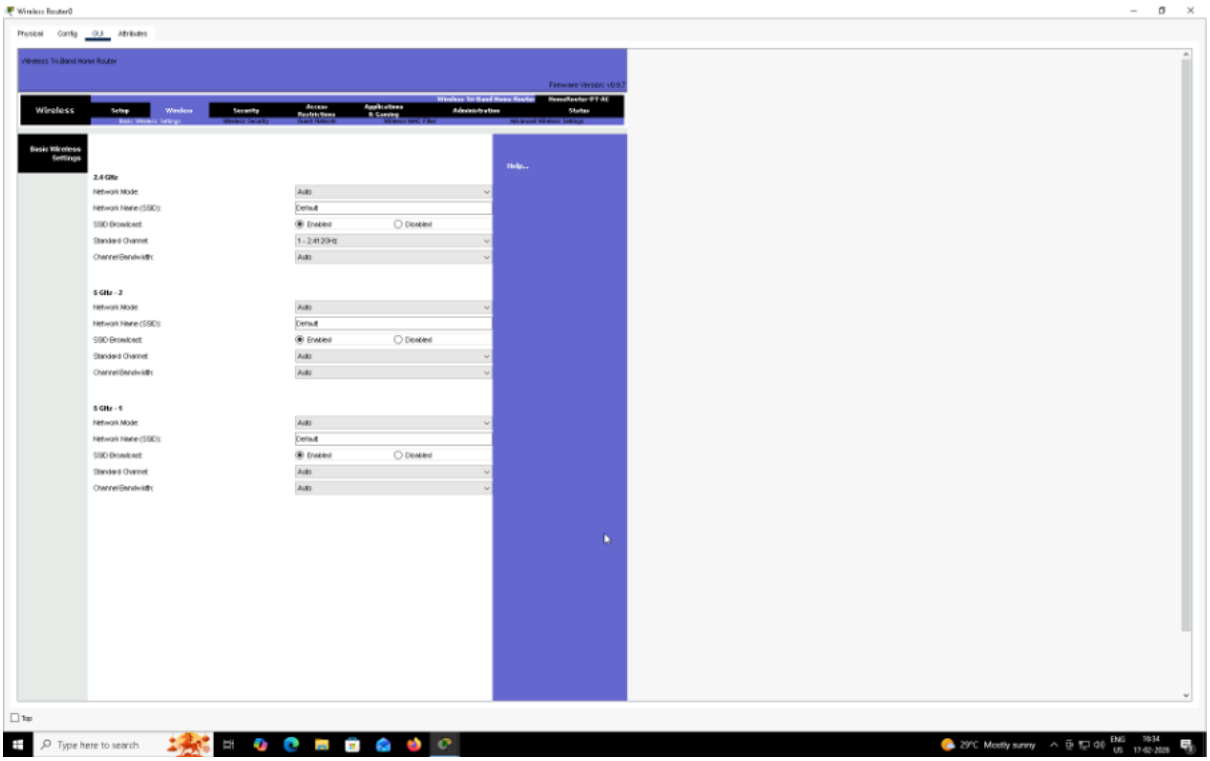


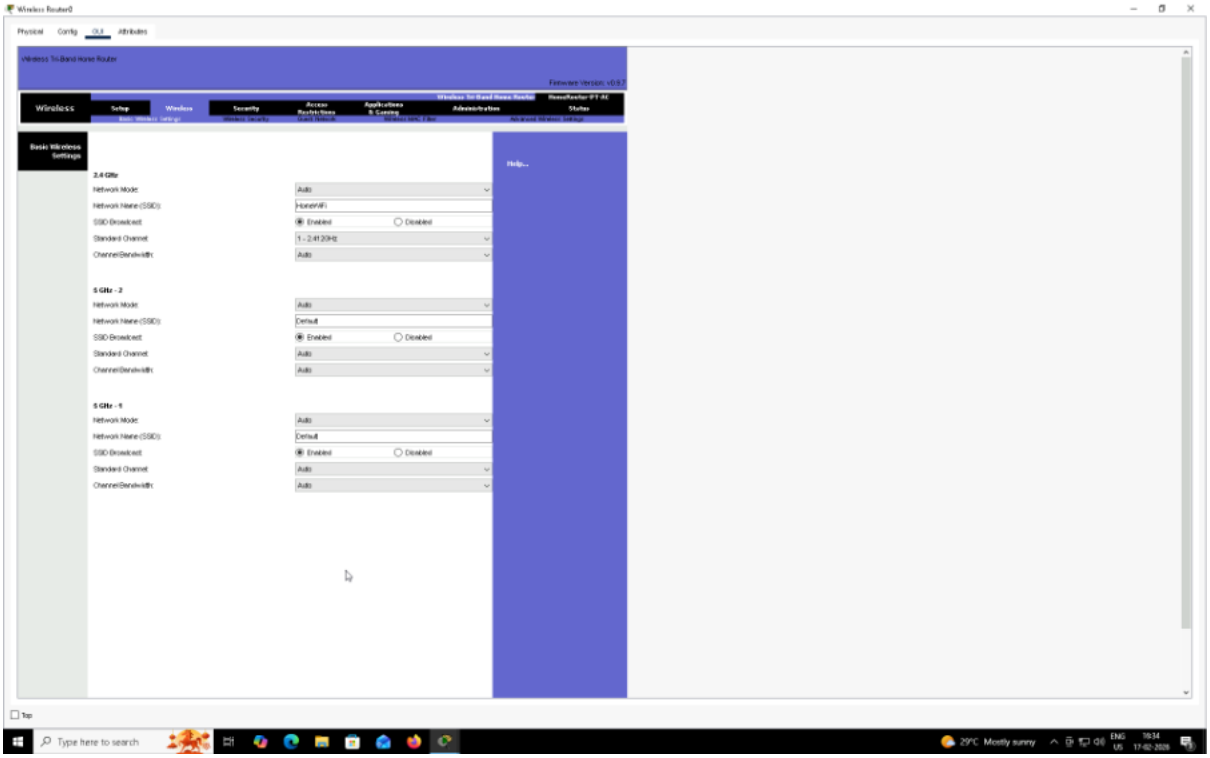
Successful ping replies confirmed proper wireless communication and IP configuration through DHCP.



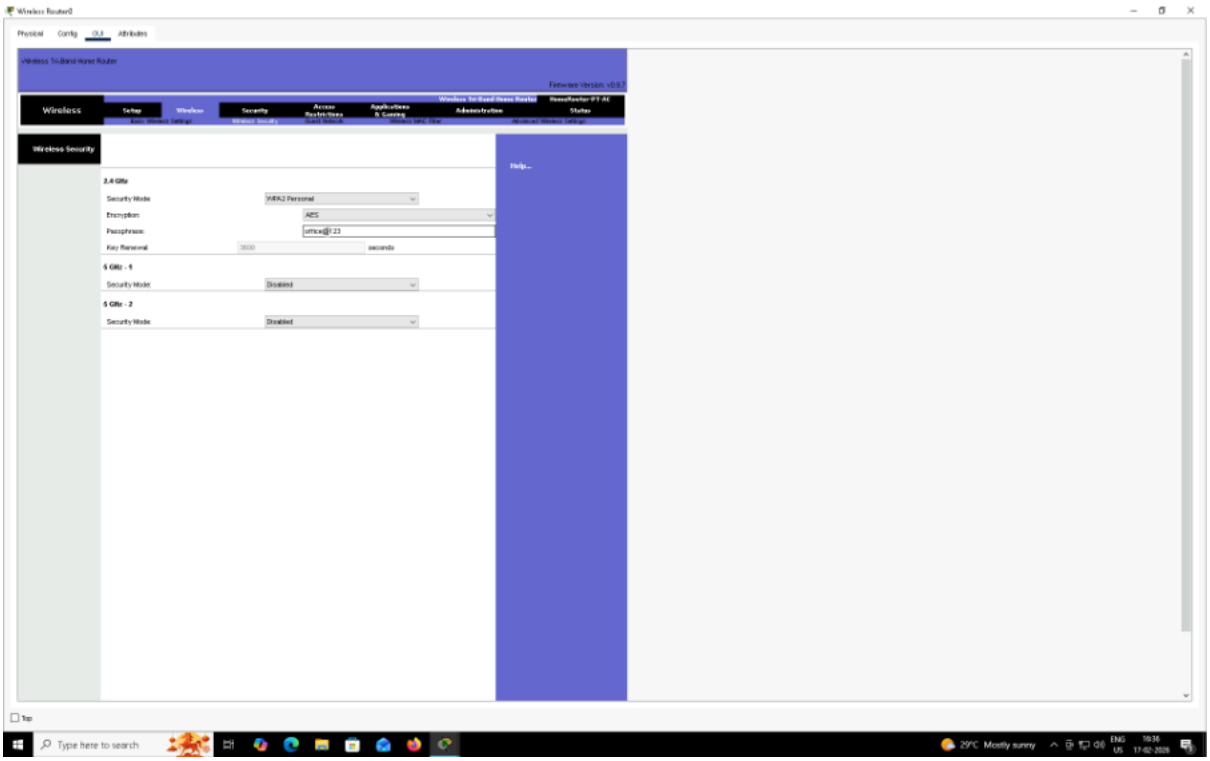


The wireless router was configured to automatically provide IP addresses and act as the gateway for the office network.

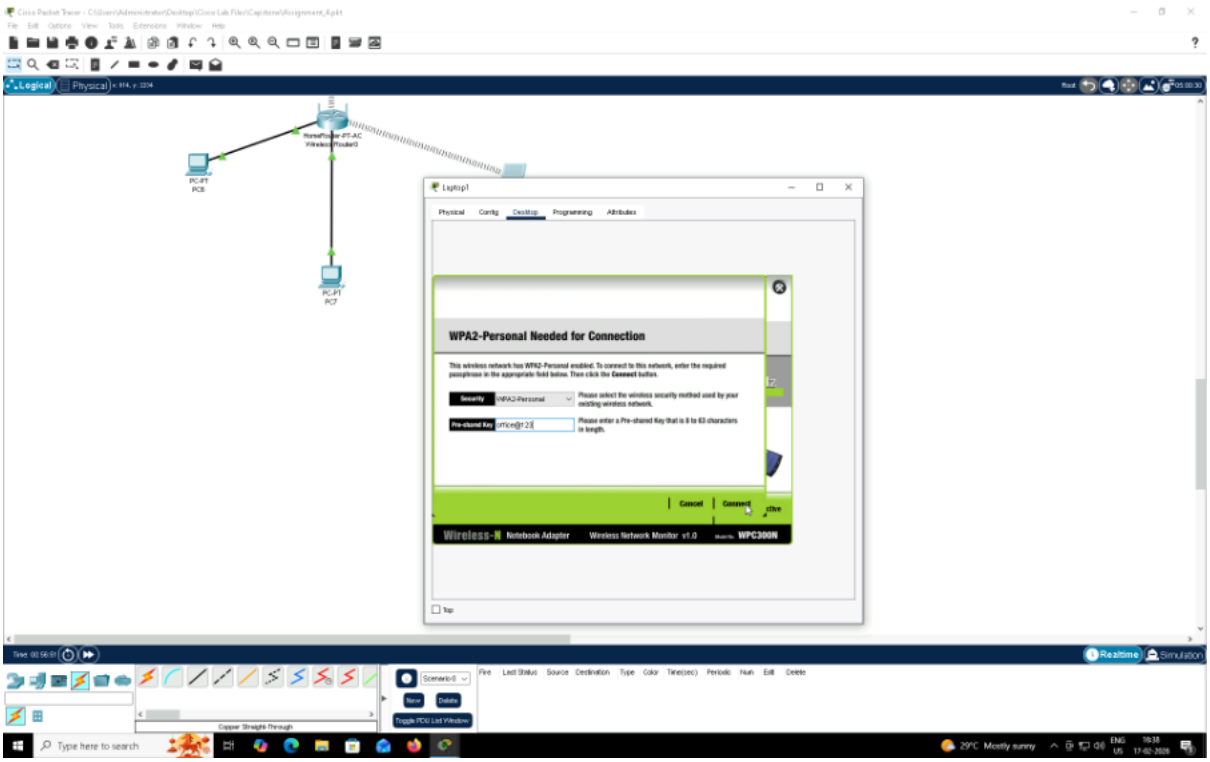


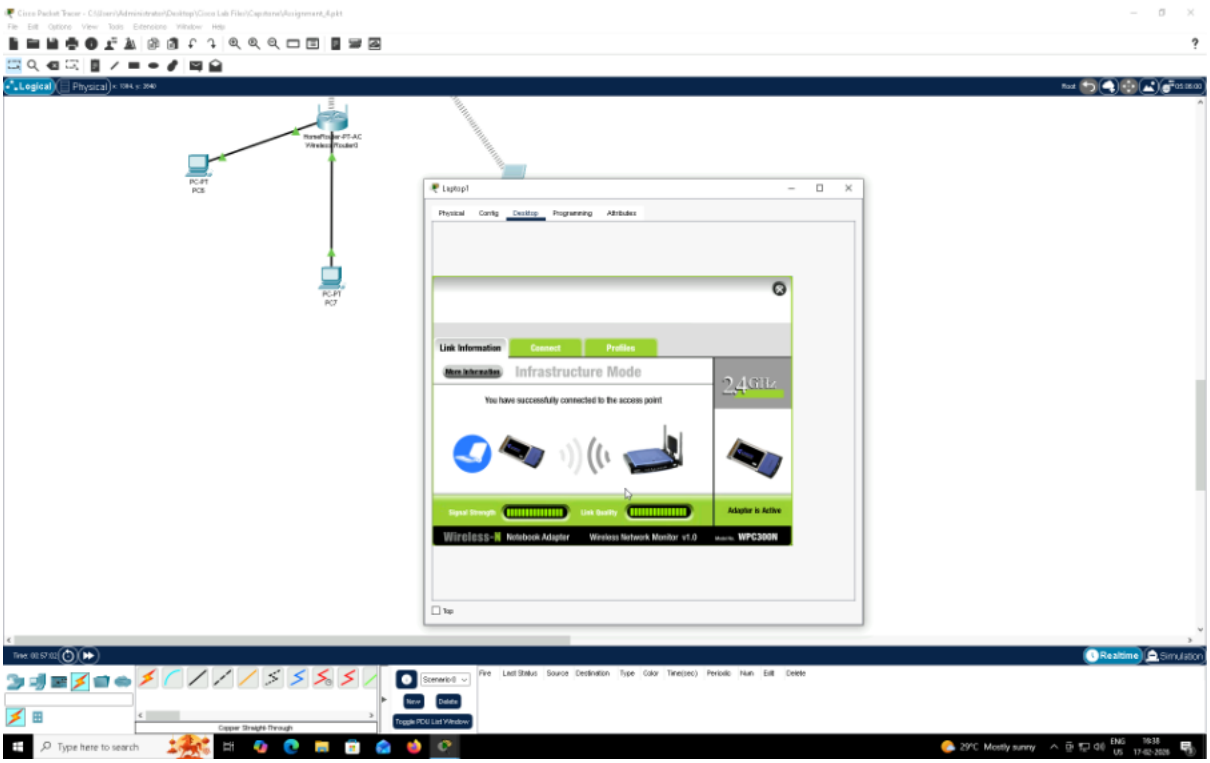


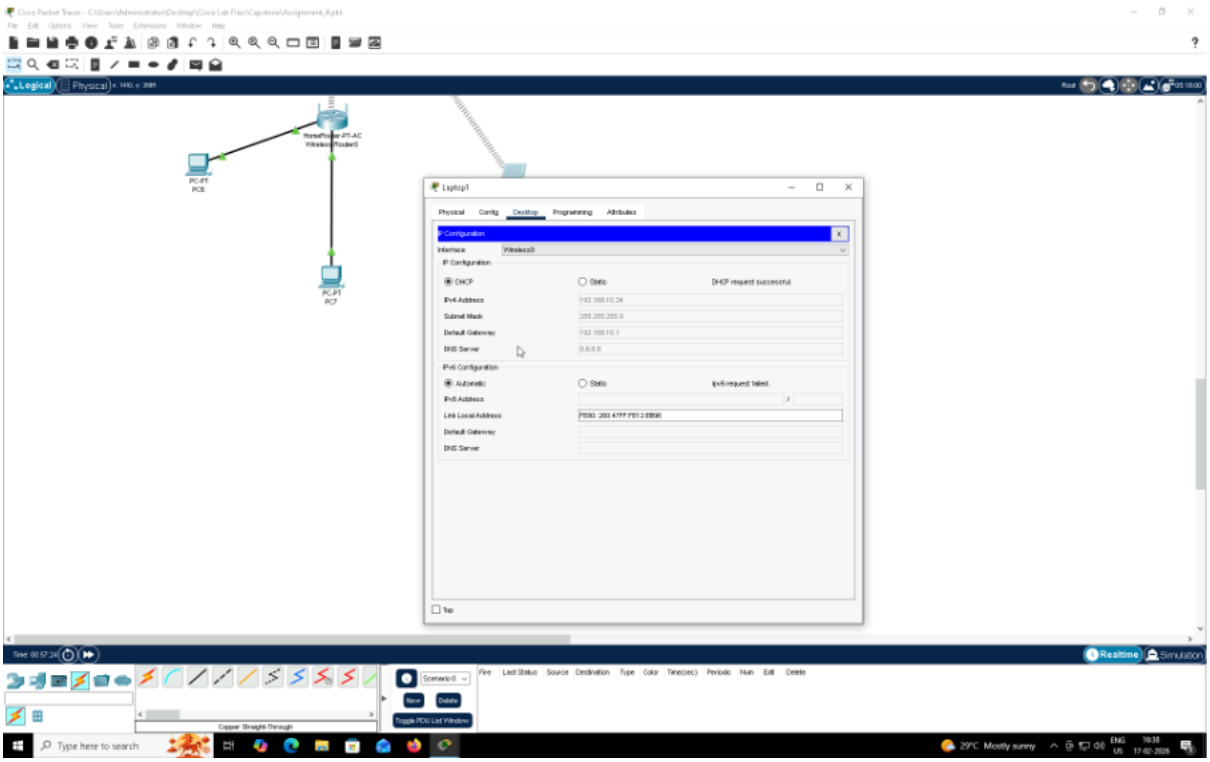
The wireless network name (SSID) was configured so that users can identify and connect to the office network.

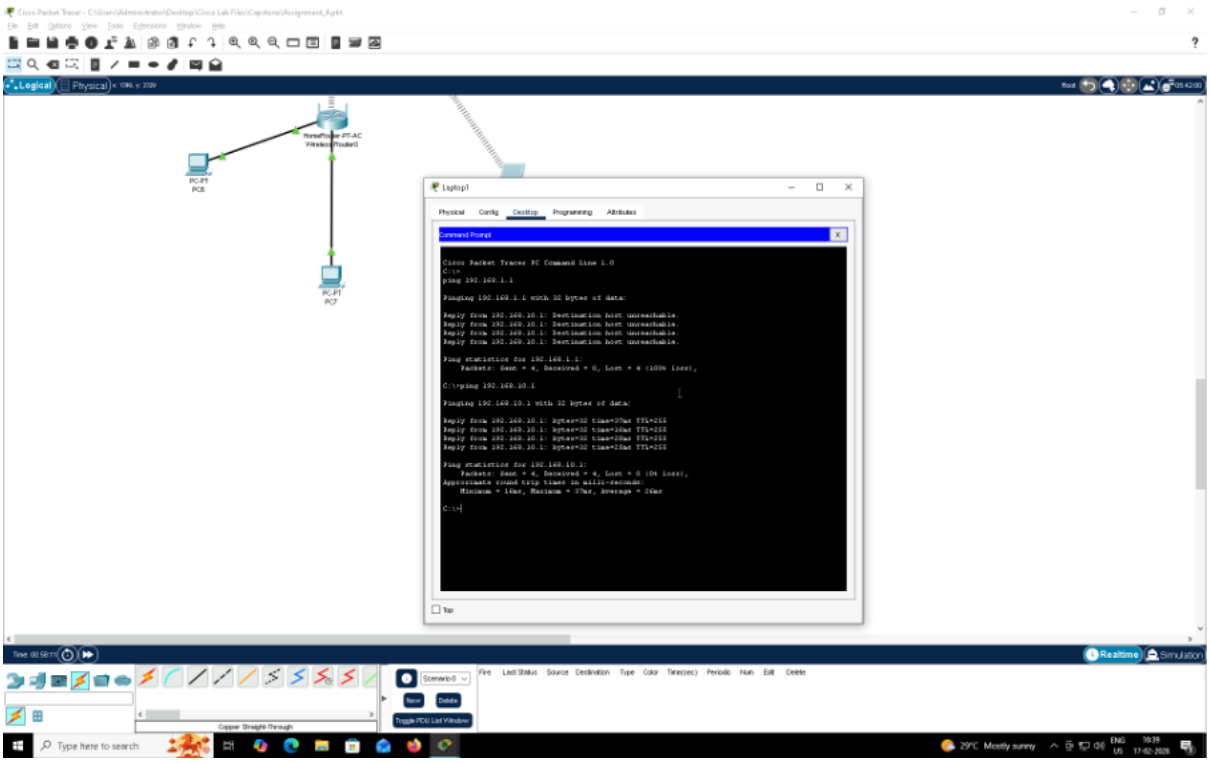


WPA2 authentication was enabled to secure the wireless network and prevent unauthorized access.






Laptop1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
Subnet Mask
Default Gateway
DNS Server
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Default Gateway
DNS Server
Top
PC-PT PC8
PC-PT PC7
Realtime
Simulation

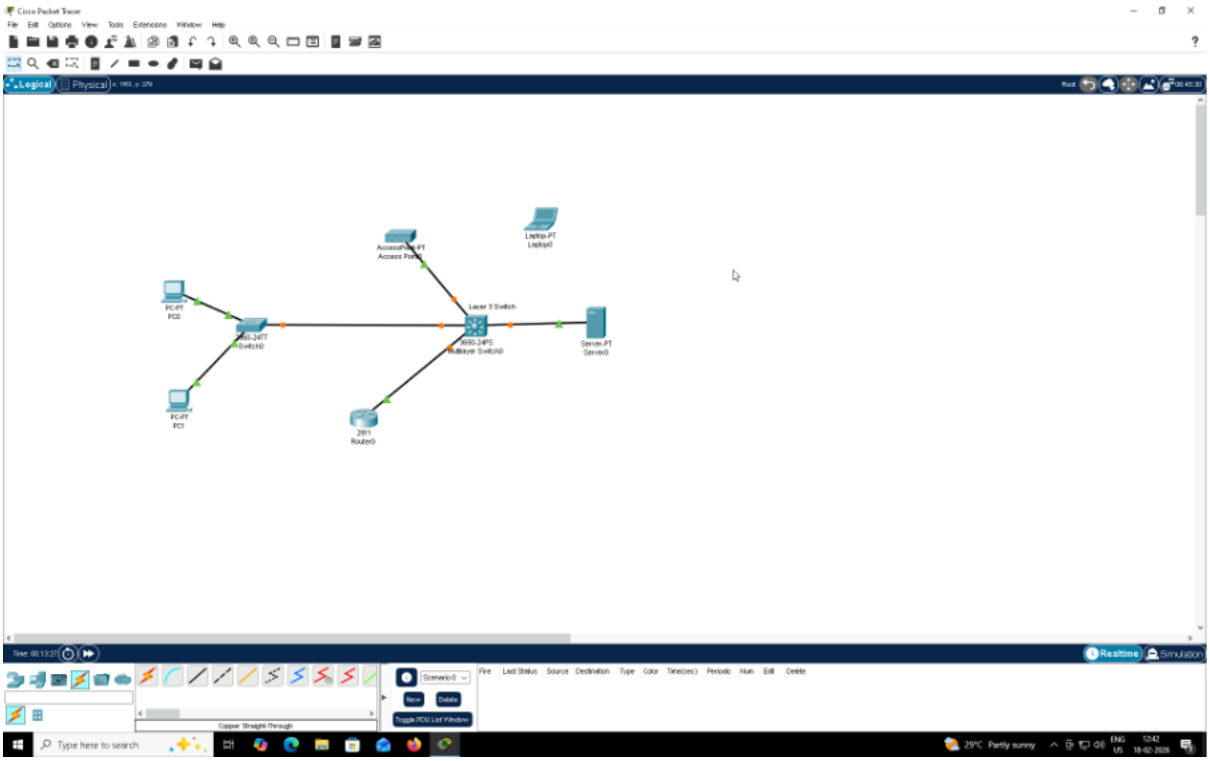

Laptop1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=27ms TTL=255
Reply from 192.168.10.1: bytes=32 time=24ms TTL=255
Reply from 192.168.10.1: bytes=32 time=23ms TTL=255
Reply from 192.168.10.1: bytes=32 time=23ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 23ms, Maximum = 27ms, Average = 24ms
C:\>
Top
PC-PT PC8
PC-PT PC7
Realtime
Simulation

The laptop was connected to the configured SSID using WPA2 authentication. The device successfully obtained an IP address via DHCP and connectivity to the router was verified using ping.

Assignment 5: Network Device Configuration Simulation

As part of this assignment, you need to complete below tasks:

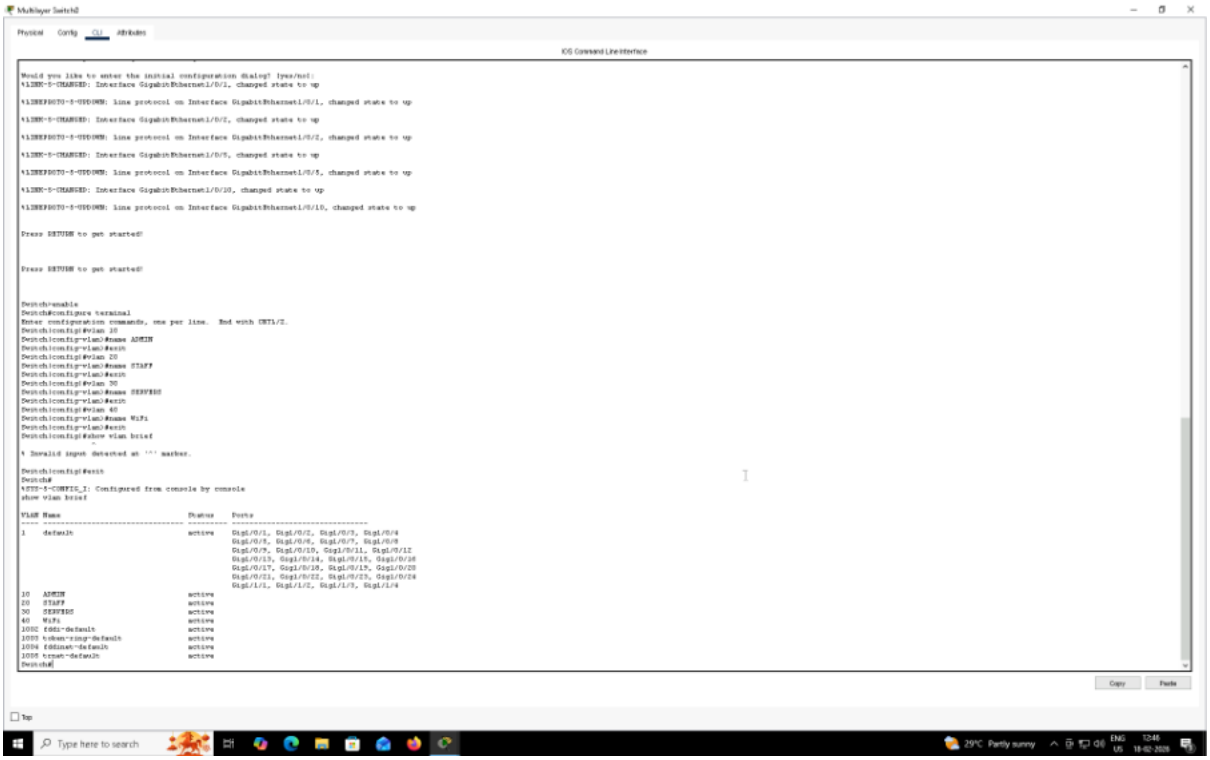
1. Configure and interconnect: Layer 2 switch, Layer 3 switch, Router, Firewall, Wireless Access Point.
2. Demonstrate device-specific functionalities.



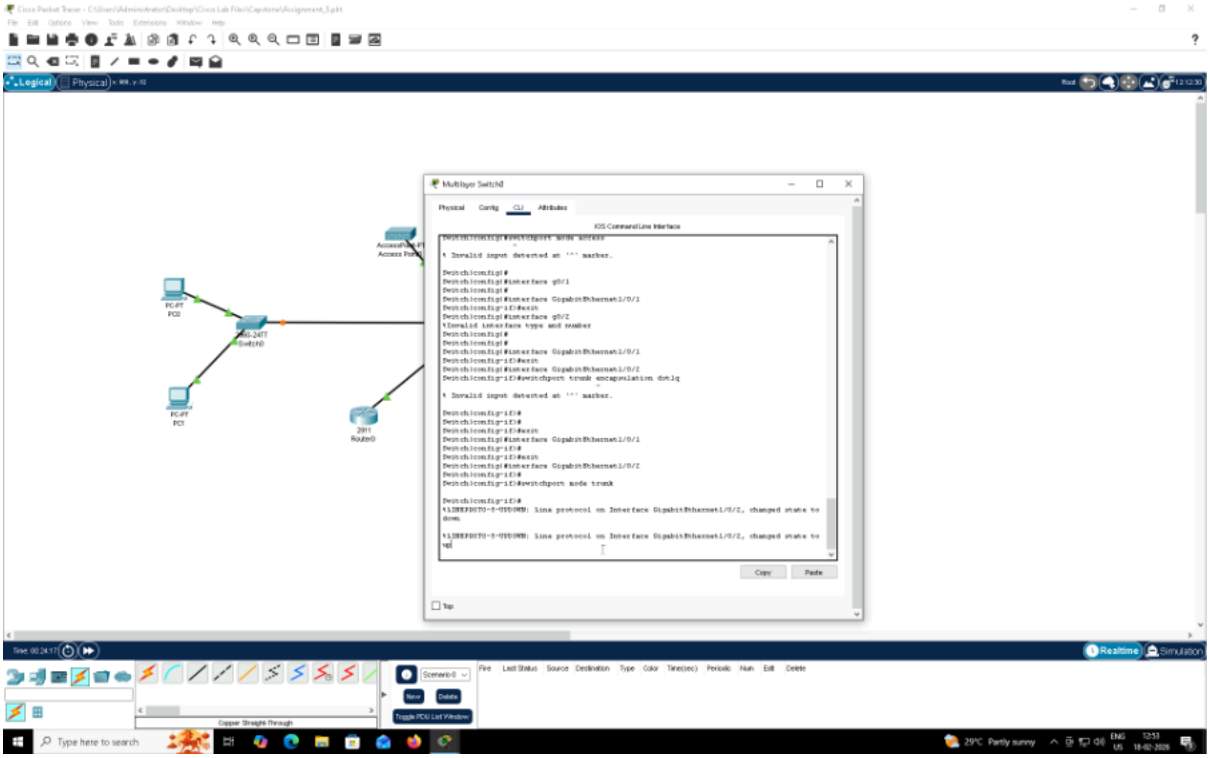
A hierarchical enterprise topology was created using a router, layer-3 switch, layer-2 switch, server, access point, and client devices to simulate real-world network infrastructure.

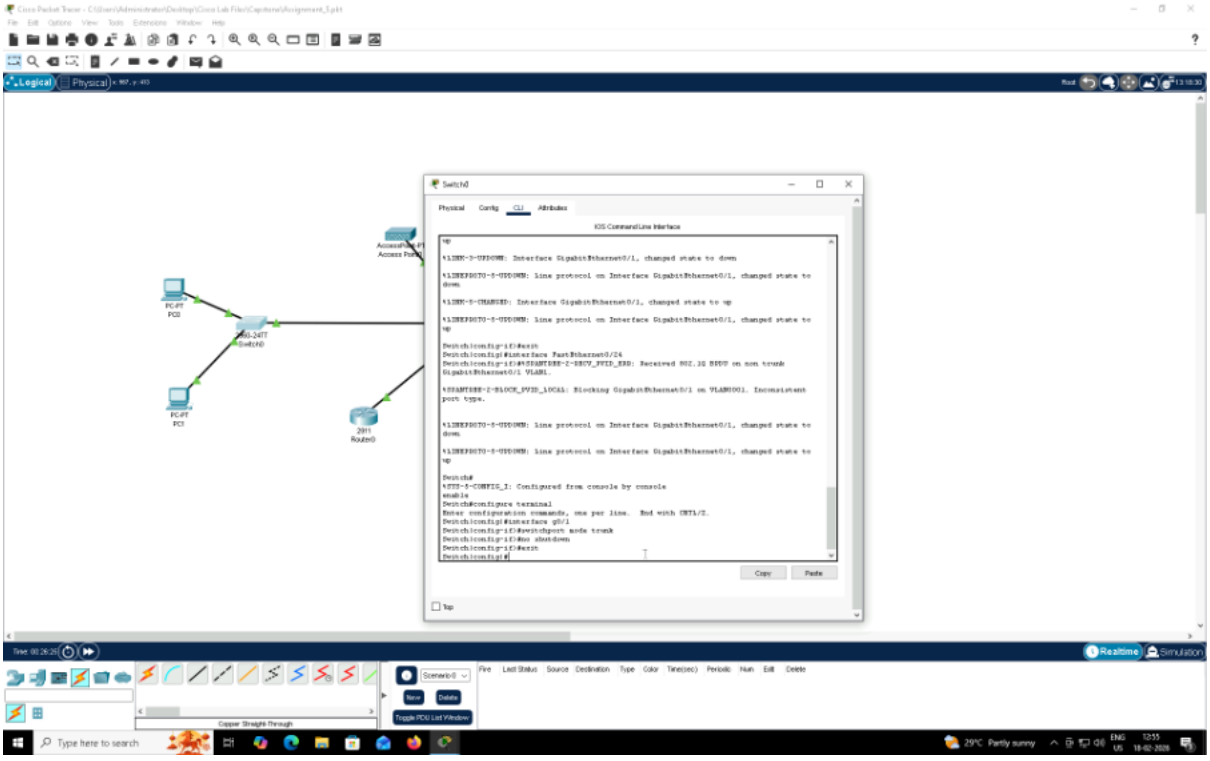
VLAN	Department	Devices
10	Admin	PC0
20	Staff	PC1
30	Servers	Server

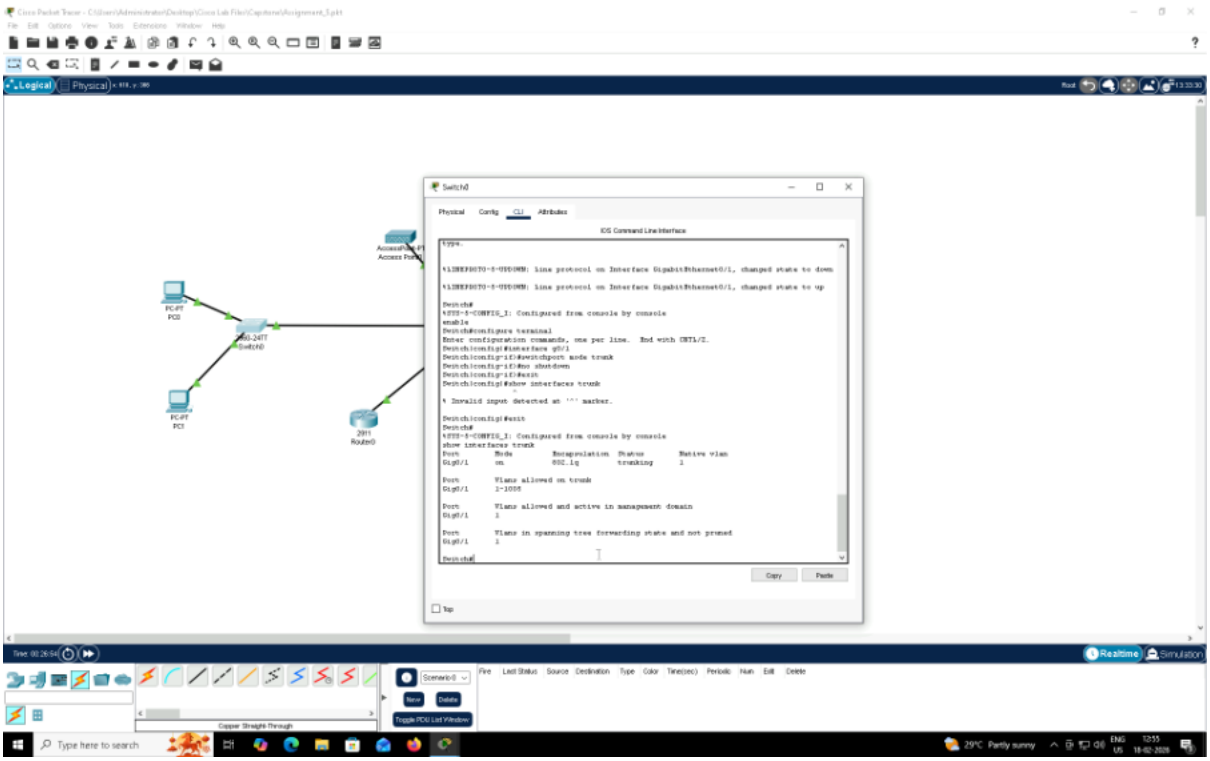
40	Wireless	Laptop / AP

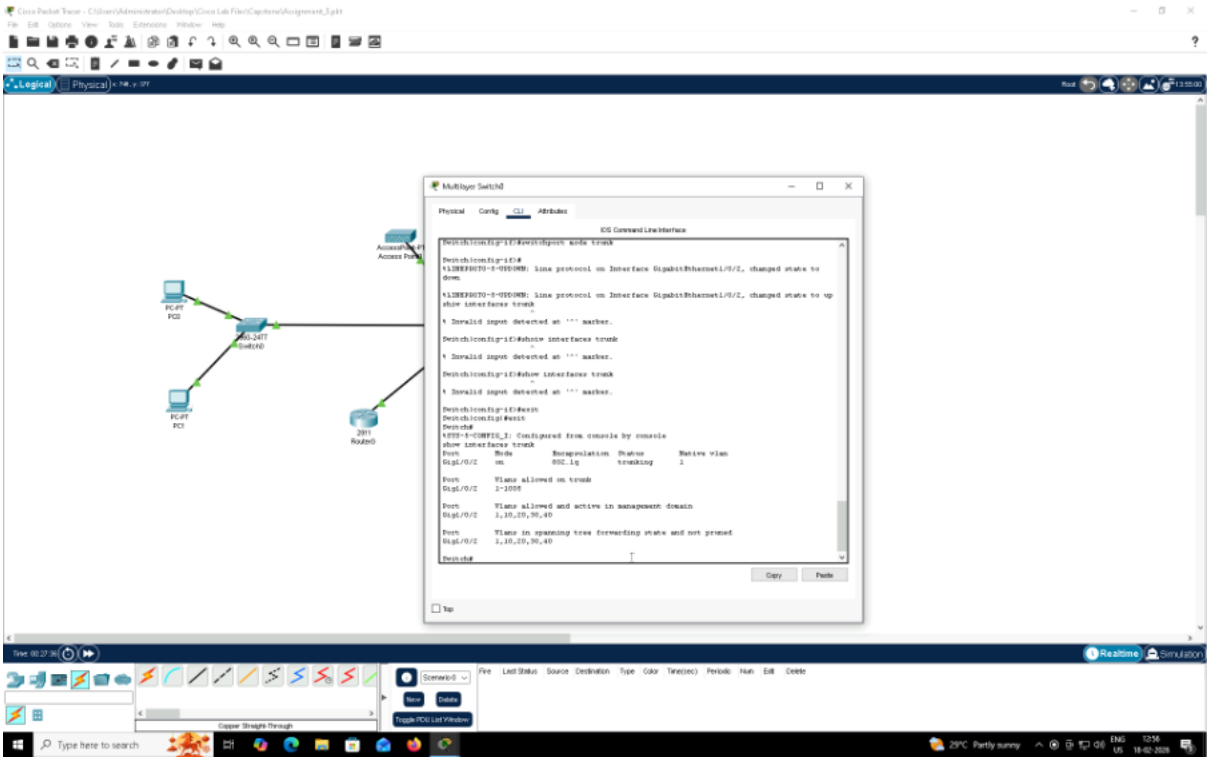


Multiple VLANs were created on the layer-3 switch to logically segment departments and reduce broadcast traffic within the enterprise network.

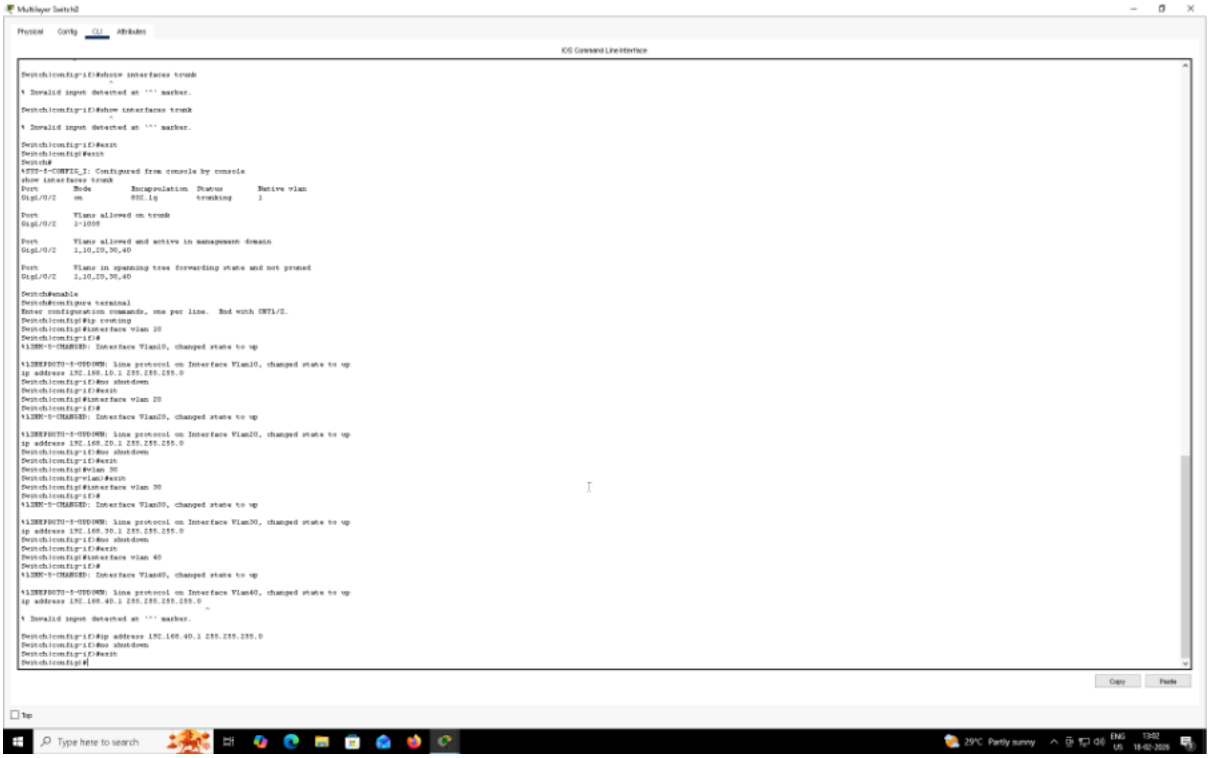

Multilayer Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config)#switchport mode access
% Invalid input detected at '^' marker.
Switch(config)#
Switch(config)#interface g0/1
Switch(config)#
Switch(config)#interface GigabitEthernet1/0/1
Switch(config-if)#exit
Switch(config)#interface g0/2
%Invalid interface type and number
Switch(config)#
Switch(config)#
Switch(config)#interface GigabitEthernet1/0/1
Switch(config-if)#exit
Switch(config)#interface GigabitEthernet1/0/2
Switch(config-if)#switchport trunk encapsulation dot1q
% Invalid input detected at '^' marker.
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface GigabitEthernet1/0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface GigabitEthernet1/0/2
Switch(config-if)#
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
vp
Copy
Paste
Top
PC-PT PC0
PC-PT PC1
2911 Router0
Realtime
Simulation
Copper Straight-Through
Type here to search
29°C Partly sunny
ENG US
12:53
18-02-2020

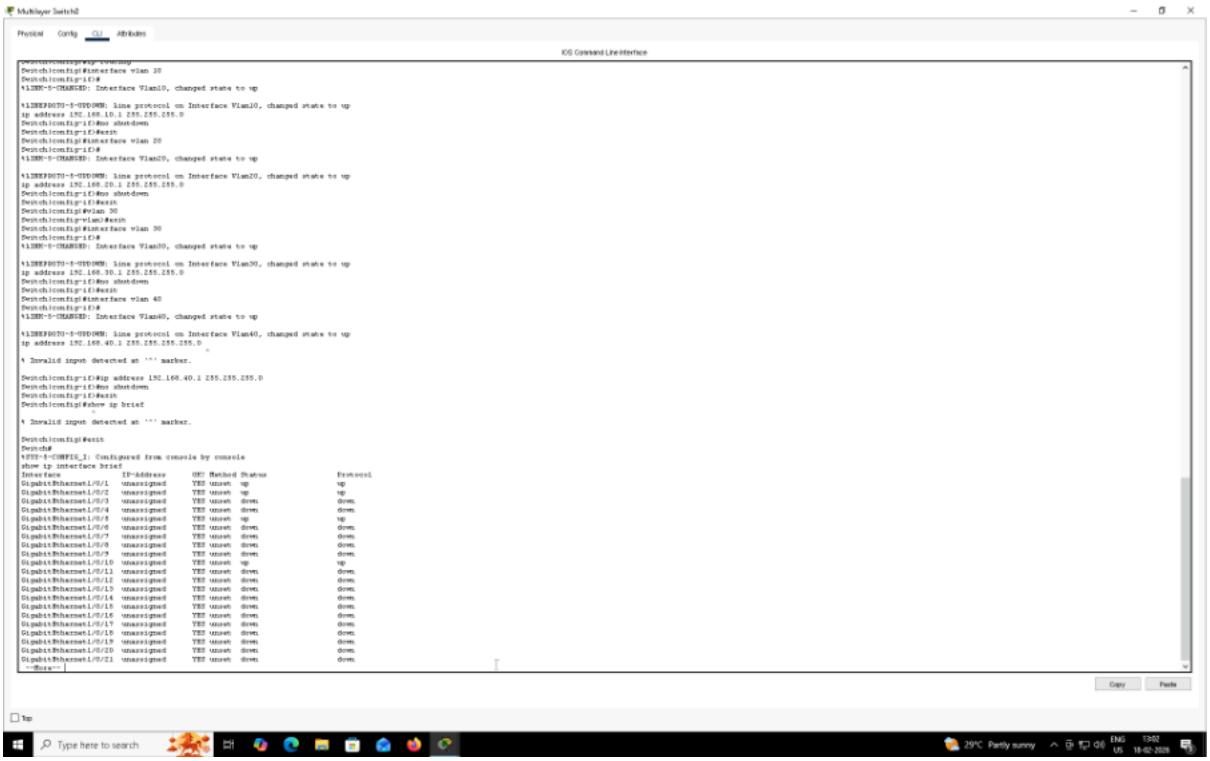

Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
vp
%LINK-5-UPDOWN: Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/24
Switch(config-if)#%GIGAMIBE-2-SECV_PVID_ERR: Received 802.1Q BPDU on non trunk GigabitEthernet0/1 VLAN1.
%GIGAMIBE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent port type.
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Switch#
%SYS-5-CONFIG_I: Configured from console by console
enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Copy
Paste
Top
PC-PT PC0
PC-PT PC1
2911 Router0
Realtime
Simulation
Copper Straight-Through
Type here to search
29°C Partly sunny
ENG US
12:55
18-02-2020

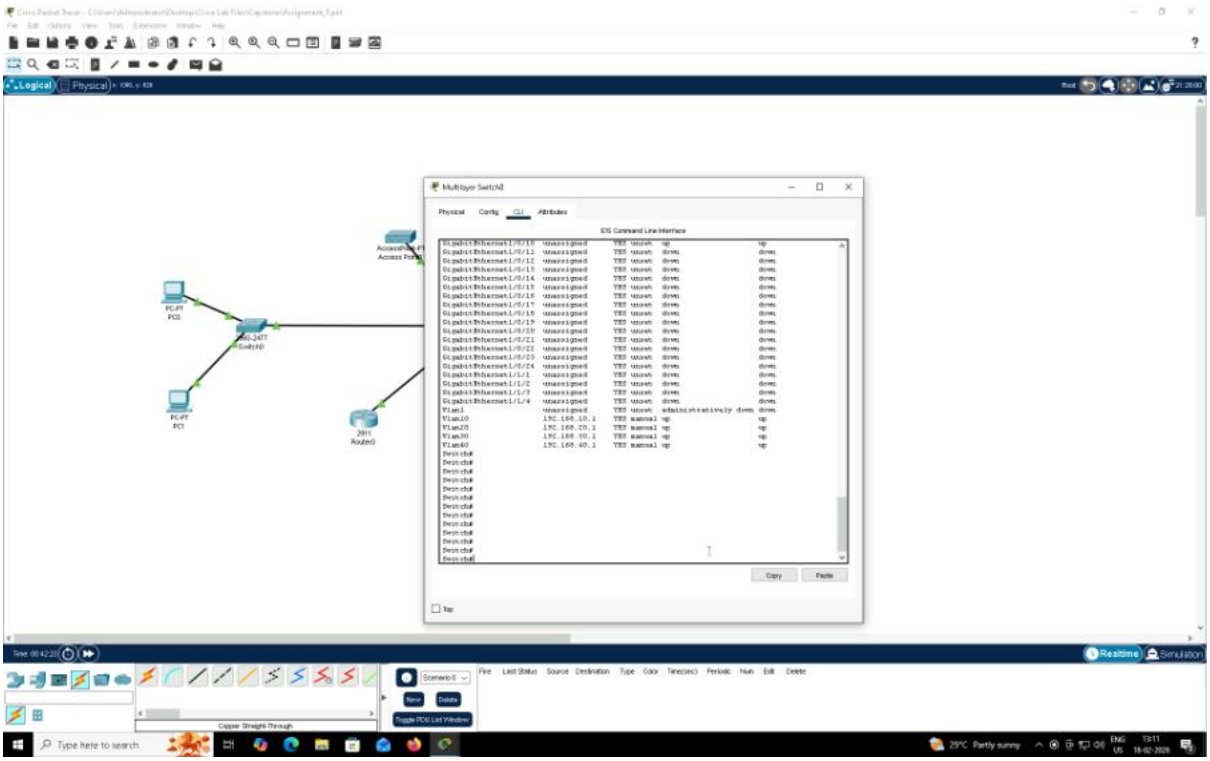




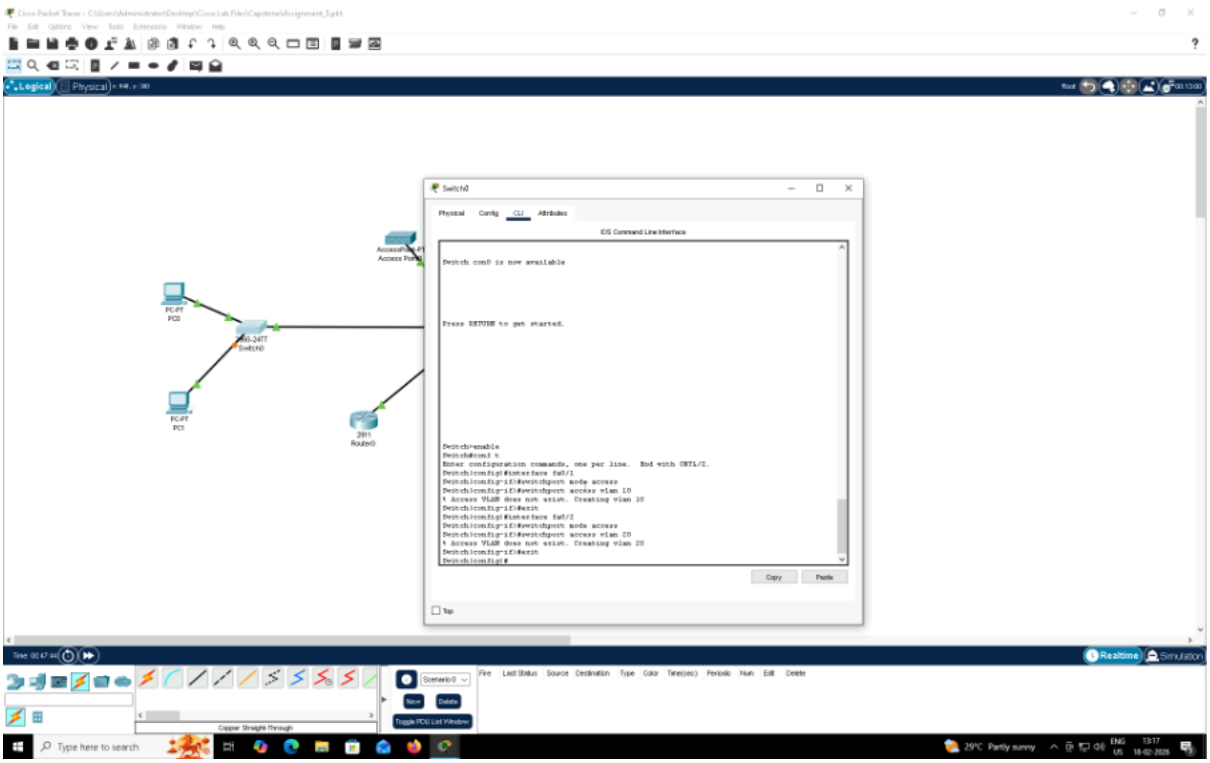
A trunk link was configured between the layer-2 switch and layer-3 switch using IEEE 908.1Q encapsulation. This allows multiple VLAN traffic to travel through a single physical connection.

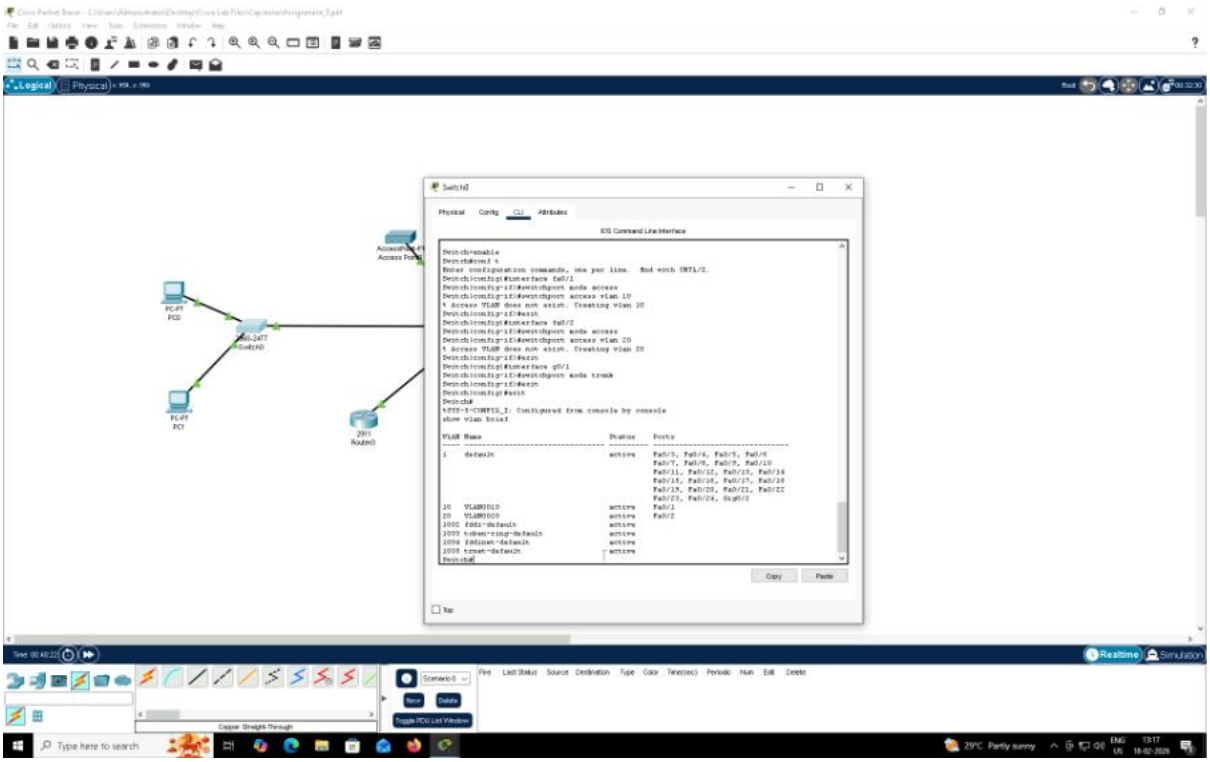

Multilayer Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config-if)#show interfaces trunk
^
% Invalid input detected at '^' marker.
Switch(config-if)#show interfaces trunk
^
% Invalid input detected at '^' marker.
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces trunk
Port Mode Encapsulation Status Native vlan
Gig0/0/2 on 802.1q trunking 1
Port Vlans allowed on trunk
Gig0/0/2 1-1005
Port Vlans allowed and active in management domain
Gig0/0/2 1,10,20,30,40
Port Vlans in spanning tree forwarding state and not pruned
Gig0/0/2 1,10,20,30,40
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#vlan 30
Switch(config-vlan)#exit
Switch(config)#interface vlan 30
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
ip address 192.168.30.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 40
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up
ip address 192.168.40.1 255.255.255.255.0
^
% Invalid input detected at '^' marker.
Switch(config-if)#ip address 192.168.40.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Copy
Paste
Top
Type here to search
29°C Partly sunny
ENG
US
13:02
18-02-2026


Multilayer Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config)#ip routing
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#vlan 30
Switch(config-vlan)#exit
Switch(config)#interface vlan 30
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
ip address 192.168.30.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 40
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up
ip address 192.168.40.1 255.255.255.255.0
^
% Invalid input detected at '^' marker.
Switch(config-if)#ip address 192.168.40.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#show ip brief
^
% Invalid input detected at '^' marker.
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet1/0/1 unassigned YES unset up up
GigabitEthernet1/0/2 unassigned YES unset up up
GigabitEthernet1/0/3 unassigned YES unset down down
GigabitEthernet1/0/4 unassigned YES unset down down
GigabitEthernet1/0/5 unassigned YES unset up up
GigabitEthernet1/0/6 unassigned YES unset down down
GigabitEthernet1/0/7 unassigned YES unset down down
GigabitEthernet1/0/8 unassigned YES unset down down
GigabitEthernet1/0/9 unassigned YES unset down down
GigabitEthernet1/0/10 unassigned YES unset up up
GigabitEthernet1/0/11 unassigned YES unset down down
GigabitEthernet1/0/12 unassigned YES unset down down
GigabitEthernet1/0/13 unassigned YES unset down down
GigabitEthernet1/0/14 unassigned YES unset down down
GigabitEthernet1/0/15 unassigned YES unset down down
GigabitEthernet1/0/16 unassigned YES unset down down
GigabitEthernet1/0/17 unassigned YES unset down down
GigabitEthernet1/0/18 unassigned YES unset down down
GigabitEthernet1/0/19 unassigned YES unset down down
GigabitEthernet1/0/20 unassigned YES unset down down
GigabitEthernet1/0/21 unassigned YES unset down down
--More--
Copy
Paste
Top
Type here to search
29°C Partly sunny
ENG
US
13:02
18-02-2026

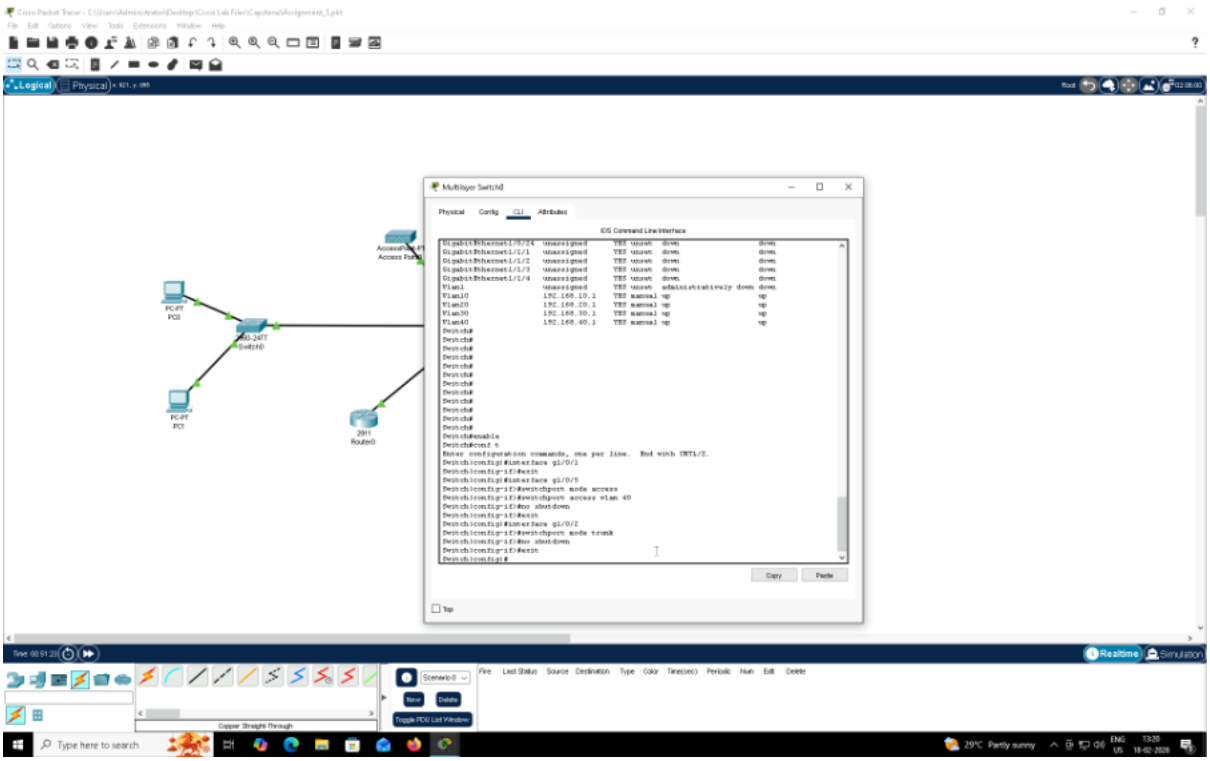


Switch virtual interfaces (SVIs) were configured on the multilayer switch and assigned IP addresses to act as default gateways for each VLAN. IP routing was enabled, allowing communication between devices in different VLANs.





Access switch ports were assigned to respective VLANs based on department segmentation. A trunk link was configured between the access switch and multilayer switch to allow multiple VLAN traffic.



The server was directly connected to the multilayer switch and assigned to VLAN 40 by configuring the corresponding switch port in access mode. This allows centralized services to be accesses by all VLANs through inter-VLAN routing.

